

Polymer Grafted Inorganic Nanoparticles, Preparation, Properties, and Applications: A Review

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The addition of inorganic nanoparticles to polymers causes changes in the properties and applications of both the nanoparticles and polymers. In this review article the functionalization of nanoparticles, grafting of polymers on nanoparticles, properties, and applications of a number of nanoparticle-polymer composites are discussed. It can be seen that in all these cases attempts have been made to uniformly disperse nanoparticles in polymer matrix for the purpose of gaining stability and processability. This review is limited to the recent developments in the field of metal/metal oxide nanoparticles only. The carbon nanotube functionalized with polymer was seen as a very large area and it is omitted here due to size constraints.

Keywords Hybrid materials, functionalization, grafting, polymers, nanoparticles

1. Introduction

Metal nanoparticles are one of the important classes of nanoparticles due to its significance in applied and theoretical fields. These nanoparticles have generated intense interest due to their fascinating applications in the field of bio-imaging, bio-medicine, bio-technology, catalysis, cross linkers, fuel cells, sensors, flame retardant, UV shielding, optical, electronic, magnetic, etc.^{1–13} However, the surface to volume ratio of nanoparticles often causes their aggregation and that would irreversibly affect their characteristic properties. But surface modification techniques have been employed to solve this problem.¹⁴ Various chemical and physical methods are used for this modification. The chemical functionalization is permanent but physical adsorption is temporary. During the past decades a lot of attention has been given to nanoparticle-polymer composites due to their extraordinary properties and wide applications. Grafting of metal nanoparticles with polymer is mainly done by two methods—“*grafting from*” (growing of polymer chains on the surface of nanoparticles) and “*grafting to*” (reaction between functionalized polymer chains and nanoparticles). In order to graft polymer chains on to nanoparticles, various surface initiated polymerization techniques such as ATRP, NMP, RAFT, SIP, cationic, anionic, free radical, frontal, and

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ring opening polymerization were employed.^{15–28} The resultant polymer grafted inorganic nanoparticles have enhanced or controlled properties for particular applications.

2. Nanoparticles

2.1. Silica Nanoparticles

Large scale production of silicon nanoparticles are done by flow methods. SNPs have broad excitation band and size-dependent optical emission, so they can be used for detection purposes. However, the surface of the nanoparticles undergoes some modification for improving surface defects and to achieve intense and stable luminescence. Due to the potential applications, the area of inorganic (silica)-organic (polymer) nanocomposite was extensively investigated.

2.1.1. With Polymethacryloyloxyethyl Phosphorylcholine. Here radical graft polymerization of MPC was used for the preparation of biocompatible polymer on silica nanoparticles surface.

The amino groups on the silica surface were obtained by the reaction of γ -APS with silanol groups.²⁹ The azo groups can be introduced into the surface of silica nanoparticle (Silica-NH₂) by the reaction of surface amino groups with ACPA.³⁰

Silica-NH₂, TDI, ACPA, DMSO, and 2-picoline (2-methylpyridine) were stirred under dry nitrogen. The silica-azo was prepared by centrifugation and then dried in vacuum. The trichloroacetyl groups on silica (Silica-COCCl₃) were introduced by the reaction of trichloroacetyl isocyanate with the amino groups on the silica surface.¹

The radical graft polymerization of MPC was done in two ways.

- a. Polymerization of MPC initiated by silica surface containing azo group
All the processes were carried out in a sealed tube under vacuum. Silica-azo, MPC, and DMSO solvent were frozen in liquid nitrogen bath and heated. The product was dispersed in DMSO and centrifuged. This dispersion and centrifugation process was repeated. The graft polymerized product was dried at room temperature.
- b. Graft polymerization of MPC initiated by Silica-COCCl₃ and Mo(CO)₆
Polymerization tube containing Silica-COCCl₃, Mo(CO)₆, MPC, and DMSO was cooled, degassed, thawed, and heated with stirring. The product was purified and isolated as before.

FT-IR spectrum of poly (MPC) grafted nanoparticles showed characteristic peaks of PO₄ group, ester bond, and methylene bond indicated the successful grafting. The density of the polymer and functional groups on the nanoparticle surface was determined by TGA and DSC analysis, respectively.

When the hydrophilic nature of the surface increases, the grafting density also increases. The grafted silica acts as stabilizer of chloroform droplets in water chloroform mixture. The poly(MPC)-grafted silica has an extremely hydrophilic nature, shows amphiphilic properties and non thrombogenic characteristics (confirmed by the Lee-White method). Poly(MPC)-grafted silica has an ability to delay the coagulation time of blood.¹⁴

2.1.2. With Polyethylene Glycol. The “Grafting to” approach was used for the synthesis of Pegylated SNPs. Silicon nanoparticles were prepared by laser pyrolysis.^{31,32} The silicon nanoparticles were dispersed in ethanol using ultrasonic tip. Water was added for the

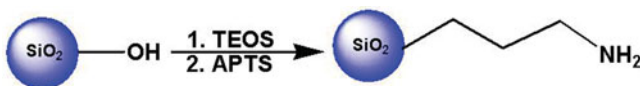


Figure 1. Preparation of amino functionalized silica.

hydrolysis. In order to start the reaction, TEOS and ammonia were stirred overnight. Then APTS was added. The particles were then recovered and purified (Fig. 1).

In order to generate azide functional group on silicon nanoparticles, first the Si-NH₂ is solubilized in dried DMSO under nitrogen. DMAP and Bromoisopropionyl bromide were added and stirred. A solution of NaN₃ in a mixture of DMF/water was transferred to the reaction medium and the stirring was continued. It was purified by centrifugation from a mixture of water and ethanol.

Two different routes were used to graft PEG chains on to the oxidized surface of nanoparticles.

The CuAAC Route: Ultrasonic tip was used for the dispersion of Si-N₃ in water. Then Cu(I)Br, PMDETA were added. Alkyne-PEG was added into the reaction mixture and stirred. Centrifugation-redispersions in water, ethanol, and ether were done for purification.

The NHS ester Route: Water dispersed amino-coated SiNPs (Si-NH₂) were added in to PEG-NHS in DMSO and stirred well. TEA was the catalyst of the reaction (Fig. 2).

The obtained nanohybrids were characterized by IR, TGA, and TEM. From these characterizations the grafting density, SNPs size, etc., were determined.

PEG functionalization fictionalization increases the potential of the silicon particles for use in bioapplications. Pegylated silicon nanoparticles were applied in bio-imaging as fluorescent tracers.³³

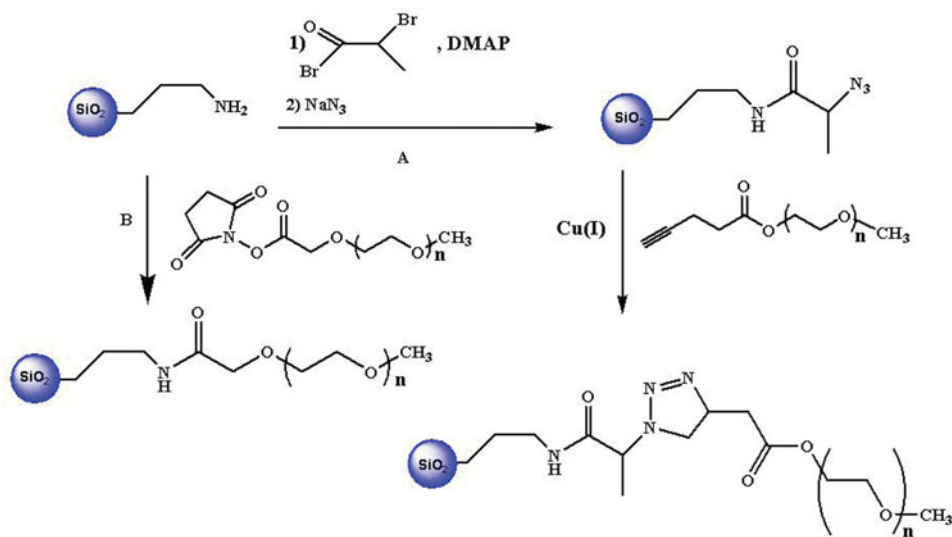


Figure 2. Pegylation via NHS ester route (A) or CuAAC route (B).

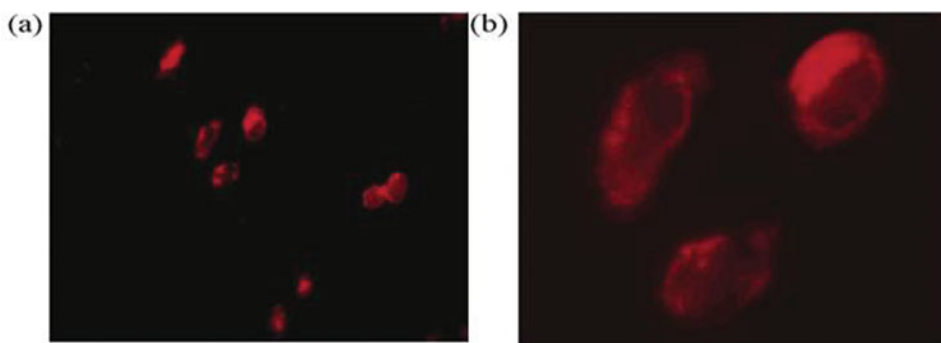


Figure 3. Fluorescent microscopy image of PAAc-Si nanoparticles labeled CHO cells. Low (a) and high (b) magnification.³⁵ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

2.1.3. With Poly(acrylic acid). UV-induced graft polymerization was used for the polymerization of water-soluble polymer acrylic acid (AAc) on silicon nanoparticles surface. Silicon nanoparticles were synthesized by Laser induced heating of silane³⁴ and then etched with a mixture of HNO₃ and HF. The resulting particle exhibits a bright red photoluminescence under UV excitation.

For the graft polymerization Si nanoparticles were introduced onto AAc in ethanol. The solution was bubbled and stirred under UV irradiation. The polymerization was done in a Rayonet Chamber Photochemical Reactor. After the reaction, a clear solution of well-dispersed nanoparticles with grafted polymers was obtained.

Covalent grafting of PAAc on silicon nanoparticles was confirmed by XPS and NMR spectroscopy. TEM and AFM analysis shows the high dispersible nature of the grafted nanoparticles.

The highly photoluminescent stable water-soluble PAAc-Si nanoparticles (Fig. 3) can be used as fluorescent bioimaging material for cells or other biological molecules. The water soluble Si quantum dot labels showed strong fluorescence images and give higher resistance in photo bleaching.³⁵

2.1.4. With Polymethyl Methacrylate. Grafting of PMMA onto silica nanoparticles were done by different polymerization techniques. Some of them are:

2.1.4.1. By Catalytic Chain Transfer Polymerization. In order to functionalize nanosilica with MPS, first it was dispersed in toluene and stirred. The mixture was then mixed with 3-MPS under nitrogen. The product, MPS functionalized nano silica, was isolated and washed with toluene.

CCTP of PMMA to functionalized nano silica was achieved by the following steps. MPS functionalized nano silica was mixed with AIBN, MMA, the prepared catalyst CoBF₃,³⁶ and toluene in a flask under oxygen free conditions. Then methanol was added into it after the reaction was quenched in ice bath. The product PMMA grafted nanosilica were isolated and washed with the toluene-acetone mixture.

The result of FT-IR, TGA, and SEM indicates PMMA was successfully grafted on to nano silica. The presence of vinyl groups on end of the polymer was confirmed by FT-Raman spectroscopy. Molecular weight of surface-grafted PMMA was calculated by GPC. TGA results show that the molecular weight of the polymer decreases with increasing CoBF concentration.³⁷

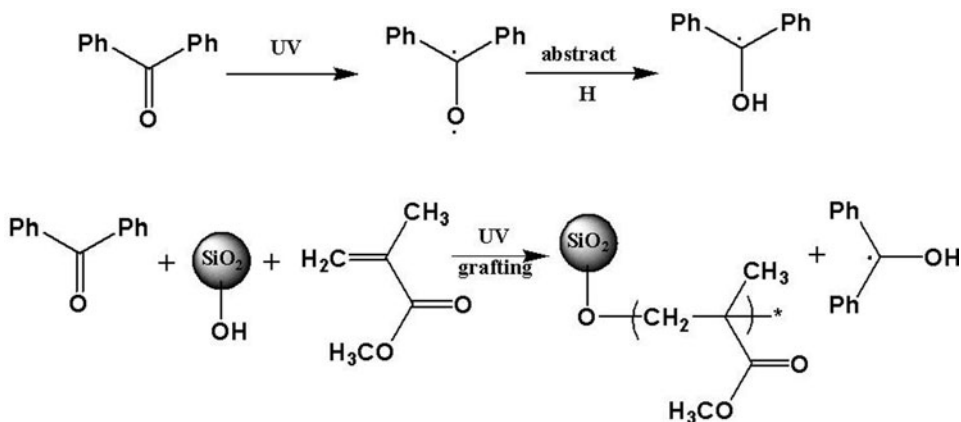


Figure 4. Grafting of PMMA onto silica nanoparticles.

2.1.4.2. By Surface-Initiated Polymerization. The spherical silica nanoparticles were prepared by sol-precipitation method using TEOS.³⁸ Silica, ethanol, and BP were charged into a glass Petri dish. The Petri dish was placed under a UV lamp and irradiated at room temperature. Benzophenone was used as an initiator; it absorbs in the far UV spectrum and forms free radicals. Free radical abstracted hydrogen from the silica surface and created a free radical center on the silica surface. After irradiation, MMA was added and irradiation was continued. Thus free radical on the silica surface initiated the growth of a grafted chain. Thus a covalent bond, Si–O–C is formed on the surface of silica. The reaction mixture was centrifuged in methanol. The sediment was redispersed in methanol and centrifuged. Thus the polymer-grafted silica nanoparticle was obtained by SIP (Fig. 4).

The indication of covalent bond between PMMA chains and silica nanoparticles was obtained from FT-Raman spectroscopy and XPS. Change in mean diameter and polymer content was determined by SEM and TGA.³⁹

2.1.4.3. By Atom Transfer Radical Polymerization. Ultrafine silica, thionyl chloride, and benzene were refluxed with stirring and dried.⁴⁰ The obtained chlorosilylated silica was mixed with dioxane, TBHP, and sodium bicarbonate (dehydrohalogenation catalyst). The mixture was stirred in the presence of nitrogen in the dark, washed with methanol, and dried in vacuum.⁴¹

Silica having peroxide groups and cyclohexanone were ultrasonicated. CuCl₂ and bipy were added into it. MMA was injected and the solution was degassed. Here the polymerization took place via reverse ATRP. The polymer-grafted silica nanocomposite was separated by centrifugation.

Grafting of polymer on the silica surface was confirmed by FTIR. The molecular weight of the grafted polymer was determined by GPC. The microstructure of the nanocomposite was studied using TEM. The surface topography of the composite was examined by AFM analysis.

SiO₂-g-PMMA hybrid nanoparticle can be used for the synthesis of block copolymer-grafted hybrid nanocomposite (SiO₂-g-(PMMA-b-PSt)).⁴²

2.1.4.4. By Microemulsion Polymerization. The functionalized silica nanoparticles were obtained by the reaction between MSMA and the silica acid precursor.^{43–45}

The microemulsion polymerization, that is, an additional monomer-adding process of functionalized silica nanoparticles was conducted by using Ming's method with some

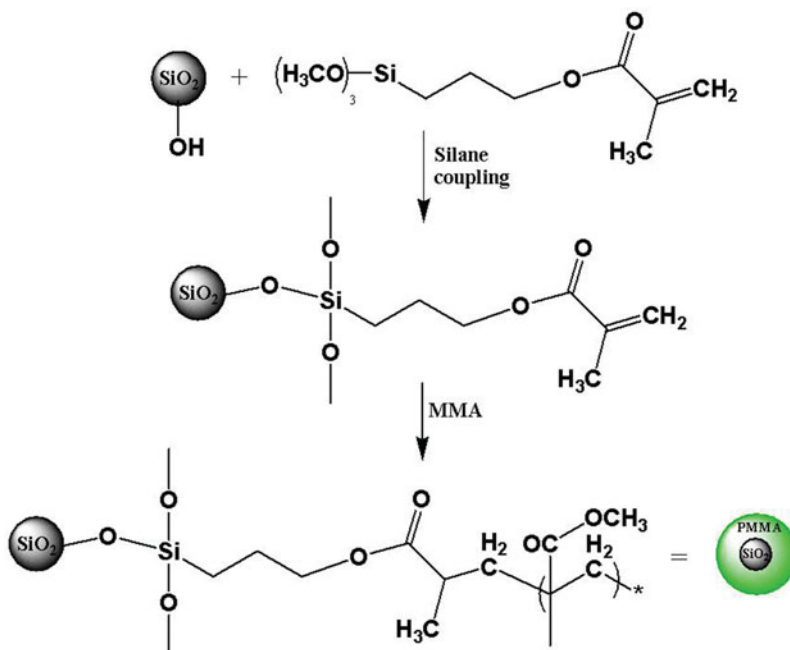


Figure 5. Synthesis of silica/PMMA nanocomposite latex.

modifications.⁴⁶ H_2O , surfactant, and MMA were mixed and the pH of the mixture was adjusted by adding NH_4OH and CH_3COOH . Then silica nanoparticles and aqueous solution of the initiator KPS were heated under nitrogen atmosphere. The final product was obtained as a polymer coated core shell nanoparticles. It is also a stable nanoscale polymer/inorganic hybrids microemulsion (Fig. 5).

The particle size of both the hybrid and nanoparticles were determined by DLS. By comparing FT-IR spectra of nanoparticles and its hybrids, successful grafting was confirmed. TEM analysis of nanohybrid was carried out in both acidic and basic condition. In the former case a core shell microspheres obtained, whereas in the latter, compounds with irregular shapes were obtained. TGA was also carried out to calculate the percentage of grafting.

The morphology and size of composites were controlled by adjusting the monomer addition and polymerization process.⁴⁷

2.1.5. With Poly Styrene. The UV surface-initiated controlled living radical polymerization was used for the grafting of polymer on nanoparticles.

The assembly of MPTES on silica nanoparticle produces methacryloxypropyl modified silica nanoparticles.⁴⁸ Methacryloxypropyl silica nanoparticles and previously prepared VBDC⁴⁹ were dispersed in toluene, irradiated using a UV lamp, and washed with methanol. Thus 4-vinylbenzyl N,N-diethyldithiocarbamate modified silica nanoparticles (VBDCSNs) were obtained.

The VBDCSNs were dissolved in toluene and styrene was dispersed in this reaction mixture with ultrasonic vibrations. The reaction mixture was then irradiated using a UV lamp. After polymerization, the polystyrene grafted silica nanoparticles were separated by centrifugation and dried in vacuum (Fig. 6).

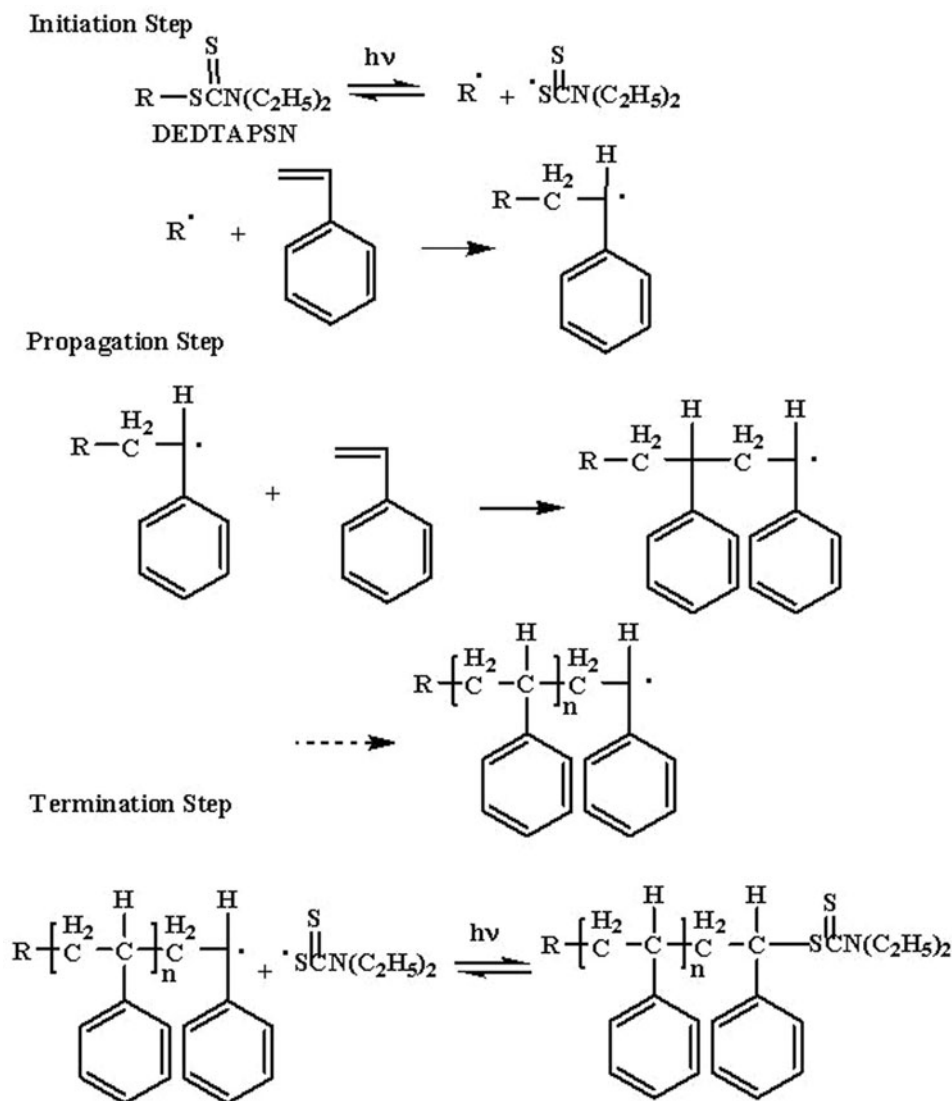


Figure 6. Photo-induced controlled “living” radical polymerization of styrene.

The vibration bands of the diethyldithiocarbamyl group, PS, and silica were all found in FT-IR spectrum of PS grafted silica nanoparticles.⁵⁰

2.1.6. With Poly(*N*-Isopropyl Acrylamide). Grafting of Poly(*N*-isopropyl acrylamide) to silica nanoparticles was done by different methods. Two of them are:

2.1.6.1. By Grafting from Technique. A two-step process was used for the synthesis of core-shell silica nanoparticles with spherical shape. This was prepared by the Stöber protocol using a sonicator.^{51–53} TEOS in ethanol and liquor ammonia were mixed by sonication followed by hydrolysis and condensation.⁵⁴ The resultant spherical core Stöber silica nanoparticles were added to a mixture containing water and CTAB and sonicated.

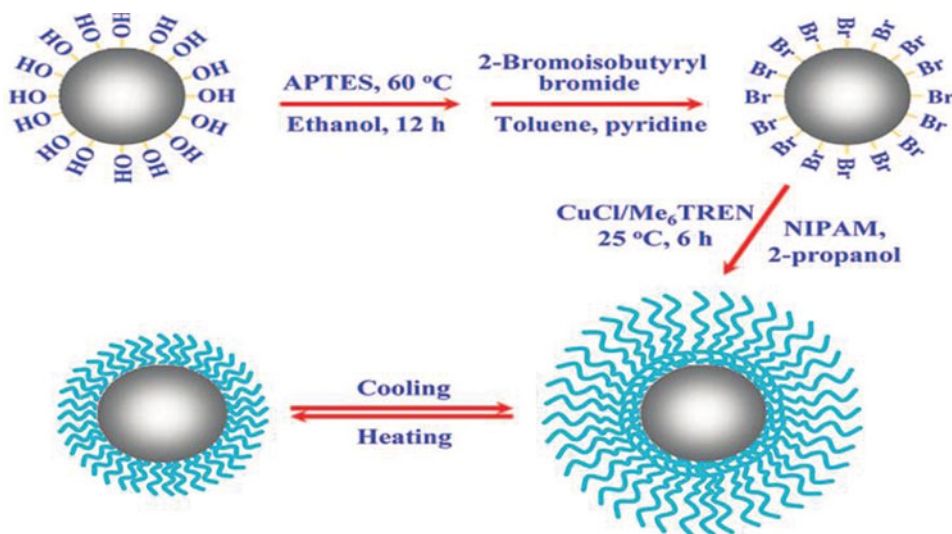


Figure 8. Synthesis of hybrid silica nanoparticles coated with PNIPAM brushes via ATRP.⁵⁶ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

The double phase transition behavior of polymer brushes can be utilized to fabricate more complex nanostructured devices.⁵⁶

2.1.7. With *Poly(2-((bromobutyryl)oxy)ethylacrylate)* and *Poly(2-dimethylaminoethylmethacrylate)*. Surface-initiated self-condensing atom transfer radical vinyl polymerization was used for the polymerization of hyperbranched polymer on the exterior surface of MSNs.

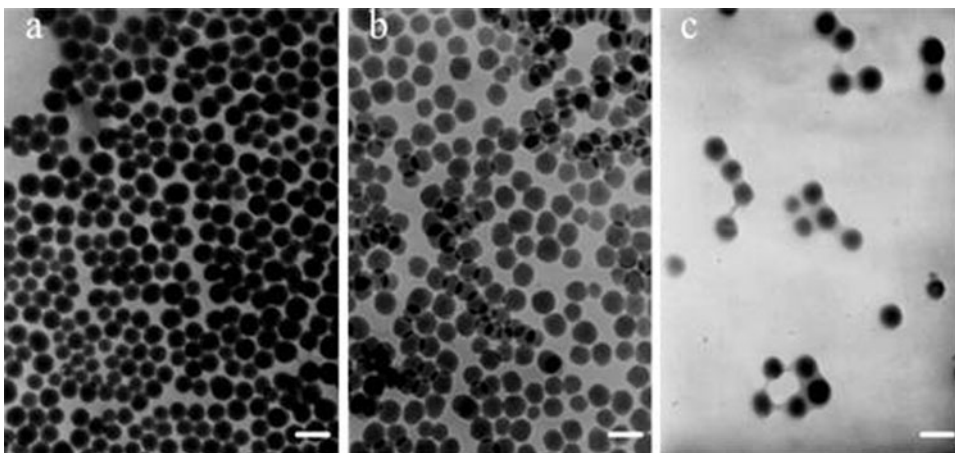


Figure 9. TEM images of (a) bare silica nanoparticles, (b) functionalized silica nanoparticles, and (c) hybrid silica nanoparticles coated with PNIPAM.⁵⁶ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

For the preparation of MSNs, sodium hydroxide solution was added to a solution of CTAB. To this mixture TEOS was added and stirred. The precipitate separated by filtration⁶⁰ was then mixed with 5,6-epoxyhexyltriethoxysilane in toluene and was refluxed. The resultant product was then filtered, washed, and dried. The obtained 5,6-epoxyhexyl coated MSNs were refluxed in methanolic solution of hydrochloric acid to convert the functional groups into 5, 6-dihydroxyhexyl units. These functionalized nanoparticles (MSNs-OH) were filtered and washed with methanol.

The functionalized nanoparticles were dispersed in dry THF and α -bromoisobutyryl bromide in THF was added into it and stirred. The resultant bromine functionalized MSNs were centrifuged washed and dried under vacuum. Then it was mixed with BBEA, CuBr, and PMDETA in toluene were mixed. After a repeated cooling and heating process, the reaction vessel was kept in vacuum and heated. The reaction mixture was diluted with THF, sonicated, and centrifuged. The product was washed with THF, methanol, and dried under vacuum.

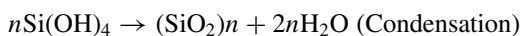
The above product was mixed with PMDETA, Cu(1)Br, DMAEMA, and THF. The polymerization tube was kept in vacuum after repeated cooling and heating cycles. It was then heated and the mixture was dissolved in THF, centrifuged, washed, and dried in vacuum.

XPS and FT-IR results demonstrated that esterification was used for the modification of exterior surface of MSNs. BET and BJH method were used for the calculation of surface area and pore volume/ pore size distribution. GPC, HRTEM, and TGA were also used for the characterization of nanocomposites.

The nanoparticles, MSN-g-HPBBEA-g-PDMAEMA showed pH-responsive property so the polymer grafted nanoparticles have applications in biomedicine and biotechnology.⁶¹

2.1.8. With Poly(ethylene terephthalate). PET-SiO₂ nanocomposites were synthesized by *in situ* polymerization.

In order to prepare a hydrolyzed silica precursor, an aqueous solution of TEOS was mixed with acetone and HCl then sonicated. The resultant product was added to the condensation polymerization system.⁶²



1,2-ethylene glycol, terephthalic acid, and antimony(III) oxide were mixed. The reaction mixture was heated and cooled. Then the silica precursor was added with stirring, heated, cooled, repeatedly washed with water, and dried under vacuum. Thus PET-SiO₂ hybrid was obtained. The crystallization performance and morphology of composites were observed by DSC and SEM respectively.

The resulting nanocomposite, PET-SiO₂, has a higher crystallization temperature, glass transition temperature (T_g), and high degree of homogeneity.⁶³

2.1.9. With Poly(sulfobetaine methacrylate). PSBMA chains grafted on silica nanoparticles were prepared via SI-ATRP of SBMA from the initiator-functionalized silica nanoparticles.

Bare silica nanoparticles were prepared via the Stober process⁶⁵ and then modified with APTMS to obtain the amino-functionalized silica nanoparticle. By the amidation of

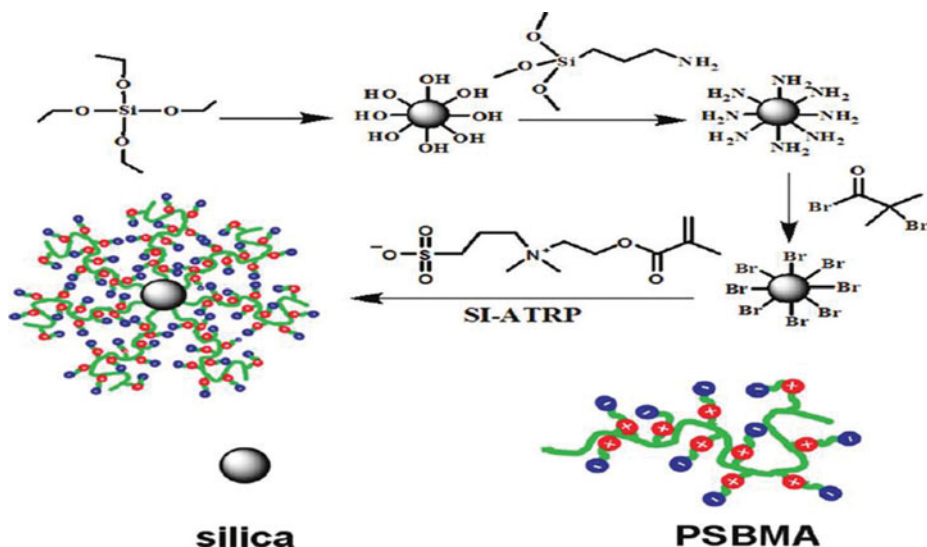


Figure 10. Preparation of Silica-g-PSBMA Nanoparticles.⁶⁴ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

the amino-functionalized silica nanoparticles with BIBB, the initiator functionalized silica nanoparticles were obtained.

Initiator-functionalized silica nanoparticle,⁵⁶ CuBr_2 , bpy, SBMA, NaCl, and methanol/water were taken in a polymerization tube. After ultrasonication the initiator particle was dispersed and the reaction mixture was deoxygenated and then CuBr was added. For the polymerization, the tube was immersed in an oil bath. The reaction mixture was dialyzed against distilled water and the product obtained was silica-g-PSBMA nanoparticles (Fig. 10).

UCST of polymer grafted silica in aqueous solution was observed by DLS and UV-visible spectroscopy. The nanocomposite concentration, the molecular weight of polymer, and the ionic strength were the factors affecting the UCST.

Polysulfobetaine-grafted silica nanoparticles have the tunable phase behavior and stability in protein solutions (Fig. 11). These properties make it to be used in adaptive surfaces, and applied in diagnostic biosensors, controlled release, smart throughput membranes, implanted devices, and other biotechnologies.⁶⁴

2.1.10. With Polypropylene. Nanosilica particles were synthesized by Stober method.⁶⁵ The prepared nanosilica particles were subjected to surface modification. In this process nanosilica particles were dispersed in dry toluene. OTS was added, refluxed, washed, and dried.⁶⁶

The above OTS functionalized nanoparticles were incorporated in PP by melt blending method. At first the samples were dried in a vacuum and then melt blended in a DSM micro twin-screw compounder.

The results of FT-IR and TGA gave a clear evidence of organo-functionality on the surface of silica. Optical microscopy and DSC were used to study its nucleating efficiency. Surface morphology was measured by using SEM. The obtained PP grafted nanosilica has enhanced nucleation effect.⁶⁷

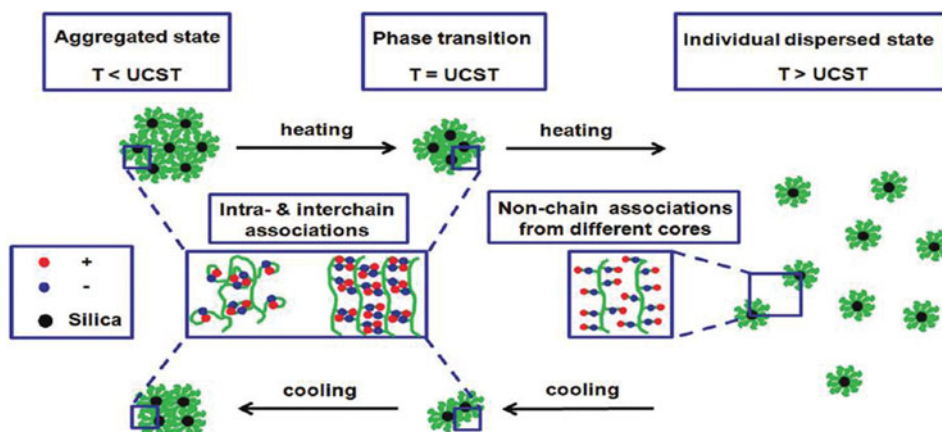


Figure 11. Model for the phase transition of the silica-g-PSBMA nanoparticles in water.⁶⁴ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rights holder.

2.1.11 With Polybutadiene. Silica nanoparticles were surface functionalized by an interfacial condensation process. In this process silica nanoparticles were sonicated in toluene solution. To the reaction mixture TDI with DBTDL was added as a catalyst and heated with stirring. The resultant slurry (SiO_2 -TDI) was dried in vacuum.

The functionalized nanosilica was dispersed in toluene and toluene solution of HTPB was added into the reaction mixture. TDI-HTPB grafting reaction started only after the addition of DBTDL. Then the resultant particles were washed with anhydrous toluene. HTPB grafted nanosilica was separated by centrifugation and dried under vacuum to obtain SiO_2 -TDI-HTPB.

The functionalization of nanosilica was characterized by titration and FT-IR spectroscopy. Dispersion of silica in polymer matrix was characterized by SEM and EDX. The good suspension stability of nanocomposites in the organic medium was proved by conducting sedimentation experiments.

The HTPB grafted nanosilica was mixed with SBR-1502. HTPB grafting on SBR-1502 improves the vulcanization characteristics and technical properties of the rubber. Also, this grafting makes good dispersion of nanosilica on rubber surface and its interaction to the rubber matrix can also be reinforced.⁶⁸

2.1.12. With Poly(amic acid). *In situ* condensation polymerization was used for the preparation of PAA grafted silica nanoparticles.

Silica nanoparticles, toluene and APTES were dispersed with ultrasonic vibrations. Then the mixture was refluxed and centrifuged. The precipitate was dispersed in ethanol and centrifuged again. The washing process was repeated.⁴⁸

APSNs and PMDA were mixed in DMAc with ultrasonic vibrations, bubbling with Helium, then stirred with diamines (DDE, DDM, and DDS) and poured into water. The resulting precipitate was dried. The PAA grafted silica nanoparticles were separated by dispersion in DMAc and centrifugation (Fig. 12)

The nanocomposites were characterized by elemental analyses, FTIR, XPS, TGA, and TEM. TEM shows its good dispersion. The grafted silica nanoparticle could be better

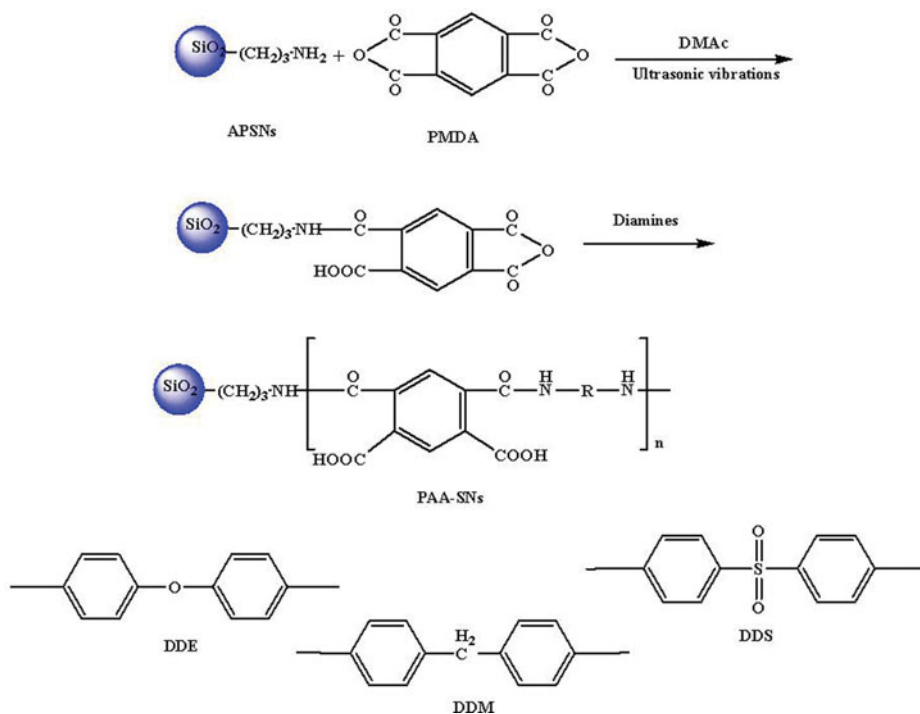


Figure 12. Synthesis of PAA grafted silica nanoparticles.

dispersed in organic solvents and their surface has many carboxylic acid groups, so it is used as reactive nanofillers or nano-crosslinkers.⁶⁹

2.1.13. With Poly(oligo (ethylene oxide) monomethyl ether methacrylate). Silica nanoparticles were functionalized by the following method. At first the silica particles were dispersed in MIBK and 1-Chlorodimethylsilyl hexyl 2-bromoisobutyrate (prepared by using hex-5-enyl-2-bromo-2-methylpropionate and 1-chlorodimethylsilylhexyl-2-Bromo-2-methylpropionate) was added into it and then refluxed and cooled. To this dispersion HMDZ was added, stirred, heated, and mixed with ethanol. The resultant nanoparticles were collected by filtration and were redissolved in THF. Again, the particles were extracted with hexane and dried under vacuum.¹⁵

The functionalized nanoparticles were then grafted with POEOMA by an ATRP mechanism. In ATRP mechanism CuBr was taken in a sealed round-bottom flask. Then a mixture of 2,2'-bipyridyl, monomer, and functionalized nanoparticles were added to the CuBr solution under nitrogen pressure. The reaction flask was placed in an oil bath after removing the nitrogen completely. When the reaction mixture was exposed to air, the polymerization stopped. The resultant product was dissolved in THF, then precipitated into cyclohexane and kept in a vacuum oven (Fig. 13).

The polymer molecular weight was determined by SEC. The hybrid composites were characterized by SAXS, TGA, and rheological measurements. Initial core flood experiments and oil displacement studies also demonstrated its wide applications.

The synthesized nanohybrids are brine-dispersible, stable, and are used in the mobilization of waterflood residual oil.⁷⁰

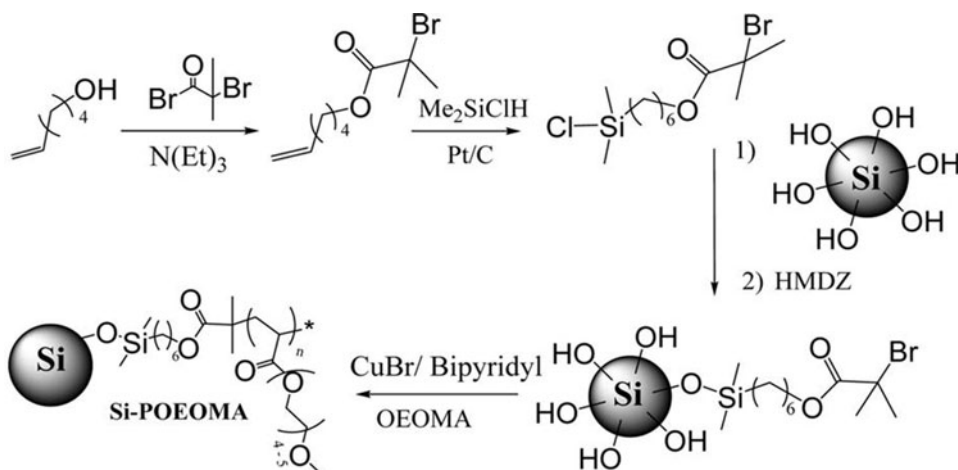


Figure 13. The preparation of POEOMA grafted silica nanoparticles.⁷⁰ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

2.1.14. With Hyperbranched Aromatic Polyamide. “One pot” melt polycondensation process was used for synthesis of hyperbranched polyamide grafted nanosilica particles.

In order to introduce amino group onto nanosilica particle, nanosilica and toluene were taken in a flask and treated ultrasonically. Then γ -APS was added and refluxed under the protection of nitrogen. The mixture was filtered and extracted with ethanol, and the product SiO_2 -APS was dried in vacuum.⁷¹

The monomer DABA and SiO_2 -APS were taken in a flask and self-polycondensation was performed with agitation under argon gas stream. Then it was cooled and extracted with DMAc in a Soxhlet apparatus. Finally, the grafted product, SiO_2 -g- HBP, was washed with methanol and dried in vacuum.

The morphology of nanosilica, before and after the polymerization was studied by using TEM. DLS test was used for analyzing its particle size. TGA was used for analyzing polymer content in the nanocomposites. NMR and FTIR were the other tools used for the characterization of nanocomposites. DSC was used for the determination of its T_g .

The hyperbranched polymer grafted nanosilica particles showed better dispersion and stronger interfacial interaction and exhibited high thermal stability.⁷²

2.1.15. With Poly(ethylene glycol) methacrylate. Both click chemistry and ATRP were used for the grafting of silica nanoparticles with PEGMA.

Using TEOS, silica nanoparticles were prepared by the Stöber method.⁶⁵ The silanization reaction was used for the preparation of amino functionalized SiO_2 . The SiO_2 - NH_2 was mixed with APTES and refluxed with stirring.

The obtained SiO_2 - NH_2 was collected by centrifugation. It was then washed and dispersed in DMF. 3-Bromopropionic acid, NHS and EDC were added to the solution while stirring. The SiO_2 - N_3 was obtained by adding NaN_3 . Azide-modified SiO_2 was separated by centrifugation and washed. It was then mixed with 3-bromopropylpropionate. The resultant solution was dispersed in DMF then CuBr, bpy ligand, and PEGMA were added and stirred under N_2 . PEGMA grafted silica particles were separated by centrifugation washed with ethanol and water and dried (Fig. 14).

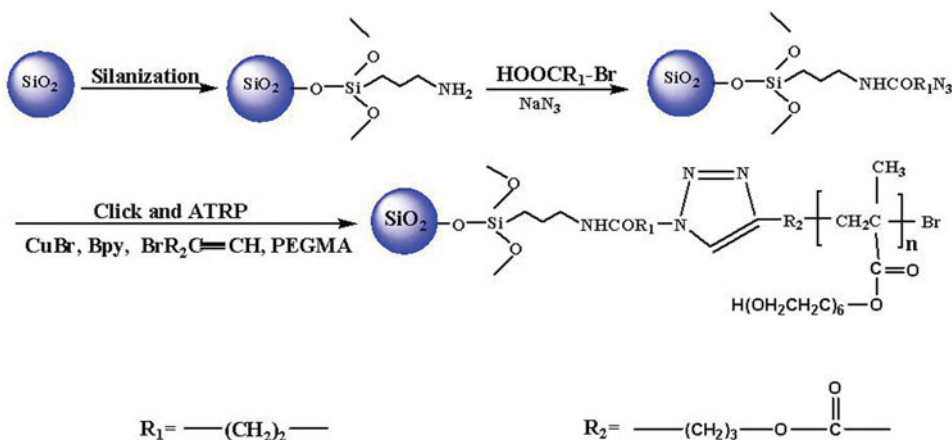


Figure 14. The steps of functionalized silica nanoparticles by click chemistry and ATRP.

TEM characterizes the morphology of SiO_2 and modified SiO_2 nanoparticles. The hybrid was characterized by XPS, FTIR, and TGA. In TGA two stages of weight loss indicate the degradation of PEGMA and that of the 1, 2, 3-triazole ring derivatives.

PEGMA grafted silica nanoparticles are able to absorb more Pb(II) ions from solution than the ungrafted silica nanoparticles. This ability is also applicable in the removal of heavy metal ions (Hg(II) , Cu(II) , As(III) , Cd(III) , etc.).⁷³

Here silica nanoparticles were grafted with large number of organic polymers like PMPC, PEG, PAAc, PMMA, PS, PNIPAM, PBBEA, PET, PSBMA, PP, HTPB, PAA, POEOMA, HBP, and PEGMA using various grafting techniques. The resultant nanocomposites have improved applications such as bio-imaging, fluorescent tracers, multiparameter immunolabeling of cells, synthesis of block copolymer-grafted hybrid nanocomposite, bacterial disease prevention, drug release systems, diagnosis, and enzyme imaging, fabrication of nanostructured devices, biomedicine and biotechnology, smart throughput membranes, implanted devices, reactive nanofillers or nano-crosslinkers, and mobilization of waterflood residual oil.

2.2. Magnetic Nanoparticles

In recent years extensive attention has been given to iron oxide magnetic nanoparticles due to their exceptional magnetic properties (superparamagnetism, high coercivity, low Curie temperature, high magnetic susceptibility, etc.), non-toxicity, biocompatibility, and low cost of production. Due to these properties it is extensively used in a range of nanotechnology applications in various disciplines.

2.2.1. With Poly(glycidylmethacrylate-methacryloxyethyl trimethyl ammonium chloride).

The Fe_3O_4 nanoparticles were prepared from Fe^{2+} and Fe^{3+} ions⁷⁴ with some modifications. FeCl_2 and FeCl_3 solution was prepared separately with deionized water and transferred into a flask. Then it was stirred and heated. NH_4OH was added in drops and heated. The precipitate formed was then isolated from the reaction mixture by magnetic decantation and washed with deionized water and finally it was dried under vacuum.

To the ethanolic solution of Fe_3O_4 nanoparticles, NH_4OH was added and sonicated. VTES was then added to it, which react with hydroxyl groups on the nanoparticles. The

obtained VTES modified Fe_3O_4 nanoparticles were separated by permanent magnet and washed with deionized water, ethanol, and dried under vacuum.

VTES-modified Fe_3O_4 nanoparticles, AIBN, ethanol, and water were taken into the flask and mixed ultrasonically. The flask was kept in a water bath, mechanically stirred under nitrogen and a mixture of GMA, MATAC, ethanol, and deionized water were added. The products were collected by magnetic separation and washed with ethanol and distilled water, extracted in ethanol, and dried.

XRD was used to examine the crystal structure of magnetic nanoparticles. The size and shape of nanoparticles was determined by TEM. The magnetic behavior of bare, modified, and grafted nanoparticles was also measured. FTIR and vibrating sample magnetometry indicates that the polymer chains grafted effectively on the surface of nanoparticles.

Poly(GMA-MATAC)-grafted Fe_3O_4 nanoparticles are used for the immobilization of biomolecules due to the presence of epoxy groups and positive electrical charges on its surface. The immobilized biomolecules had high thermal stability than the free one.⁷⁵

2.2.2. With Chitosan. The Fe_3O_4 nanoparticles were prepared using a high temperature decomposition process.^{76,77} Methyl 3-mercaptopropionate ($\text{HSCH}_2\text{-CH}_2\text{COOCH}_3$) was used for the modification of Fe_3O_4 nanoparticles. It was attached to nanoparticles surface by Fe-S bond.⁷⁸ The $-\text{OCH}_3$ group was converted to an NH-NH_2 functional group by hydrazinolysis and this group allows subsequent conjugation of doxorubicin drug (DOX).

Functionalized Fe_3O_4 nanoparticles were dispersed in methanol containing acetic acid to form a colloidal solution. Then doxorubicin was added while stirring. The resultant solution was centrifuged; the precipitate was again dispersed in methanol. The DOX-loaded Fe_3O_4 nanoparticles were dried.

Encapsulation of chitosan-g-poly(NIPAAm-co-DMAAm) with DOX-loaded Fe_3O_4 nanoparticles was done in two steps. First, the smart polymer, chitosan-g-poly(NIPAAm-co-DMAAm) was prepared.^{79,80} Second, the prepared polymer functionalized nanoparticles and 1,6-diaminohexane were dispersed in DMSO and sonicated. The encapsulation occurs due to the reaction between 1, 6-diaminohexane and 4-nitro-phenyl chloroformate groups of polymer.⁸⁰ After sonication, the encapsulated product were centrifuged and washed with methanol and then dried.

TEM, FTIR, and TGA were carried out to analyze the polymer grafted nanoparticles. Drug release of this nanocomposite was also measured with respect to time.

This magnetic nanoparticle drug carrier helps for the controlled drug release with respect to the change in external temperature or pH.⁸¹

2.2.3. With β -cyclodextrin-polyurethane. Magnetic nanoparticles (Fe_3O_4) were prepared via an improved chemical co-precipitation method.⁸² The Fe_3O_4 nanoparticles have many hydroxyl groups on the surface due to the contact with aqueous phase. In order to graft the polymer into the surface, first, nanoparticles were dispersed in DMF by sonication and then hexamethylene diisocyanate in dry DMF was added to the reaction mixture, and stirred. Hexamethylene diisocyanate bonded to the surface by the reaction of hydroxyl group on the surface.

In order to graft β -cyclodextrin-polyurethane polymer into the surface of Fe_3O_4 nanoparticles, β -cyclodextrin dissolved in dry DMF was added in drops to the reaction mixture and then stirred. The precipitate was separated by magnetic decantation followed by washing with water and acetone. β -CDPU-MNPs were dried in vacuum.

The results obtained from SEM, TEM, and VSM shows that magnetic nanocomposites were superparamagnetic and mean diameter of 59 nm. The grafting was confirmed by FTIR and TGA. Atomic absorption measurements were also performed.

The β -cyclodextrin-polyurethane polymer grafted nanoparticle used as a solid-liquid phase transfer catalyst for nucleophilic substitution reactions of benzyl halides. It is water insoluble and its easy availability, good stability, and excellent catalytic activity make it as an alternative to the water soluble β -cyclodextrin.⁸³

2.2.4. With Poly(acrylic acid). SI-ATRP was used for the preparation of polyacrylic acid grafted magnetic nanoparticles.

Magnetite nanoparticles were prepared by thermal decomposition method.⁸⁴ In this method iron (III) acetylacetonate ($\text{Fe}(\text{acac})_3$) and benzyl alcohol were mixed by magnetic stirring under nitrogen flow. The mixture was heated and the precipitant was removed using an external magnet. It was then washed with ethanol and CH_2Cl_2 repeatedly to remove benzyl alcohol. The precipitated MNPs were dried at room temperature.

In order to prepare oleic acid-coated MNPs, the dried nanoparticles were then introduced into an oleic acid solution in dried toluene and ultrasonicated. BTPAm is then immobilized on the surface of oleic acid-coated MNPs by mixing them in TEA containing toluene. The mixture was stirred under nitrogen. By magnetic separation the BTPAm-modified MNPs were obtained and were dried in vacuum.

BTPAm-immobilized MNPs were added to a mixture containing dioxane, PMDETA, and t-BA. The mixture was stirred under nitrogen blanket. The mixture was then heated; ATRP polymerization took place at this stage. The resulted poly (t-BA)-grafted MNPs were magnetically separated and dried under vacuum. Poly (t-BA)-coated MNPs were then hydrolyzed in TFA and diluted with CH_2Cl_2 solution and repeatedly precipitated in cold hexane to obtain acrylic acid functional groups on MNP surfaces.

Poly(AA)-coated MNPs were dispersed in NHS and EDC.HCl. The mixture was kept in dark and the separated particles were washed and dried. The particles were then added to a mixture containing EDA-FA and EDC in anhydrous DMSO. It was then agitated and the recovered folic acid immobilized Poly(AA)-coated MNPs were washed in DMSO and dried.

Poly(AA) grafted MNPs act as a cancer cell targeting agent. FTIR was used to monitor this reaction progress. The results of PCS indicate that as increasing the pH, the hydrodynamic radius was decreased. NMR, GPC, DLS, VSM, and UV-visible spectroscopy were used to analyze the polymer nanohybrid. The cytotoxicity test proved that the nanohybrid were non-toxic.

This magnetically guidable nanocomplex is used as a good drug delivery vehicle in cancer treatment.⁸⁵

2.2.5. With Polystyrene. ATRP procedure was used for the preparation of polystyrene grafted Fe_3O_4 nanoparticles.

Magnetite (Fe_3O_4) nanoparticle, extra dry toluene, and CTCS, were mixed in an ultrasonic bath at room temperature. It was subsequently washed with THF and then dried under vacuum. Thus functionalized CTCS-MNs were obtained.⁸⁶

The polymerization of styrene was performed with CuBr/bpy as the catalyst,⁸⁷ TsCl as the free initiator, CTCS on magnetic nanoparticles as the initiator, and DPE as the solvent. A solution that contained deoxygenated styrene, CuBr , Bip, DPE, TsCl , and CTCS-MNs was sealed under a nitrogen atmosphere and heated. The styrene grafted magnetite nanoparticle was obtained.

The water contact-angle measurement shows the hydrophobic nature of composite. A variety of techniques such as GPC, AFM, TGA, and FTIR were used to characterize the nanocomposites⁸⁸

2.2.6. With Poly(*poly(ethylene glycol) monomethacrylate*). The PEGMA grafted MNPs were prepared via a solvent-free atom transfer radical polymerization.

The Fe₃O₄ magnetic nanoparticle was prepared by the following method. Here Fe(III) acetylacetonate, 1,2-hexadecanediol, oleic acid, oleylamine, and benzyl ether were mixed, magnetically stirred, refluxed, and cooled. Then ethanol was added to the mixture, and the precipitate was separated and dissolved in hexane in the presence of oleic acid and oleylamine. The product, Fe₃O₄ nanoparticles, was then precipitated with ethanol, collected by centrifugation, and dried.^{89,90}

Under the protection of argon, the oleic acid-stabilized nanoparticles were dispersed in CPA in hexane and stirred. The resulting precipitate was separated and washed with hexane. The nanoparticles were then dried.

The CPA-stabilized nanoparticles dissolved in PEGMA monomer in a Pyrex tube to form a transparent brownish solution. The solution was purged with argon; then CuCl and bpy were added. After purging with argon the Pyrex tube was sealed and kept in a water bath under stirring. After the reaction was complete, the mixture was diluted with THF. The dilute solution was centrifuged, added into diethyl ether, and the precipitate was dried under vacuum.⁹⁰

Grafting of polymer on nanoparticle surface was confirmed from XPS and FTIR analysis. MRI of polymer grafted nanoparticles shows that the magnetic resonance signal enhanced with increasing nanoparticles concentration in water. TEM, TGA, and GPC were used to characterize the nanocomposites.

The PEGMA grafted MNPs performs as T₂ –type contrast agents in magnetic resonance imaging.⁹¹

2.2.7. With Polymethylmethacrylate and Poly Acrylic Acid. Here emulsion polymerization was used for the preparation of polymer grafted magnetic nanoparticles.

Iron oxide magnetic nanoparticles were prepared by co-precipitation of Fe²⁺ and Fe³⁺ salts solution^{92, 93} with different molar ratio of Fe²⁺:Fe³⁺. (In sample 1, Fe²⁺:Fe³⁺ = 1:1, sample 2, Fe²⁺:Fe³⁺ = 1:2) MMA and AAc (co monomers) were coated with iron oxide nanoparticle by emulsion polymerization.⁹³ The surfactant was added into the dispersed iron oxide nanoparticles suspension and dispergation was repeated. Then, the co-monomer mixture was added to the dispersion and mixed in an ultrasonic bath. The mixture was transferred to the Schott Duran reactor, initiator KPS was added and shaken at constant temperature. The coated magnetic nanoparticles were washed with a mixture of methanol and water. These particles were then immersed in H₂SO₄ solution. Polymer-coated nanoparticles were further washed with milli-Q water.

Approximate particle size was determined by SEM. To examine the molecular structure of nanocomposites, FTIR was utilized.⁹⁴

2.2.8. With Poly(*glycidylmethacrylate*). NH₄OH was added in drops to the aqueous solution of ferrous chloride to precipitate metals as hydroxide gel. It is then washed and stirred under nitrogen. During the reflux process, the gel network breaks into small crystalline iron oxide.⁹⁵ The product Fe₃O₄ nanoparticles obtained were filtered and dried.

Poly(glycidylmethacrylate) was telomerized by dissolving glycidylmethacrylate and MPS in benzene followed by the addition of AIBN and it was degassed using argon gas and

then sealed under vacuum. Then diethyl ether was added and the precipitated copolymer was dried under vacuum. Thus SiPGMA was obtained. It was then silanized onto Fe_3O_4 nanoparticles in toluene / methanol and refluxed. The silanized particles were washed and dried in vacuum.

The molecular structures of Fe_3O_4 -SiPGMA and SiPGMA were determined by FTIR spectroscopy. NMR was used to characterize the structure of nanocomposites. The effect of heating on the structural changes of both bare and grafted nanoparticles was determined by using TGA. XRD was used to evaluate the mean size of crystallites. Nanocomposites size obtained from TEM is larger than the crystalline size obtained from XRD. This reveals the polycrystalline nature of nanocomposites.

The silanized particles were then used for fabrication of enzyme electrode. For this, cysteamine decorated gold electrode was immersed in Fe_3O_4 -SiPGMA solution containing phosphate buffer. The enzyme HRP was then attached to modified gold electrode using the free epoxy groups on the polymer arms of Fe_3O_4 . The resulting HRP biosensor was then washed and was used for the determination of phenolic compounds. PGMA grafted Fe_3O_4 nanoparticles could be used to immobilize other enzymes to construct a range of biosensors.⁹⁶

2.2.9. With Polymethylmethacrylate. PMMA can be grafted to magnetic nanoparticle by different polymerization techniques. Some of them are:

2.2.9.1. By Grafting from Method. Magnetic nanoparticles were prepared by dissolving ferrous sulphate and anhydrous ferric chloride using ultrasonicator. To precipitate the magnetic nanoparticles the mixture was added to NH_4OH in an inert condition and stirred. The precipitate was washed and centrifuged, and dried under vacuum.

Magnetic nanoparticles was dispersed in toluene and sonicated. To this mixture MPTMS was added, stirred, and kept under argon atmosphere. The product thiol functionalized MNPs (MNPs-SH) were centrifuged, washed, and dried in vacuum.⁹⁷

MNPs-SH and toluene were mixed in a round bottom flask and sonicated. To the reaction mixture BL and MMA were added and heated with stirring. The polymerization reaction was quenched by placing them in an ice bath, and diluted with THF. The mixture was precipitated with hexane and dispersed in THF and then centrifuged. The product was dried and thus red brown PMMA-g-MNPs nanocomposites were obtained.

The morphology of nanocomposite was confirmed by TEM. TGA and GPC were used respectively to estimate the amount of polymer on the nano surface and to determine the molecular weight of polymer. The other techniques such as FT-IR, NMR, XPS, and EDX were also used to characterize the nanocomposites. PMMA-g-MNPs nanocomposites have a superparamagnetic character. These grafted magnetic nanoparticles show good dispersibility in organic solvents.⁹⁸

2.2.9.2. By In situ Polymerization. $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and sodium oleate were added into a mixed solvent (ethanol, deionized water, and hexane) and then heated and refluxed. After that the organic phase (containing iron oleate complex) was washed with deionized water and evaporated. Later this complex and oleic acid were dissolved in dioctyl ether, heated, and cooled. Magnetic nanoparticles were separated after the addition of ethanol. Nanoparticles were then re-dispersed in hexane and ethanol was added. The precipitate thus formed was centrifuged and the Fe-oxide nanoparticles were dispersed in hexane and stored.

The colloids containing well-dispersed Fe-oxide nanoparticles and MMA monomer were mixed. Using the initiator AIBN the polymerization was carried out. The obtained PMMA/Fe-oxide nanocomposites were very transparent.

TEM, XRD, TGA, and DSC were used to characterize the PMMA grafted magnetic nanoparticles. TEM results show that nanoparticles have small size and narrow size distribution. XRD results show that nanoparticles have undergone some changes in its crystallinity, when it forms nanocomposites.

These nanocomposites are used in magneto-optical instruments and shielding applications. PMMA/Fe-oxide nanocomposites prepared by the *in situ* method shows enhanced thermal stability^{99,100} than the one prepared by direct blending.¹⁰¹

2.2.10. With Poly(ethylene glycol) monomethacrylate. Magnetic nanoparticles were synthesized from alkaline solution of Fe(II) and Fe(III) ions. Ammonia solution was added to the solution of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$ in distilled water with stirring and then heated. The magnetic nanoparticles were washed and dispersed in toluene. The dispersed MNPs, BTPT (prepared chain transfer agent),¹⁰² and toluene were added into a flask and heated with stirring. MNPs-BTPT was washed with toluene and DMF.

The RAFT polymerization technique was used for the grafting of PEGMA on to magnetic nanoparticle surface. For this, the prepared MNPs-BTPT, PEGMA, AIBN, and DMF were mixed. The mixture was heated with stirring after replacing oxygen with nitrogen. The centrifuged PEGMA grafted magnetic nanoparticles were washed with distilled water and dispersed in water ultrasonically (Fig. 15).

TEM was used for analyzing the size and morphology of MNPs before and after grafting with polymer. The chemical structure of nanocomposites was determined with the help of FT-IR analysis. XPS and TGA were also used for characterizing the nanohybrid.

The polymer P(PEGMA) decreases the nonspecific proteins adsorption on magnetic nanoparticle. P(PEGMA) grafted MNPs lengthen the circulation time in drug delivery and release.¹⁰³

2.2.11. With Poly(styrene-itaconic acid-divinylbenzene). The inverse-phase suspension polymerization method was employed to prepare Poly(styrene-itaconic acid-divinylbenzene) microspheres. The continuous phase consists of castor oil and span 80. St, DVB, and N,N -Methylene-bisacrylamide were dissolved in DMSO and the organic phase

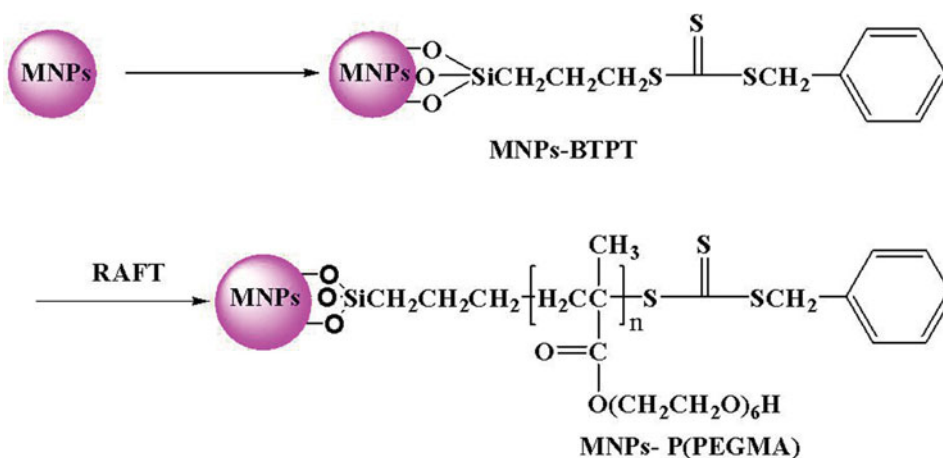


Figure 15. The process of grafting P(PEGMA).

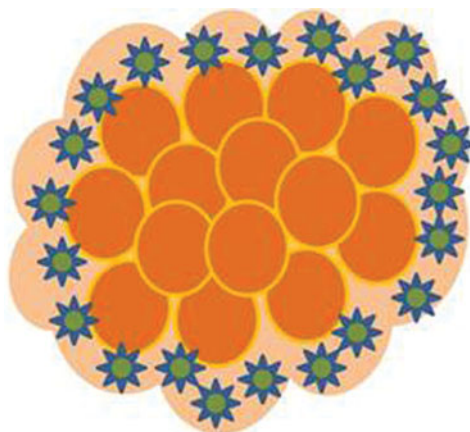


Figure 16. Schematic representation of TMR nanoparticle.

was added into the flask. The initiator APS was added with constant stirring heated and centrifuged. The centrifuged microspheres were washed with diethyl ether and water.

The purified P(St-IA-DVB) microspheres were dispersed in deionized water. Fe_3O_4 nanoparticles were added to the solution. The product was washed with plenty of water and centrifuged. The magnetic nanoparticles grafted P(St-IA-DVB) microspheres thus obtained were freeze dried.

The grafting of polymer onto nanosurface was confirmed by FTIR. The crystallographic structure of these microspheres was determined using XRD. The surface morphology was obtained from SEM analysis.

These microspheres were stable and mono-disperse which shows great potential application in doxorubicin delivery. The doxorubicin loaded microspheres used as an anticancer agent, which implies the great application in magnetic targeted tumor therapy.¹⁰⁴

2.2.12. With Stimuli Sensitive Polymers. A temperature sensitive and alternative magnetic field responsive mode of drug delivery was targeted for treating cancers. Here we have employed two temperature responsive block copolymers, NIPAM and DMAEMA, for entrapping magnetite nanoparticles inside the polymer pores to form a core-corona structured hybrid nanoparticles. The hybrid nanoparticle formed had an inner magnetic core (Fe_3O_4) surrounded by a shell of drug loaded polymeric corona. Thermo-Magneto-Responsive nanoparticles were spontaneously formed when magnetite nanoparticles were treated with a homogeneous solution of NIPAM and DMAEMA separately.^{105–107} After a thorough homogenization of this mixture, a broad spectrum anticancer drug 5-fluorouracil was successfully added to load the TMR nanoparticles. The outer shell of the TMR nanoparticle served as an adsorbent for 5FU to fill within the remaining pores. The so formed TMR nanoparticles were schematically represented in Fig. 16. Optimum conditions were investigated for the TMR nanoparticle formation and were characterized using TEM, SEM-EDS, and XPS for the preliminary studies. Figure 17 demonstrates TEM image of the core-corona structured TMR nanoparticles, whereas Fig. 18 depicts the SEM image showing morphological and polydispersive nature of the same. SEM-EDS analysis showed the presence of constitutive elements within the TMR nanoparticles.

Cytotoxicity of the free copolymers was analyzed to conclude the biocompatibility of the two copolymers employed in the TMR nanoparticle synthesis. Alamar blue assay

was performed using human cortical neurons (HCN) and glioblastoma (G1) cells. The percentage of cell viability for each of the cell lines were obtained at different concentrations of NIPAM and DMAEMA and graphs were plotted.

Core-corona structured polymer coated magnetic nanoparticles were successfully synthesized with the drug inside. Preliminary characterization showed that the size of the hybrid nanoparticle formed was around ~ 20 nm with drug inside. Cytotoxicity assays revealed that NIPAM was less cytotoxic towards normal cell lines, HCN and it showed a dose dependent killing of G1cells (Fig. 19). In case of DMAEMA, both cell types were drastically affected by the toxic effects of the polymer showing the high cytotoxic nature of the same (Fig. 20).

Further studies are needed to find out the magnetic behavior, drug loading capacity, drug release profile, and the TMR nature of the hybrid nanoparticles. The toxicity of the polymers can probably be reduced by modifying them with a functional group. This may enhance the biocompatibility.¹⁰⁸

Iron oxide nanoparticles were grafted with suitable polymers like P(GMA-MATAC), Chitosan, β -CDPU, PAA, PMMA-PAAc, PS, PEGMA, PGMA, PMMA, PEG, and P(St-IA-DVB). The resulting nanocomposites have many potential applications in various fields.

2.3. Titanium Dioxide Nanoparticles

Titanium dioxide is the naturally occurring oxide of titanium. It exhibits various properties such as a high refractive index, UV light absorption, and photocatalytic activity. Nowadays, the modification of TiO_2 particles with various chemicals enables them to be used as a photocatalyst.

2.3.1. With poly methyl methacrylate. PMMA can be grafted to TiO_2 nanoparticle by different methods. Two of them are:

2.3.1.1. By both Suspension and Bulk Polymerization Techniques. Titania nanofibers were prepared by a direct sol-gel process in supercritical carbon dioxide.¹⁰⁹ In order to functionalize the titania nanofiber with acid functional group, 2-propanol was used and into which calcined titania powder was dispersed by sonication. It was then reacted with methacrylic acid. The product was filtered, washed, and dried under vacuum condition.

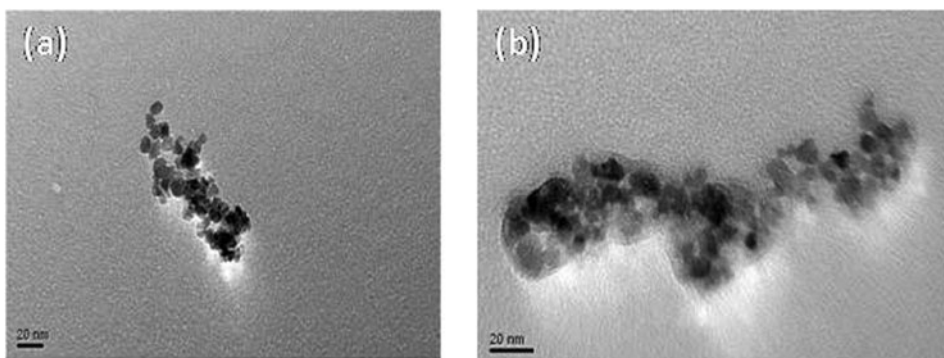


Figure 17. TEM images of magnetic nanoparticles (a) and DMAEMA coated magnetic nanoparticles (b).

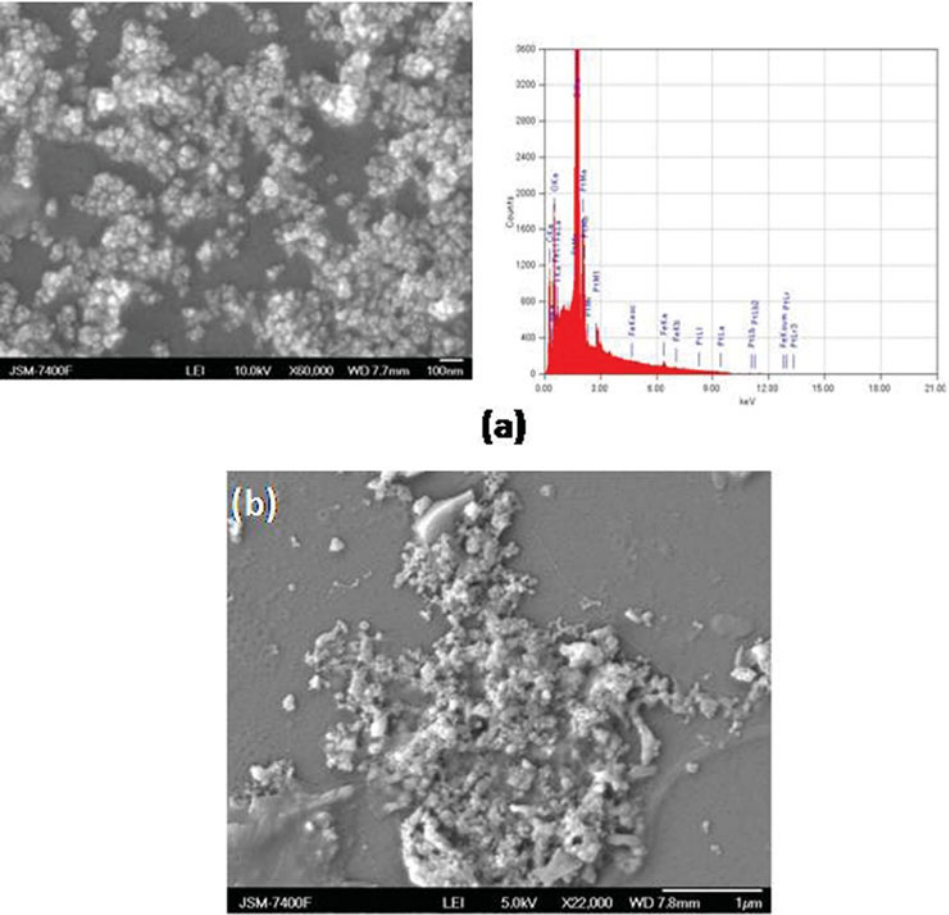


Figure 18. (a) SEM image and EDS data of DMAEMA coated magnetic nanoparticles. (b) SEM image of NIPAM coated magnetic nanoparticles.

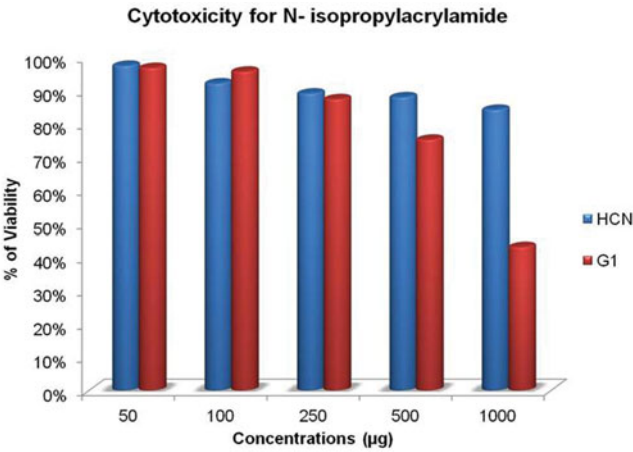


Figure 19. Cytotoxicity assays in the case of NIPAM copolymer.

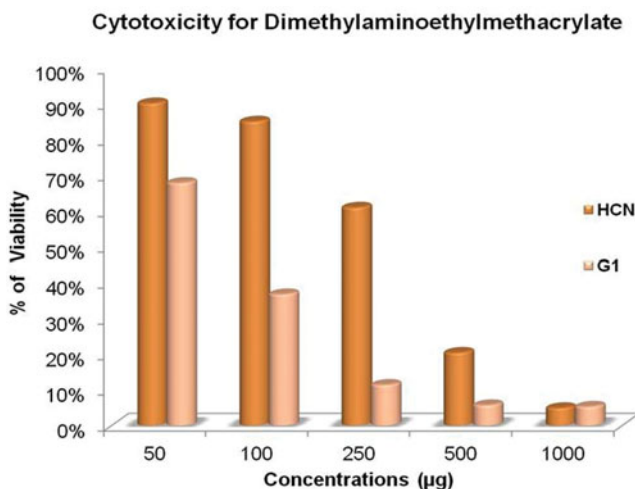


Figure 20. Cytotoxicity assays in the case of DMAEMA copolymer.

Both suspension and bulk polymerization techniques were used for the synthesis of TiO₂-PMMA (TiO₂-poly methyl methacrylate) nanocomposites.

In the suspension polymerization PVA was dissolved in deionized water with constant stirring. Then nanotitania, BPO, and MMA monomer were added to it. The polymerization step was performed under N₂ atmosphere.

In bulk polymerization, the nanocomposite was prepared by using PMMA powder and MMA liquid. The powder consists of PMMA beads and BPO and the liquid contains MMA, a dimethacrylate cross-linker, and an activator. The process was done at room-temperature. The viscosity, polymerization shrinkage, and heat of reaction were controlled by liquid MMA monomer.^{110,111} Ultrasonic agitation was used for the dispersion of functionalized n-TiO₂ into the liquid monomer portion. The resulting materials were injected into a polysiloxane mold. After curing, the products were taken out of the mold and aged at room temperature (Fig. 21).

Powder X-ray diffraction studies of the nano-TiO₂ helps to analyze the effect of crystallinity of nanoparticle on the properties of the polymer nanocomposites. FTIR spectral studies confirm the functionalization of nanotitania. Experiments like monomer/water partitioning were conducted to explore the hydrophilic nature of functionalized nanotitania. SEM and TEM analysis were also carried out to examine the microstructures and nanofiller distribution within the nanocomposites. STEM and elemental analysis were done to study the nanostructure of the polymer nanocomposites (Fig. 22). The thermal stability of the composites was observed through DSC and TGA.

The functionalized titania nanofibers and the PMMA matrix produced have higher values of elastic moduli compared to the composites with non-functionalized titania nanofibers. Mechanically they are better reinforcers than nanospheres. TiO₂ fiber-PMMA nanocomposites exhibit a unique combination of radiopacity.

TiO₂ fiber-PMMA nanocomposites are applied in dental composites, bone cement, and implant biomaterials.¹¹²

2.3.1.2. By Surface-Initiated Atom Transfer Radical Polymerization. TiO₂ was dispersed in water; a white homogeneous dispersion was obtained after ultrasonication and stirring. Then the initiator DOPA¹¹³ was dissolved in water and added into the dispersion in

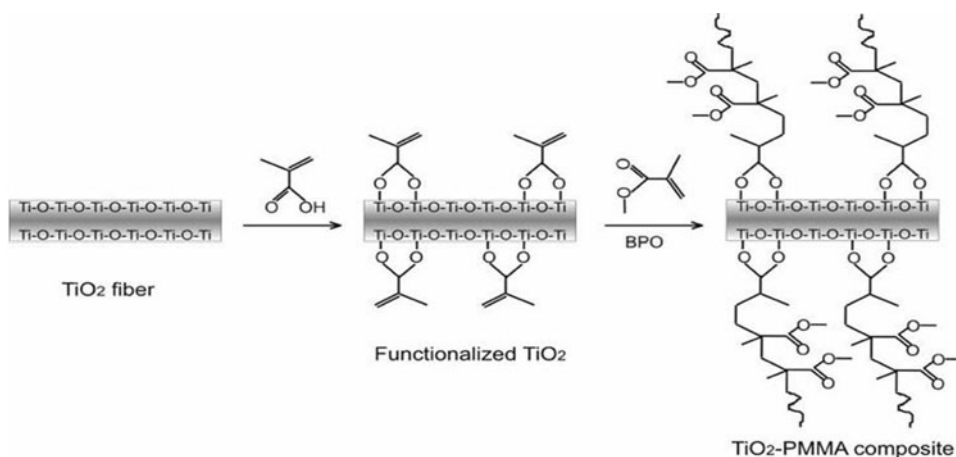


Figure 21. Synthesis of TiO₂-PMMA nanocomposite.¹¹² © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

drops; it turns yellow on stirring. The yellow precipitate obtained was then re-dispersed in water, filtered, and washed. The formed complex was collected and dried in vacuum.

TiO₂-initiator complex was dispersed in DMF by ultrasonication until a yellow mixture was obtained. After purging under Ar, CuBr, and PMDETA were added. Under Ar protection MMA was injected via a syringe. The samples were centrifuged and the product was re-dispersed in DMF and centrifuged again. It was washed with THF and acetone. The

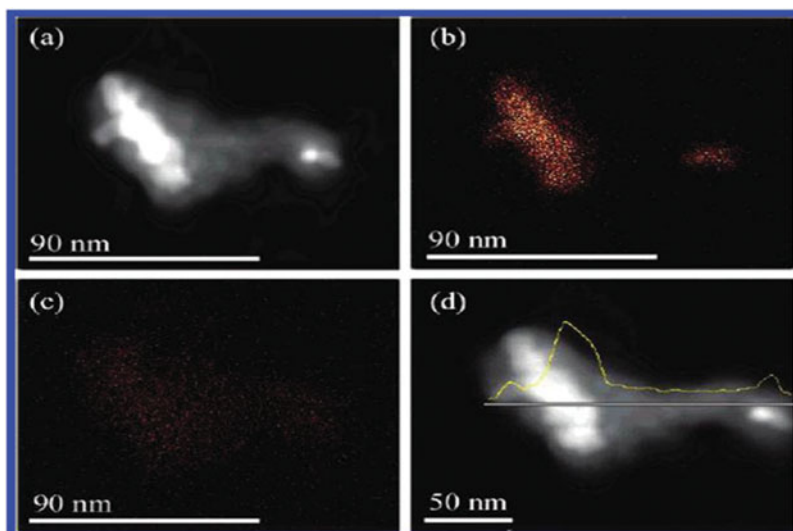


Figure 22. (a) STEM of TiO₂-PMMA nanocomposite; (b) EDS of titanium mapping; (c) EDS, oxygen mapping; (d) STEM and EELS landscape.¹¹² © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

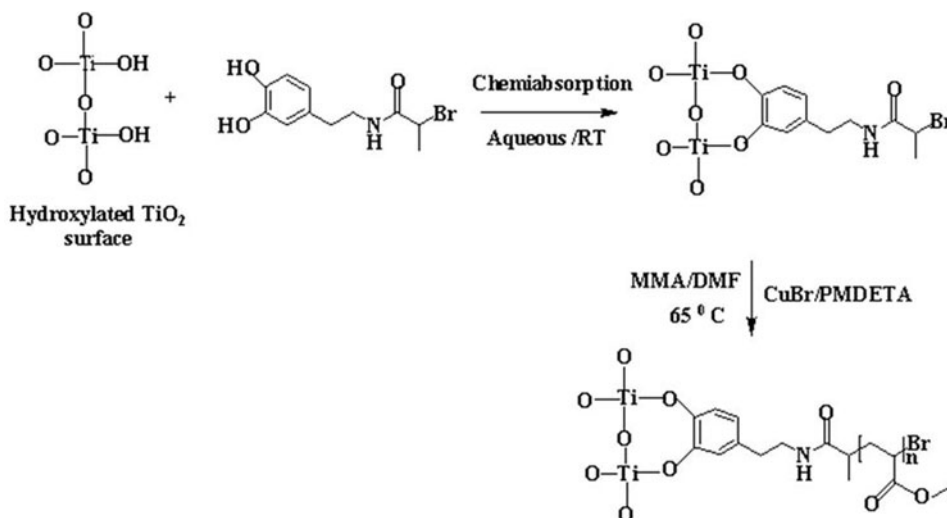


Figure 23. Initiator immobilization and SI-ATRP.

polymer grafted TiO_2 was dispersed in toluene and precipitated into cold methanol and filtered (Fig. 23).

FT-IR and UV-Vis studies proved the sufficient grafting of initiator onto the surface of TiO_2 nanoparticles. TGA was carried out to quantify the content of the initiator present in TiO_2 -initiator complex. FT-IR and TGA were also carried out to examine the grafting of PMMA from nanotitania. The grafting was confirmed by TEM analysis. The molecular weight and polydispersity index of the polymer grafted from the nanoparticle was observed by GPC analysis.

This method is used for modification of nanofillers (metal oxides) and in the synthesis of functionalized polymer-matrix nanocomposites and ultrathin films.¹¹⁴

2.3.2. With poly(methyl methacrylate) and poly(tert-butyldimethylsilyl methacrylate). The surface of TiO_2 nanoparticles was modified with MPS by two methods: In the first method¹¹⁵ ethanolic solution of titanium dioxide powder was prepared. Ammonia and MPS were added with stirring. The obtained dispersion was then purified by centrifugation and the product was dried in vacuum. In the second method¹¹⁶ titanium dioxide nanoparticle along with MPS were dispersed in acetone by vigorous stirring. After that, diluted maleic anhydride was added as a catalyst, refluxed, and purified by centrifugation. The product was dried in vacuum. Thus MPS modified TiO_2 nanoparticle was obtained.

Grafting of MMA and MASi polymers to the surface of MPS-modified titanium dioxide nanoparticles involves the “grafting through” method using the RAFT process. Toluene solution of the monomer (MMA and MASi), titanium dioxide modified by MPS, CPDB the chain transfer agent, and AIBN were stirred in a flask. The solution was then degassed, stirred, and heated. After the reaction, it was cooled by nitrogen liquid. Then the solution was centrifuged and washed with toluene and dried in vacuum. Thus the polymer grafted TiO_2 nanoparticle was obtained (Fig. 24).

The surface modification and grafting of titanium nanoparticles were confirmed by FTIR spectroscopy. The amount of coupling agent and polymer on the nanoparticle surface

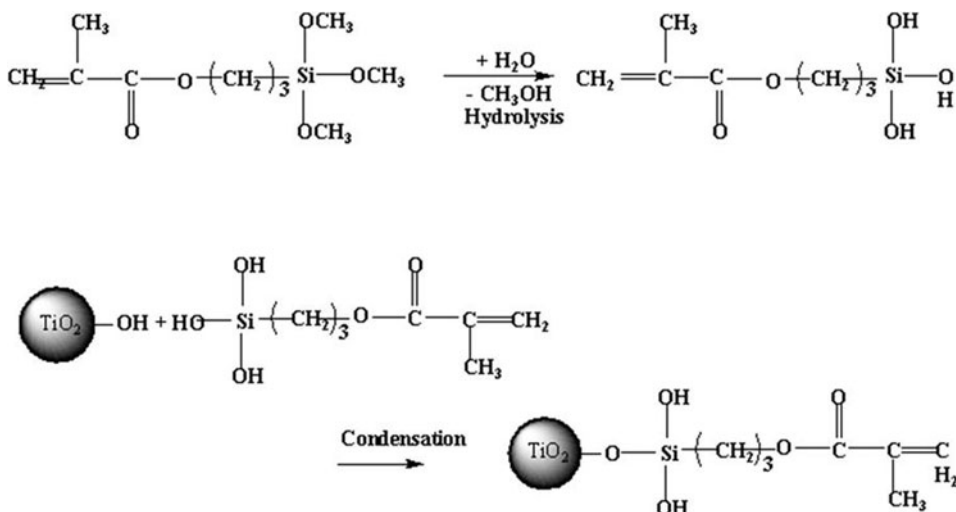


Figure 24. Functionalization TiO_2 nanoparticles.

was determined by using TGA. Surface morphology of the resultant nanocomposites was studied by HRTEM. The livingness of the polymerization was verified using ^1H NMR.²⁷

2.3.3. With Poly(methyl methacrylate)-*b*-polystyrene. In order to graft poly(methyl methacrylate)-*b*-polystyrene to TiO_2 , the nanoparticle can be functionalized by two methods, namely:

2.3.3.1. By RAFT Functionalizing Agent. The RAFT agent 4-cyano-4- (dodecylsulfanylthiocarbonyl) sulfanyl pentanoic acid was prepared.¹¹⁷ The RAFT agent and $n\text{-TiO}_2$ were dispersed in THF and stirred well. The resultant particles were recovered by centrifugation, dissolved in THF, and reprecipitated by centrifugation. The solid product, RAFT agent functionalized TiO_2 , was dried.

MMA, RAFT functionalized TiO_2 , and AIBN were dispersed in THF and stirred. The products were redissolved in THF and collected by centrifugation. The TiO_2 /PMMA nanocomposite product was dried (Fig. 25).

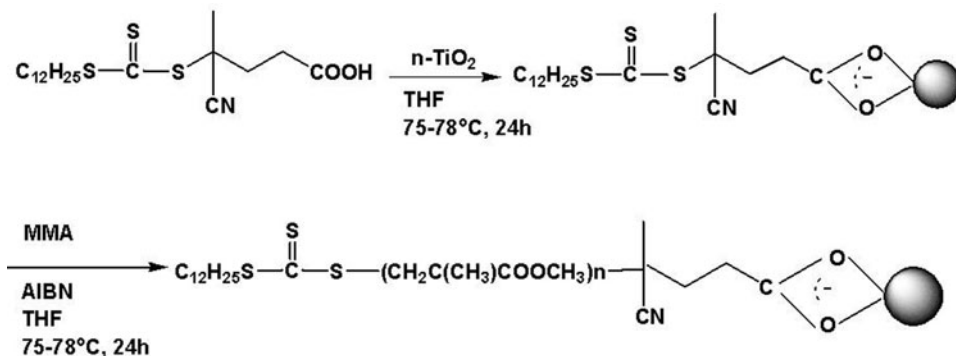


Figure 25. Preparation of PMMA- TiO_2 nanocomposite.

Organic phase/water partitioning studies were carried out to study the hydrophilicity of nanotitania before and after functionalization. ^1H NMR technique was used to verify the livingness of the polymerization. FTIR and XPS analysis were conducted to examine the functionalization of the nanoparticle. Dispersion of nanofiller in the polymer was determined by electron microscopic analysis.

The polymer grafted nanocomposites exhibited minimum agglomeration of nanoparticles and enhanced the filler-matrix interaction.²⁶

2.3.3.2. By Sodium Ethyl Xanthate and 3-(chloropropyl)triethoxy silane (WD-30).

The TiO_2 nanoparticles were modified by diblock copolymers, PMMA-b-PS, via RAFT polymerization followed by preparation of TiO_2 /Epoxy nanocomposites.

Silane coupling agent WD-30 can be used to modify the surface of TiO_2 nanoparticles. From the resultant solution the $\text{TiO}_2\text{-Cl}$ was obtained by repeated washing with toluene. The prepared $\text{TiO}_2\text{-Cl}$ solution was added to a mixture of sodium ethyl xanthate and anhydrous ethanol. It was then washed with anhydrous ethanol. The obtained RAFT initiator based TiO_2 ($\text{TiO}_2\text{-SC(=S)OC}_2\text{H}_5$) was dried.

The synthesized $\text{TiO}_2\text{-SC(S)OC}_2\text{H}_5$ was mixed with MMA, AIBN, in DMF and was placed in an oil bath. After polymerization, the reaction mixture was filtered through a PTFE membrane, washed with THF and precipitated in methanol. The resulted $\text{TiO}_2\text{-PMMA}$ was dried.

The $\text{TiO}_2\text{-PMMA}$ hybrid nanoparticles initiate the polymerization of styrene, and the procedure was the same as that of the graft polymerization of MMA. The final product was $\text{TiO}_2\text{-PMMA-b-PS}$ (Fig. 26).

Modification of TiO_2 nanoparticle by successful grafting of functional groups and polymers were examined by FT-IR spectroscopy. The presence of diblock copolymers on the surface of nanotitania was also evidenced from TGA. The surface morphology of the synthesized nanocomposite was observed by SEM.

The mechanical properties, thermal stability, flexural strength, and impact strength were found to be higher in $\text{TiO}_2\text{-PMMA-b-PS}$.¹¹⁸

2.3.4. With Poly(glycidyl methacrylate). Using a modified solvothermal method TiO_2 nanoparticles were synthesized.¹¹⁹ In this process oleic acid, oleylamine, cyclohexane, and $\text{Ti}(\text{OBu})_4$ were mixed. The TiO_2 nanoparticles were collected by precipitation and

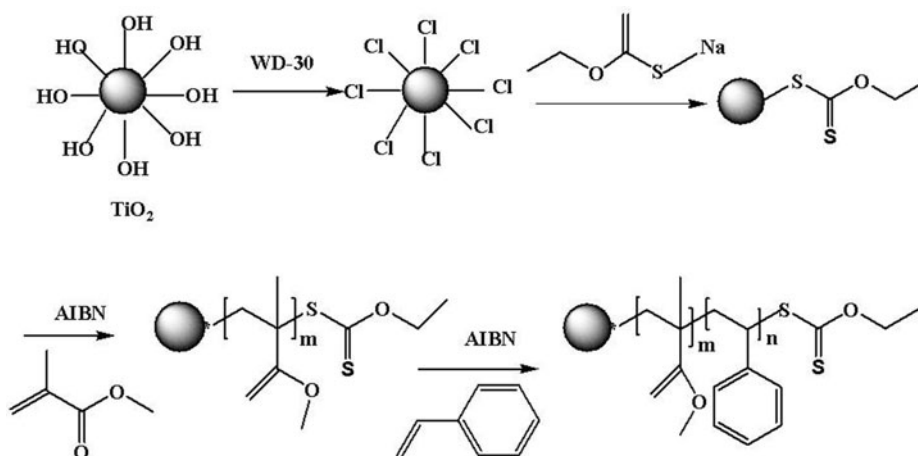


Figure 26. The modification of TiO_2 .

centrifugation. Phosphate-azide ligands were prepared by a three-step process.¹²⁰ The synthesized phosphate-azide ligands and TiO₂ nanoparticles were refluxed in chloroform. The particles were washed, centrifuged, and then dispersed in THF.

Reversible addition-fragmentation chain transfer polymerization was used for the synthesis of alkyne terminated PGMA. A solution of glycidyl methacrylate, THF, prop-2-ynyl 4-cyano-4-(phenyl carbonothioylthio) pentanoate, and AIBN were mixed in a reaction tube. The mixture was degassed and filled with nitrogen again. The polymerization was stopped by cooling and the polymer was precipitated in hexane. It was then recovered by centrifugation and dried under vacuum.^{22,23} The modified TiO₂ nanoparticles were mixed with alkyne terminated PGMA, CuBr and PMDETA ligand. The reaction mixture was dispersed in THF and degassed. The polymer grafted nanoparticles were precipitated in methanol and separated by centrifugation.

The diameter and dispersion of nanotitania was determined by TEM analysis. XRD technique was used for characterizing the crystalline nature of nanoparticles. FTIR spectral analysis was carried out to examine the surface modification of TiO₂ nanoparticles. The grafting density of polymer onto nanoparticle was calculated by TGA. The surface profile of the composite was investigated using AFM. Transparency of the nanocomposite was tested using UV-Vis spectroscopy.

The synthesized TiO₂-PGMA has high transparency and refractive index.¹²¹

2.3.5. With Polystyrene. Nitroxide-mediated radical polymerization initiator was prepared.¹⁸ A suspension of nitroxide-mediated radical polymerization initiator and TiO₂ was prepared by sonicating them in THF. The suspension was sonicated with cooling, centrifuged, decanted, and rinsed with ethanol. Nitroxide-mediated radical polymerization initiator attached TiO₂ nanoparticles were thus obtained.

The modified TiO₂ nanoparticles were mixed with styrene monomer and free initiator (4-hydroxy-1((2'-hydroxy-1'-phenylethyl)oxy)-2,2,6,6,-tetramethylpiperidine). Polymerization took place after degassing and heating. The monomer to initiator ratio determines the rate of polymerization. By centrifugation and filtration the dispersed TiO₂ nanoparticles were separated. The product, PS-grafted-TiO₂ was reprecipitated in methanol and collected by filtration. The obtained PS-grafted-TiO₂ with PS powder was spin coated through PTFE filter on a silicon wafer. The films were then dried under vacuum. Thus PS-grafted-TiO₂-PS hybrid thin film was obtained (Fig. 27).¹²²

Mass of polymer grafted on TiO₂ nanoparticle was examined by GPC analysis. TGA was used to determine the graft density of polymer on nanoparticle. Polystyrene chains on the surface of nanoparticle were characterized by WAXD. The effect of grafting on the optical properties and dispersion behavior of nano was studied from UV-Vis absorption measurements. TEM analysis was carried out to observe surface morphology. The average size of nanocomposite was determined by DLS. Wear resistance of the synthesized composites were revealed by friction tests.

TiO₂ nanoparticles with polystyrene brush were dispersed in good solvents such as THF, toluene, and ethyl acetate. Dispersion of polystyrene grafted-TiO₂ particles showed a very big difference in its absorption maxima in ultra-violet and visible regions of electromagnetic spectra. The good dispersion of TiO₂ nanoparticles in PS-grafted- TiO₂ films made them to exhibit high transmittance. PS-grafted- TiO₂ films also exhibited good wear resistance.¹²³

2.3.6. With Poly(L-lactide). Grafting of L-lactic acid oligomer onto TiO₂ nanoparticle was achieved by a condensation reaction. Stannous octanoate was used as catalyst for the reaction. First, a dispersion of TiO₂ nanoparticles in tetrahydrofuran was prepared by

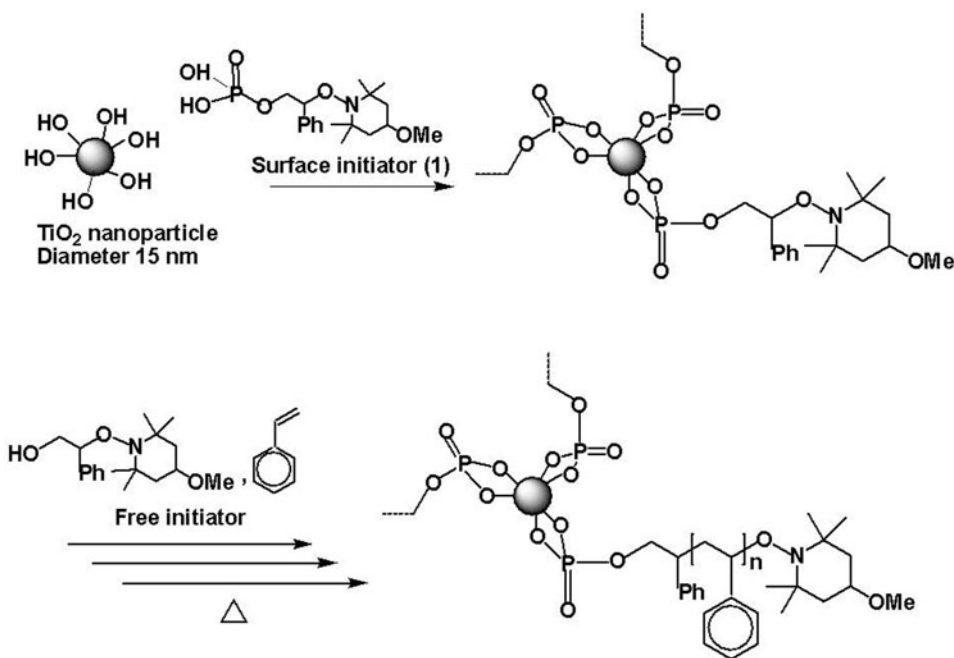


Figure 27. Preparation of PS-grafted-TiO₂-PS Hybrid film.

an ultrasound and L-lactic acid was added into it with stirring. Then stannous octanoate containing L-lactic acid was added as a catalyst and heated under vacuum. The resultant L-lactic acid oligomer grafted TiO₂ nanoparticles were collected and washed with chloroform. The precipitated PLLA- g-TiO₂ was dried in vacuum.

A chloroform solution of PLLA was prepared¹²⁴ and a mixture of g-TiO₂ and TiO₂ nanoparticles was added to it. The mixture was then placed in the mold. After complete evaporation of the solvent, the films of g-TiO₂/PLLA and TiO₂/PLLA composites were removed and dried under vacuum.

The modification of nanotitania was evaluated by FTIR. TGA was carried out to calculate the amount of polymer grafted on TiO₂ surface. The dispersibility of nanoparticles was investigated by TEM analysis. SEM analysis was conducted to study the surface morphology of the synthesized nanocomposites.

Compared with the pure PLLA, the g-TiO₂/PLLA and TiO₂/ PLLA nanocomposites exhibit a higher tensile strength and lower ductility. But the g-TiO₂/PLLA nanocomposites exhibited better mechanical properties than that of the TiO₂/PLLA nanocomposites.¹²⁵

2.3.7. With poly(dimethyl siloxane). By a two-phase thermal synthesis approach¹²⁶ titanium dioxide nanoparticles were prepared. In this process titanium (IV) butoxide was added to aqueous solution of *tert*-butylamine in a stainless steel pressure vessel. Oleic acid and toluene were homogeneously mixed with it. The reaction mixture was then heated and cooled. The obtained TiO₂ nanoparticles were recovered by centrifugation.

TiO₂ nanoparticles were then grafted with PDMS.¹²⁰ For this modified PDMS¹²⁰ was added to TiO₂ nanoparticles in chloroform and refluxed. The resultant grafted nanoparticles were precipitated by using ethanol and dispersed in chloroform.

The grafted TiO₂ nanoparticles were placed in silicone matrix. It was stirred and the solvent was removed by vacuum drying. Methylhydrosiloxane–dimethylsiloxane copolymer (cross linking agent) was added to the reaction mixture with stirring. The grafted TiO₂ NP/silicone nanocomposite was then dried in vacuum.

The crystalline nature of nanoparticle was characterized by XRD. The size and distribution of nanotitania was determined from TEM analysis. Surface modification of TiO₂ nanoparticles was obtained from ³¹P NMR spectrum. TGA was used to evaluate the density of polymer grafted on nano. Optical properties of the nanocomposites were checked by UV–Vis absorption measurements. Parametric phase diagram was used to predict the dispersion state of the grafted nano particles at equilibrium.

The prepared TiO₂ nanocomposites are used as novel light-emitting diodes (LED) encapsulant materials to significantly improve its light-extraction efficiency. These nanocomposites can be applicable in UV-filtering optical coatings also.¹²⁷

Here the Titanium Dioxide nanoparticle was grafted with various organic materials like PMMA, PMMA-PMASi, PMMA-b-PS, PGMA, PS, PLLA, and PDMS. The obtained polymer grafted TiO₂ nanoparticles have improved properties and are used in the preparation and modification of bone cement, implant biomaterials, nanofillers, dental composites, and ultrathin films.

2.4. Zinc Oxide Nanoparticles

Nano-ZnO has been gaining attention due to its significant physical and chemical properties such as high luminous transmittance, chemical stability, antibacterial properties, high catalytic activity, and intensive UV and IR adsorption. The ZnO polymer nanocomposites can enhance the stable dispersion of nano ZnO, which enhance its action in coatings, plastics, sealants, and fibers.

2.4.1. With Polychloromethyl Styrene. The synthesis of polymer grafted ZnO nanoparticles by SCVP via SI-ATRP. The SCVP of the “inimer” has both a polymerizing group and initiating group in the same molecule.¹²⁸

Bromoacetyl bromide was added in drops into chloroform solution of APTES and pyridine. Then the initiating compound solution was stirred and refluxed. The ZnO nanoparticle and initiating compound solution were irradiated ultrasonically and refluxed with stirring. The BrA-ZnO nanoparticles were filtered, washed, and dried.

BrA-ZnO, CMS, Cu(I)Br, 1,10-phenanthroline, and toluene were mixed and irradiated. It was then bubbled with nitrogen and refluxed with electromagnetic stirring. The ZnO nanoparticles grafted with hyperbranched polymer were separated and extracted. Then the product was washed to remove the copper.¹²⁹ The HP-ZnO was dispersed, stirred, and dried (Fig. 28).

Modification of the nanoparticle was confirmed by FTIR spectroscopy. The percentage of grafting of polymer on nanosurface was examined by TGA and elemental analysis. In order to elucidate the structure of grafted hyperbranched polymer ¹H NMR spectral analysis was carried out. The synthesized nanocomposites morphology was studied by using TEM. Other methods like UV–Vis absorption spectroscopy, XPS, etc., were used for the characterization.

The HP-ZnO with chlorobenzyl groups used as functional devices.¹³⁰

2.4.2. With Polymethacrylic acid. Radical polymerization was used for the polymerization of MAA. In a four-necked flask, water, MAA (monomer), KPS (initiator), isopropanol

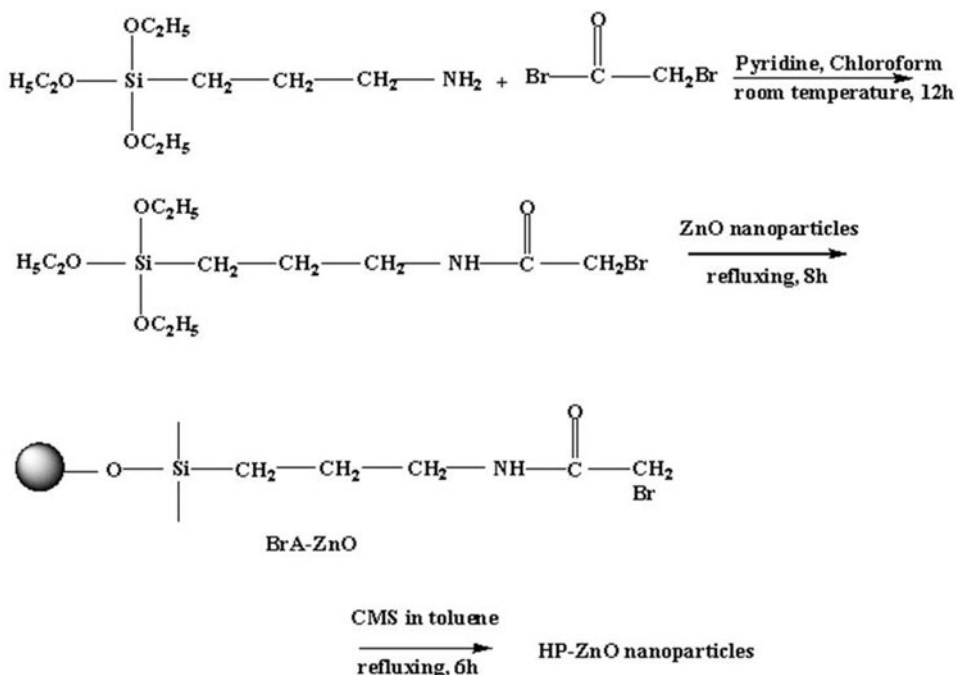


Figure 28. The preparation of HP-ZnO nanoparticles.

(chain transfer agent), and NaOH (neutralizer) were added and refluxed with Nano-ZnO and dispersed.

The functionalization of the nanoparticle was confirmed by FTIR spectroscopy. The polymer content on the functionalized nanoparticle was analyzed by TGA. X-ray diffraction studies were carried out to examine the effect of grafted polymer on the crystallinity of nano-ZnO. The dispersion as well as the size of the nanoparticle in the polymer was determined by TEM. A particle size analyzer was used to observe the mean size and the size distribution of the nanoparticles.

PMAA increased the dispersibility of nano-ZnO particles in water. This modification did not alter the crystalline structure of the nano-ZnO.¹³¹

2.4.3. With Polystyrene. Nano-ZnO's were grafted by PS using *in situ* emulsion polymerization in the presence of MPTMS as a coupling agent. ZnO particles were mixed with methanol or methanol-water mixture and MPTMS. This mixture was dispersed, heated, cooled, and diluted with n-propanol, centrifuged, and dried to obtain the MPTMS grafted nanoparticles.

Modified nano-ZnO was added to the system containing water solution of polyoxyethylene nonylphenyl ether (OP-10) (surfactant). The monomer and initiator were added followed by dispersion and heating to obtain the polystyrene grafted nanoparticle.

The morphology of ZnO nanoparticle and the nanocomposite were analyzed by TEM. The presence of functional groups on the nanoparticle surface was confirmed by FTIR. XPS analysis was done in order to understand the surface composition of the nanocomposite. Elevation in Tg determined by DSC helps to confirm the grafting of polymer on nanoparticle. SEM was used to analyze the dispersion morphology of the composite on coating latex

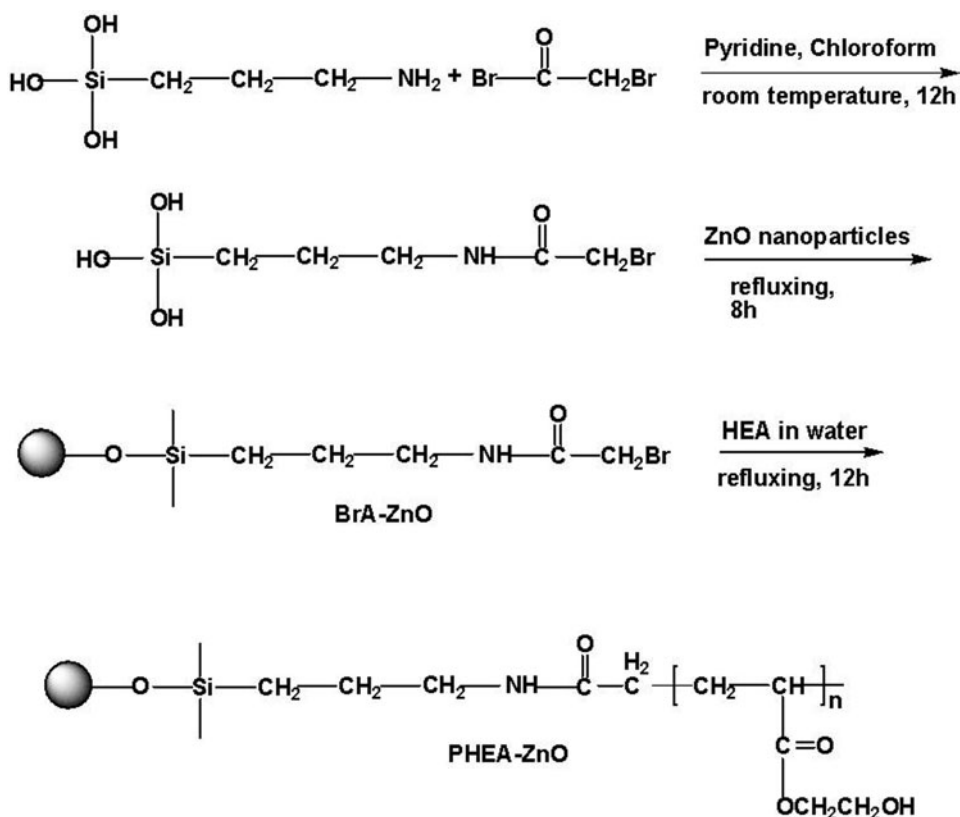


Figure 29. The preparation of PHEA-ZnO nanoparticles.

matrices. The optical properties of the synthesized nanocomposite were studied by UV–Vis absorption measurements.

Poly(styrene) zinc oxide composite particles applied to UV-shielding materials such as cosmetics, coatings, etc.¹³²

2.4.4. With Polyhydroethyl acrylate. SI-ATRP technique was used for the polymerization of polyhydroethyl acrylate grafted ZnO nanoparticles. The reaction between APTES and bromoacetyl bromide, using pyridine as catalyst produces an initiating compound. The mixture was stirred and refluxed. The ZnO nanoparticles and the initiating compound in chloroform were irradiated and refluxed. The bromoacetamide functionalized ZnO (BrA-ZnO) was separated washed and dried.

In a flask BrA-ZnO, Cu(I)Br, 1,10-phenanthroline, HEA, and water were mixed, irradiated, and refluxed. Then the mixture was poured into ethanol and the precipitate was dried. The products were washed thoroughly with water.¹³³ Using acetylacetone–ethanol solution, the powder was separated.¹³⁰ The product was then filtered and washed (Fig. 29).

The surface modification of the nanoparticle was confirmed by FTIR. XPS was also used to evaluate the surface modification. The polymer content on the functionalized nanoparticle was analyzed by TGA. TEM analysis was carried out to study the surface morphology of the nanocomposite.

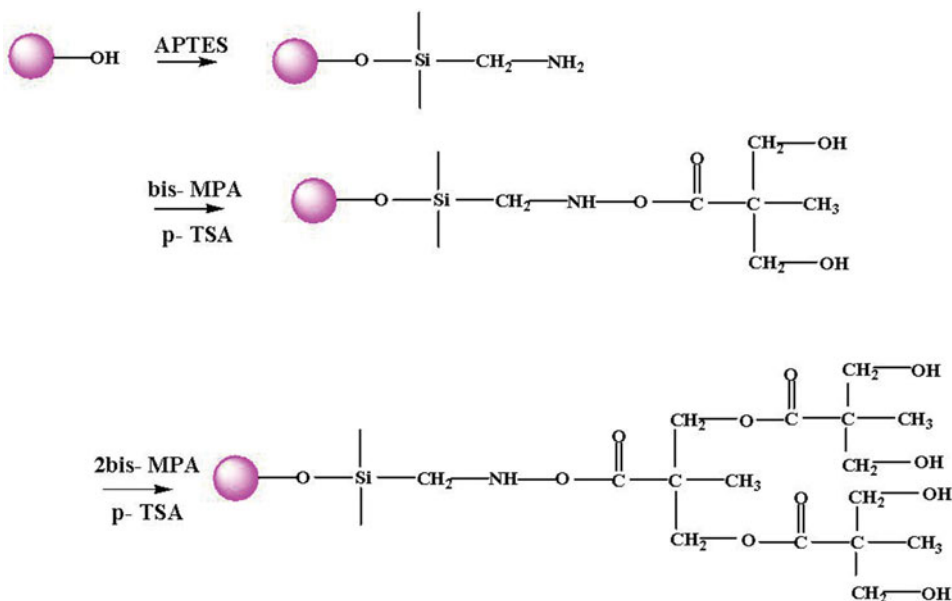


Figure 30. Preparation of HAPE- Al_2O_3 nanoparticles.

PHEA-ZnO nano composites have the ability to protect us from harmful radiations. It has better dispersibility. It is also expected to be used in ZnO semiconductor devices.¹³³

ZnO nanoparticles have unique catalytic, electrical, electronic, optical, and antimicrobial properties. So, it has widespread applications in many fields like UV-protection and antibacterial action, water-based coating, ceramic, and texture finishing agent, etc. In order to prevent the agglomeration the surface of nanoparticle should be modified. Here different polymers like polychloromethyl styrene, PMAA, PS, and PHEA were used. The resultant polymer nanocomposites are used in functional devices, semiconductor devices, and for UV-shielding.

2.5. Alumina Nanoparticles

Aluminum oxide nanoparticles were synthesized by the laser ablation method. Al_2O_3 nanoparticles have been broadly investigated due to their excellent wear resistance, and high thermal and mechanical stability with low cost.

2.5.1. With Polyester. Al_2O_3 nanoparticles were mixed with APTES, ammonia, and ethanol. The mixture was then irradiated with stirring, agitated, and heated. The product amino- Al_2O_3 was dried.

Amino- Al_2O_3 was polycondensated with bis-MPA (AB_2 -type monomer) to prepare hyperfunctional Al_2O_3 .¹³⁴ For this amino- Al_2O_3 was mixed with bis-MPA and p-TSA (catalyst). The mixture was then refluxed with stirring. After the addition of DMF the mixture was cooled and irradiated ultrasonically. It was then washed with deionized water and HAPE- Al_2O_3 was dried (Fig. 30).

The introduction of a functional group and polymer was examined by FTIR. The distribution of the nanoparticle and the surface morphology of the composite was analyzed by TEM. TGA was used to quantify the polymer content on nanoparticle. The successful

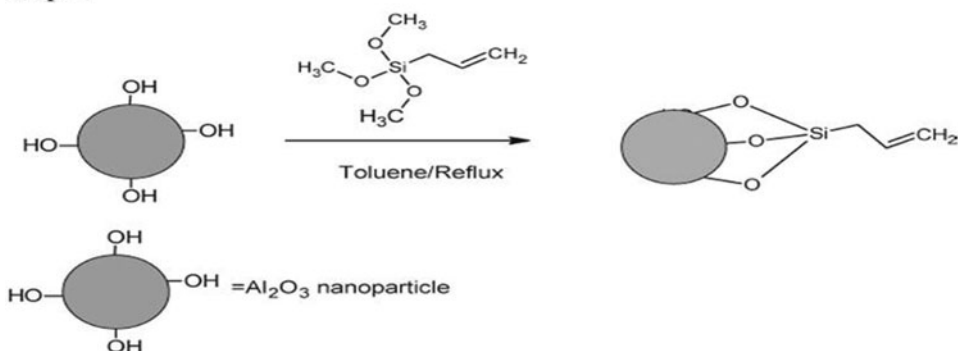
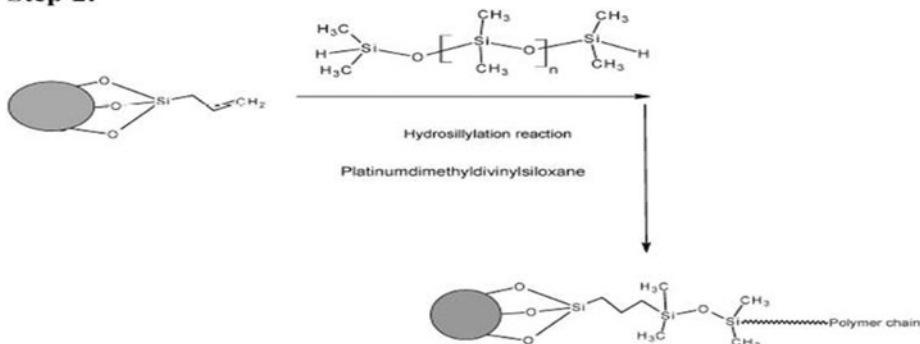
Step-1:**Step-2:**

Figure 31. (Step 1) Surface functionalization of alumina and (Step 2) hydrosilylation.¹³⁷ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

grafting of the polymer on the nanosurface was evaluated by XRF analysis. The effect of grafting on the crystallinity of nanoparticle was analyzed by XRD.

HAPE- Al_2O_3 is expected to be used as reactive fillers for polymer composites.¹³⁵

2.5.2. With Polydimethylsiloxane. First, γ -alumina nanoparticles were dried and then dispersed in dry toluene by sonication. ATMS was added into and refluxed under argon atmosphere. Toluene was then added and the precipitated allyl functionalized nanoparticles were dried under vacuum.¹³⁶

The allyl functionalized aluminum nanoparticles were mixed with the polymer matrix and dispersed using a magnetic stirrer. One drop of Karstedt catalyst was added to the mixture and stirred well. The reaction was carried out under high vacuum. The product obtained was the Polydimethylsiloxane grafted alumina nanoparticles (Fig. 31).

FTIR was used to examine the surface modification of nanoparticles. The surface modification was confirmed from XPS spectra and BET analysis. The functional groups grafted on the surface of nanoparticle were quantified by TGA and elemental analysis. Deviation in T_g determined by DSC helps to confirm the grafting of polymer on nanoparticle. Incorporation of polymer on to the modified nanoparticles was confirmed by FTIR (Fig. 32). The morphology of the synthesized nanocomposites was studied using SEM. The nanoindentation technique was used for the primary characterization of the composite. The

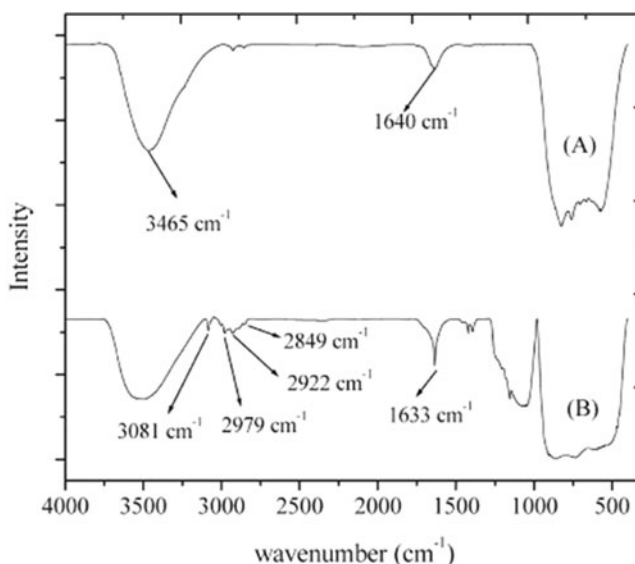


Figure 32. FTIR spectra of (A) pristine and (B) functionalized γ -alumina.¹³⁷ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

stability of the composite was tested at different pH values and their densities were also calculated.

Polydimethylsiloxane grafted alumina nanoparticles are excellent insulators. They have good thermal stability and mechanical elasticity so they are used for encapsulating organic photovoltaic devices and field-effect transistors.¹³⁷

2.5.3. With Polystyrene and Polypropylene. The surface of the γ - Al_2O_3 nanoparticles were modified by the following method: First, the γ - Al_2O_3 nanoparticles were dried and dispersed in xylene. Then, TMSPM/ xylene solution was added and heated. The mixture was cooled, filtered, and then dried. It was then put into styrene in methanol. The resulting solutions were flushed and then irradiated using γ -rays. Finally, the polystyrene grafted γ - Al_2O_3 was washed with hot benzene and dried (Fig. 33).

Using a lab-scale Brabender instrument polypropylene nanocomposites were prepared. It was done by blending polypropylene pellets, γ - Al_2O_3 -g-PS, and 1, 4-BDDA (cross linking agent), followed by e-beam irradiation.

The modification of the nanoparticle as well as its surface modification was evaluated using FTIR. As the surface modification makes the nanoparticle hydrophobic, it was confirmed by conducting a floating test on water. TGA was used to determine the weight of polymer grafted on the nanoalumina. Surface morphologies of the synthesized composite were studied by SEM.

E-beam irradiation and the dispersion of γ - Al_2O_3 -g-PS into the polypropylene matrix enhanced the mechanical properties of the nanocomposites. By increasing the amount of the cross linking agents the tensile strengths were improved.¹³⁸

2.5.4. With Polystyrene and Polyacrylamide. The nanosized alumina surfaces were functionalized by the reaction of silane. Alumina nanoparticles and γ -methacryloxypropyl

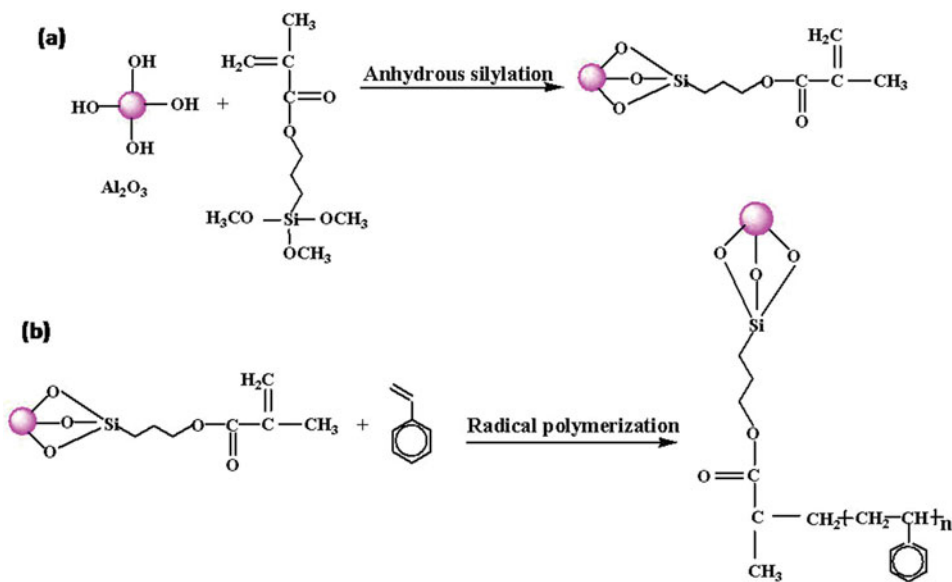


Figure 33. (a) Silylation with TMSPM and (b) grafting with styrene.

trimethoxy silane in alcohol solution were charged in to flask equipped with a reflux condenser and stirred. The alumina was centrifuged. The precipitate was extracted with alcohol and air-dried (Fig. 34).¹³⁹

The graft polymerization reactions for both styrene and acrylamide (vinyl monomers) were performed in an open flask equipped with a condenser. First, the modified alumina and the solvent were put in the temperature-controlled reactor with stirring. Then the monomer and the rest initiator were added in two equal batches after the alumina reacted with the first batch of initiator for a while. After the graft polymerization, the resultant suspension was centrifuged, washed, and dried. The precipitate was extracted with a suitable solvent (butanone for PS and water for PAAM) to isolate the polymer-grafted alumina and dried it under vacuum.

The modification of nanoalumina was observed by FTIR. Appearance of bands responsible for the silane coupling agent confirms the modification. Polymer grafting was

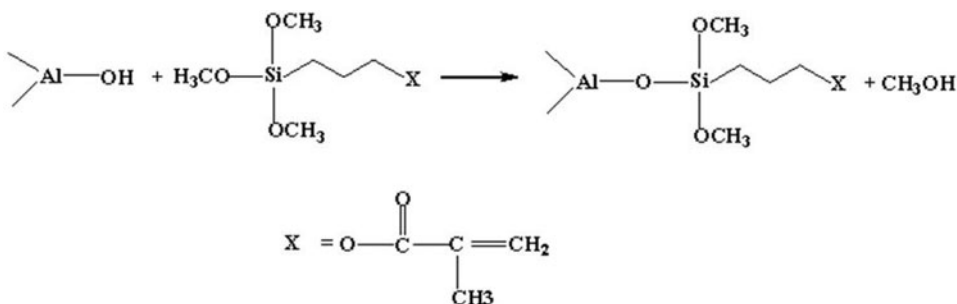


Figure 34. The reaction of silane coupling agent with the hydroxyl groups of alumina.

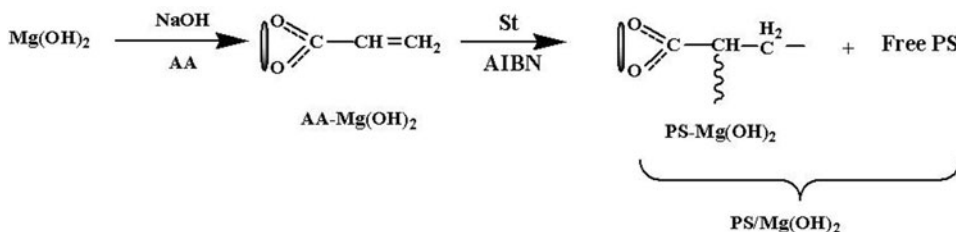


Figure 35. The procedure of PS/Mg(OH)₂ nanocomposite.

also confirmed by FTIR. Thermal stability of the composite was studied using TGA. GPC was used to find out the molecular weight of the grafted polymer.

By changing the reaction condition, the structural characteristics of the grafted polymers can be tailored. In polymer grafted nanoparticle, the hydrogen bonding between the grafted polymers and the nanoparticles was stronger than the interaction between the grafted polymers and the solvent. Therefore, a higher concentration of the monomer and the initiator was favorable to the grafting process. Prolonged reaction time helped to promote the grafting reaction of PS. But it reduced the percentage of grafting and the grafting efficiency of PAAM due to degradation of the grafted polymer.¹⁴⁰

The composites of alumina have been used in photovoltaic devices and field-effect transistors. They are good insulators and reactive fillers. This was achieved by the grafting of polymers like PE, PDMS, PS, PAAM, and PS-PP on to it.

2.6. Magnesium Hydroxide Nanoparticles

Magnesium hydroxide, Mg(OH)₂ is a metal hydrate. It can be used as a flame retardant. Nano magnesium hydroxides surfaces are hydrophobic. In order to disperse them in organic polymers, its surface was modified with different surface modifiers such as stearic acid, silicone-oil, acid coatings, silane, and titanate coupling agents.

2.6.1. With Polystyrene. In order to graft polystyrene with magnesium hydroxide nanoparticles, first, the surface of the nanoparticle is modified by different functionalizing molecules.

2.6.1.1. Using Acrylic Acid. Radical copolymerization was used for the preparation of PS/Mg(OH)₂. Functionalized magnesium hydroxide nano-needles were prepared by mixing aqueous solution magnesium chloride, AA, and sodium hydroxide. The resultant acrylic acid functionalized magnesium hydroxide nano-needles were filtered and dried.

Acrylic acid functionalized magnesium hydroxide nano-needles were mixed with styrene, toluene, and AIBN. During the polymerization process N₂ was bubbled. It was then dispersed and heated with stirring. After cooling the reaction mixture was poured into methanol. The precipitated polystyrene/magnesium hydroxide nanocomposite (PS/Mg(OH)₂) was dried in vacuum (Fig. 35).

The needle like morphology of Mg(OH)₂ was confirmed from TEM images. The surface functionalization of nanoparticle as well as polymer grafting to modified nano was examined by FTIR analysis. Thermal stabilities of the composite were investigated. From DSC analysis the variation in T_g of the nanocomposite was observed which also indicate the grafting of polymers.

Polymer grafted nano needles have improvable thermal properties, that is, a lower decomposition temperature. So it used as an environmentally-friendly flame retardant.¹⁴¹

2.6.1.2. Using α -bromoisobutyric acid. Magnesium chloride was dissolved in water and mixture containing α -bromoisobutyric acid and sodium hydroxide were added with stirring. Bromoisobutyric acid functionalized magnesium hydroxide nano-needles BA-Mg(OH)₂ obtained was filtered and washed.

For the graft polymerization, the functionalized nano-needles BA-Mg(OH)₂ were dispersed in toluene and a mixture of 2,2'-bipyridine and Cu(I)Br were added. It was evacuated and purged with nitrogen. Then the degassed St was injected into the solution. The SI-ATRP product, polystyrene grafted magnesium hydroxide nano-needles were separated by centrifugation, washed with toluene, ethanol, and water and then dried in vacuum (Fig. 36).

The grafting of macroinitiator and polymer was confirmed by FTIR. Grafting was again confirmed by elemental analysis. The polydispersity index and molecular weight of the polymer was determined using GPC. TEM was used to study the surface morphology of the composite.

By the organic phase, polystyrene grafted magnesium hydroxide nanoneedles have better wettability.¹⁴²

Here Mg(OH)₂ nanoparticles were dispersed in PS by the surface modifiers, viz. acrylic acid and α -bromoisobutyric acid. The resultant nanocomposites can be used as nanofillers and flame retardant in polymeric materials.

2.7. Gold Nanoparticles

Among the noble metal nanoparticles, gold nanoparticles are capable of existing in the unoxidized state at the nanoscale and are widely used due to their exceptional electronic, optical, biomedical, and catalytic properties.

2.7.1. With Polystyrene. HAuCl₄ aqueous solution was heated under magnetic stirring. Trisodium citrate solution was added and cooled. Gold nanoparticles (AuNPs) were obtained.

Polystyrene microspheres were prepared by dispersion polymerization. PVP in isopropanol was heated and AIBN in styrene monomer was added. The mixture was cooled and resultant PS microspheres were obtained. PS microspheres were centrifuged, redispersed, and diluted.

Gold nanoparticles-coated polystyrene composite was prepared by blending PS microspheres and AuNPs. The aqueous dispersion of PS microspheres was diluted with water. PVP, hydrochloric acid, sodium hydroxide, methanol, ethanol, or isopropanol was added into it. The AuNPs were finally added, homogenized, and stirred.

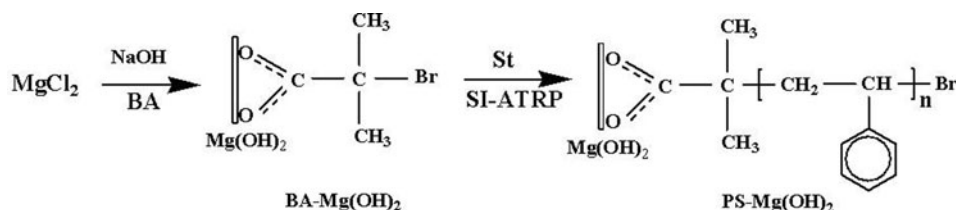


Figure 36. The procedure of PS-Mg(OH)₂ nanocomposite.

The raspberry-like morphology of the nanocomposite was obtained by techniques like SEM and TEM. The amount of polymers grafted was quantified by TGA. Using a zeta potential analyzer the composites zeta potential and dispersion at different pH was calculated.

Variation of pH in the system shows a corresponding change in the aggregation of AuNPs on the surface of PS microspheres. This work was helpful for the controllable preparation of the AuNPs-coated PS composites with different morphology. The synthesized nanocomposites are applied in catalysis. They can be also applied in surface enhanced Raman scattering.¹⁴³

2.7.2. With Poly(N-isopropylacrylamide) and Polystyrene. The “Grafting to” technique was used for the preparation of hydrophilic PNIPAM and hydrophobic PS grafted gold nanoparticles. PNIPAM and PS were synthesized by using RAFT polymerization. NIPAM monomer, RAFT agent (4-cyanopentanoic acid dithiobenzoate) and AIBN were dissolved in dioxane. The mixture was degassed and polymerized. The polymer was precipitated in diethyl ether, purified by repeated precipitations, and dried under vacuum. After polymerization, the reaction mixture was dissolved in THF and PS was precipitated in cold methanol, purified by repeated precipitations, and dried in a vacuum.^{144–146}

THF solution of Hydrogen tetrachloroaurate(III) hydrate ($\text{HAuCl}_4 \cdot x\text{H}_2\text{O}$), PNIPAM, and PS were mixed. Then the THF solution of lithium triethylborohydride ($\text{LiB}(\text{C}_2\text{H}_5)_3\text{H}$) was added in drops to the stirred solution in an ice bath. After centrifugation the supernatant was collected. The purple crude product was dispersed in ethanol and purified by ultrafiltration. Ethanol was used as an eluent to remove the ungrafted PNIPAM. The product was washed using diethyl ether and dried in vacuum (Fig. 37).

Grafting of polymers onto the nanosurface was evaluated by FTIR. HRTEM was used to examine the size distribution of nanoparticles (Fig. 38). The mean diameter of nano gold in the polymer was obtained using DLS. TGA was used to calculate the amount of polymer grafted on the nanoparticle. UV-Vis absorption measurements were carried out to measure the surface plasmon band. The monolayer properties of gold nanoparticle were studied in the air-water interface. Homogeneous distribution of polymers was observed from the contact angle measurements using XPS. SEC was used to determine the average molecular weights and polydispersity index of polymers on the surface of nano Au.

The phase separation occurs between chemically different PNIPAM and PS on the surface of the gold cores.¹⁴⁷

2.7.3. With Polyethylene glycol. To prepare the PEGylated Gold Nanoparticles, citrate-coated gold nanoparticles were added to aqueous solution of PEG (Fig. 39). The mixture was stirred, incubated, and purified by centrifugation. The supernatant was decanted and resuspended in PBS.

In the next step gold nanoparticles with mixed monolayers of PEG and RME-Peptides (Receptor-mediated endocytosis) were synthesized. The mixture was stirred, washed, and centrifuged. The decanted supernatant and the resultant pellets (Gold Nanoparticles with Mixed Monolayers of PEG) were resuspended in PBS. PEG length and particle size are two important parameters for creating stable peptide/gold nanoparticle conjugates.¹⁴⁸ No aggregation of nanoparticle took place during modification; it was cleared from the UV-visible spectrum (Fig. 40).

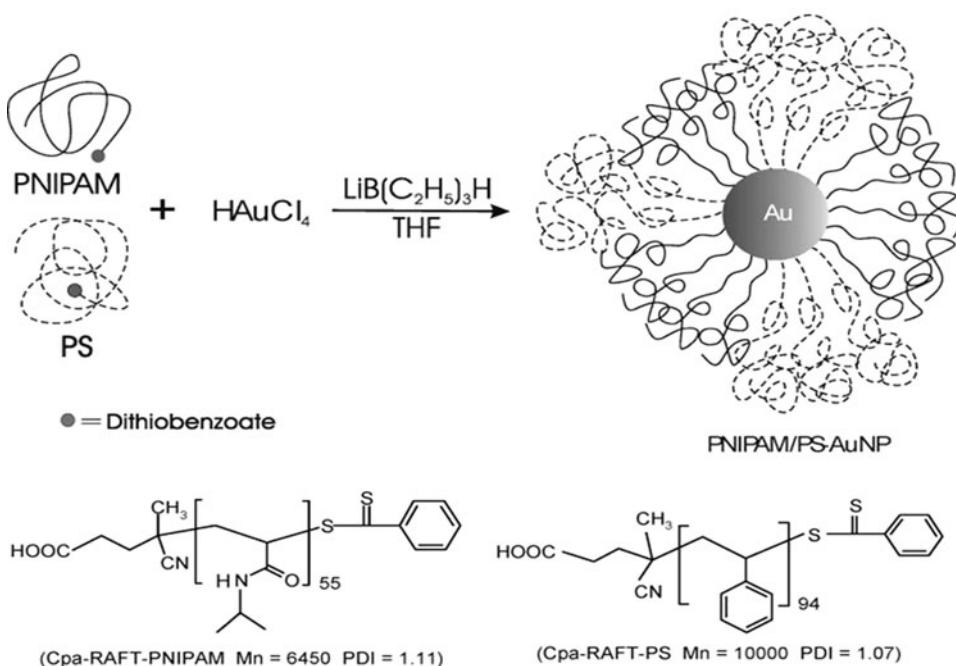


Figure 37. Preparation of gold nanoparticles grafted with PNIPAM and PS chains.¹⁴⁷ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

Gel electrophoresis and DLS were used to confirm modification of gold nanoparticles. Critical coagulation concentrations were measured to study the stability of nanoparticle achieved by binding with the polymer. Nonspecific binding of protein to modified nanoparticle was determined by fluorescence assay. Assembly of mixed monolayers on

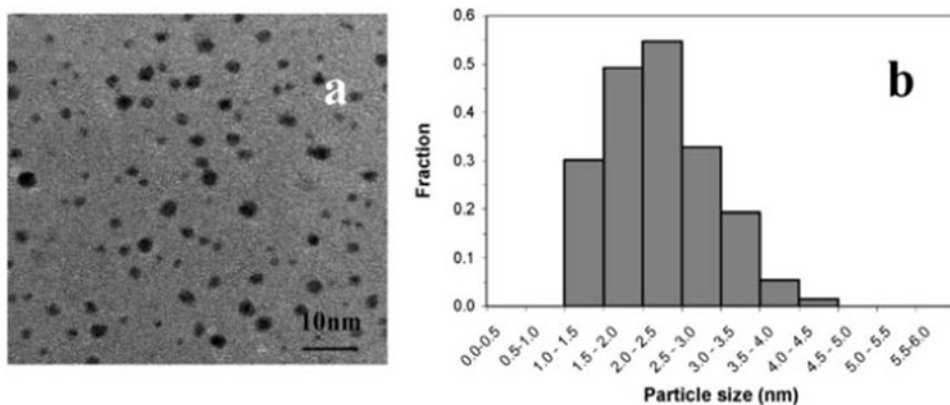


Figure 38. HRTEM images (a) and corresponding size distributions (b) for AuNP.¹⁴⁷ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

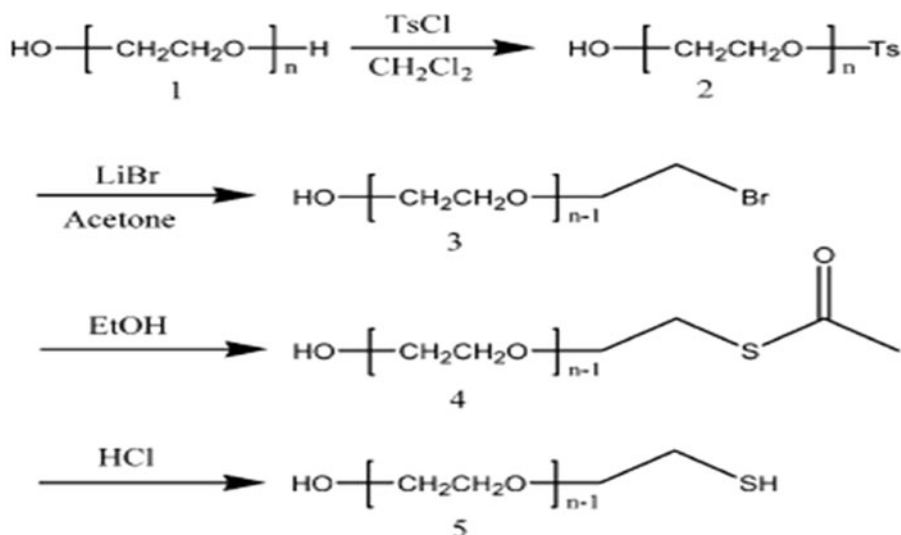


Figure 39. Synthesis of PEG 900 (n = 20) and PEG 1500 (n = 34) molecules.¹⁴⁸ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

the nanoparticle surface was investigated by UV-visible absorption spectroscopy and gel electrophoresis. Fluorescence spectroscopy was used to calculate the quantity of peptides adsorbed on the surface of nano Au. Inductively coupled plasma optical emission spectrometry was used to study nanoparticle adsorption to the cell membrane.

The synthesized nanocomposites can be applied in selective cell targeting and internalization.¹⁴⁸

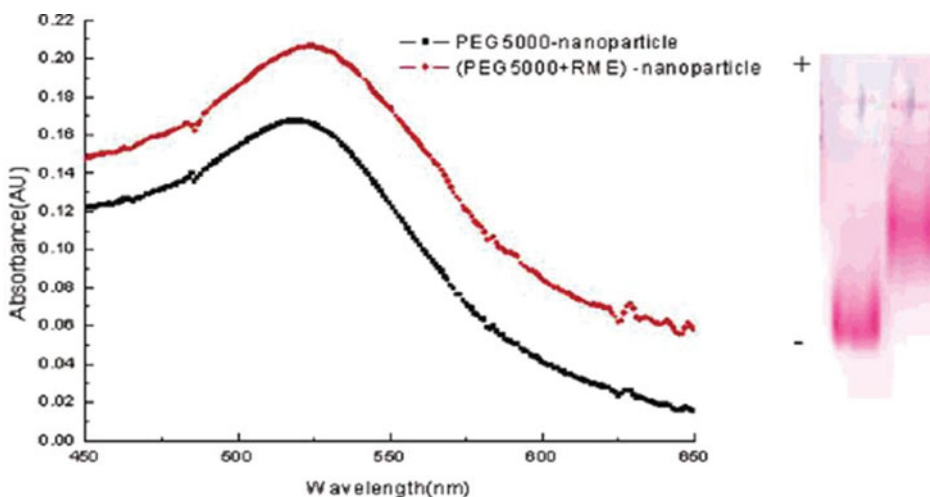


Figure 40. UV-visible absorption spectra of gold nanoparticles modified with PEG 5000 and PEG 5000 + RME peptide.¹⁴⁸ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

2.7.4. *With Poly(styrene-*b*-2-vinylpyridine)*. Au nanoparticles are stabilized by thiol-terminated PS (PS-SH) in order to incorporate it into the PS-*b*-P2VP block copolymer template. Living anionic polymerization using benzene as solvent was used for the synthesis of PS-SH. Sec-Butyllithium was used as the initiator. The resultant polystyryl anions were titrated using ethylene sulfide and then the sulfide ends were protonated by acidic methanol. Thiol-terminated P2VP (poly-2-vinylpyridine) was also prepared by the same method described above.

The two-phase method was used for the preparation of PS-coated Au¹⁴⁹ conducting separation, concentration, centrifugation, filtration, redispersion, and membrane filtration for the purification of PS-coated Au.

The one-phase method was used for the preparation of P2VP-coated Au nanoparticles. HAuCl₄ and PVP were mixed in various ratios of P2VP, dissolved in THF, and mixed under N₂. By adding super hydride reducing agent dropwise under N₂, Au nanoparticles were prepared.¹⁵⁰ The reducing agent was added until no more gas evolved. The product was purified by membrane filtration using DMF.

In order to prepare the nanocomposite, block copolymer solution in dichloromethane was mixed with PS-coated Au nanoparticles (PS-Au). A particle/block copolymer composite was prepared by solvent casting a mixture of PS-Au nanoparticles and PS-*b*-P2VP block copolymer in dichloromethane onto an epoxy substrate. Samples were annealed under a solvent and dried.

TEM was used to calculate the average particle diameter of gold nanoparticles. TEM analysis also gave information about the dispersion of gold nanoparticles in the polymer matrix (Fig. 41). TGA and elemental analysis were carried out to calculate the weight fractions of polymer chains on the nano surface. The chain areal density of polymers on the surface of the nanoparticle was also calculated.

This method is applicable to all hybrid materials by changing the position of nanoparticles and the areal density of the polymer.¹⁵¹

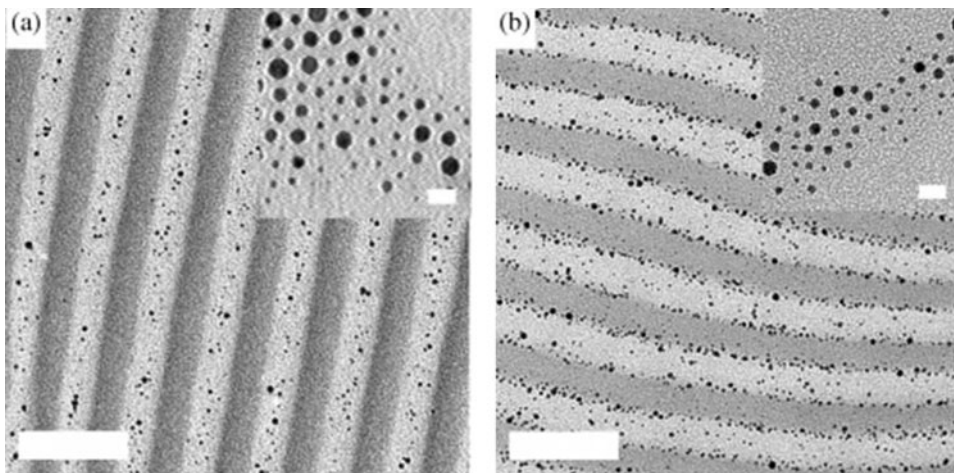


Figure 41. Effect of PS grafting densities on particle location in PS- *b*-PVP block copolymer template by ((a) one-phase method using THF; (b) two-phase method (water/toluene)).¹⁵¹ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

2.7.5. With Polymethacrylic acid. Gold nanoparticles were synthesized from chloroauric acid (HAuCl_4) using trisodium citrate as reducing agent.¹⁵² The reaction was carried out in aqueous solution and the prepared gold nanoparticles were stored under cold condition.

The prepared nanoparticles were then embedded on the MIP-coated electrodes. The procedure for the reaction is as follows: First, gold-coated silicon wafers were ultrasonically rinsed with acetone, water, and ethanol and then dried under argon. These Au slides were chronoamperometrically grafted with 4-mercaptophenyl layer and the obtained modified surface (represented as E-SH) was sonicated in ACN, DCM, and distilled water and then dried under argon. The E-SH electrodes were immersed in Au colloid solution to yield E-SH-np slides. Then a photoinitiator was grafted onto the electrode by electrochemical reduction of 4-benzoyl benzene diazonium salt to get the modified E-np-BP. This served as macro-photosensitizer for preparing MIP films by SIPP in the presence of dopamine (template). In this process MAA was the functional monomer (F), EGDMA was the cross-linker (C), DMA was the hydrogen donor, and the reaction was carried out in a mixture of methanol and chloroform. First, MAA and DMA were added to the solvent to allow the formation of a prepolymerization complex. Then EGDMA with DMA was added. The modified slides were immersed in this reaction mixture, degassed and exposed to UV light. In order to remove the unreacted monomers the dopamine-imprinted polymer films were rinsed with ACN, then CHCl_3 , and stored under argon. The E-np-MIP samples were then rinsed in distilled water and acetone.

The synthesized nanoparticles were characterized by UV-Vis absorption measurements. SEM was used to study the size and morphology of gold nanoparticles. Polymer coated gold surfaces were investigated by AFM. XPS was used to observe the changes in the chemical composition of the gold electrode by electrografting.

The diazonium compounds are of great interest in designing high performance ultrathin biomimetic molecularly imprinted sensing layers. It is expected that the synthesized molecularly imprinted polymer grafts with gold nanoparticles will be used in sensor devices. This is due to the enhanced surface area and catalytic effect of the nanoparticles.¹⁵³

In order to improve the properties of gold nanoparticle it is functionalized with different groups. Here gold nanoparticles are grafted with polymers like PS, PNIPAM-PS, PEG, PS-b-P2VP and PMAA. The resulting nanohybrids have many applications. Most important uses of them are in ultrathin biomimetic molecularly imprinted sensing layers and sensor devices.

2.8. Silver Nanoparticles

Physical vapor deposition and ion implantation were used for the synthesis of silver nanoparticles. Silver nanoparticles were used in numerous technologies and incorporated into a wide range of consumer products due to their optical, diagnostic, antibacterial, and conductive properties. Polymers can effectively support the nanoparticles stabilization process.

2.8.1. With Cotton Fabric. Amino functional silver nanoparticles were prepared by the reaction of silver nitrate and amino-terminated hyperbranched polymer.¹⁵⁴ Both the reactants were mixed with stirring and heated to obtain a yellow solution. It was then cooled and deionized water was added in order to overcome the loss of water by the heating process.

An aqueous solution of sodium periodate was prepared and cotton fabric was dipped in it. The cotton fabric had been oxidized; it was then washed with water and dipped in Ag NPs polymer hybrid solution. It was again washed and the resulting sample was then air dried for obtaining silver grafted oxidized cotton fabric (Ag-GOCF) (Fig. 42).

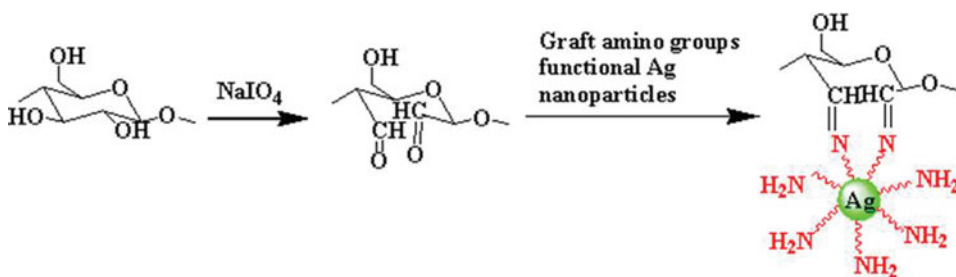


Figure 42. Preparation of Ag-GOCF.

The size distribution of amino functionalized silver nanoparticles was obtained by TEM. The size and polydispersity of modified nanoparticle was examined by HPPS. FTIR studies confirm the surface functionalization and grafting. EDS was used to study elemental composition. SEM was used to study the surface morphologies of the grafted cotton fiber. AES was used to determine the amount of Ag in the cotton fabric. The silver content was responsible for the antibacterial property of fabric. The breaking strength of the fabric was also measured.

Silver grafted oxidized cotton fabric exhibit good antibacterial properties. Ag-GOCF exhibit laundering durability which make it suitable for various commercial applications.¹⁵⁵

2.8.2. With Polystyrene. Polystyrene can be grafted to the silver nanoparticle either by functionalizing the polymer or nanoparticle.

2.8.2.1. By Functionalizing Silver Nanoparticle. The surface of the silver nanoparticles was modified by mixing NAg and APTMS in octane and refluxed with stirring. The solid obtained by decantation was dried.

Encapsulation of the silver nanoparticle was done by miniemulsion polymerization techniques using a co-stabilizing agent (HD). The polymerization was done in a micellar solution prepared by mixing a surfactant, CTAB and water. A mixture of initiator AIBN, modified Ag nanoparticles, and styrene monomer were dispersed in this solution ultrasonically. The polymerization process was carried out under nitrogen atmosphere. In order to enhance the stability and compatibility of the solution methacrylic acid was added to it. The residue polystyrene grafted silver nanoparticles was separated and washed with deionized water and toluene.

Surface functionalization and grafting of polymer to functionalized nano was confirmed by FTIR. Spherical morphology of nanosilver inside the polymer was obtained from SEM. The antimicrobial activity of the synthesized composite was also studied.

The synthesized PS-NAg nanocomposites showed a good biocidal potential against *E. coli* and *S. aureus*. So these particles are applied in clinical paints and in the manufacturing of coating materials used for biomedical applications.¹⁵⁶

2.8.2.2. By Functionalizing Polystyrene. Thiolate/thiol-functionalized polystyrene was prepared as follows: A macroinitiator solution in THF was taken in a reaction tube and warmed. Styrene and ethylene sulfide were added in drops into it. The product PS-SH was obtained and stored under cold conditions.

Polystyrene-stabilized silver nanoparticles were prepared by the following methods: (a) The solution of silver trifluoroacetate in THF was added to the functionalized polystyrene. The product was precipitated in methanol, washed with water, and dried in vacuum. (b) In super hydride reduction method a solution of thiolate-functionalized polystyrene in THF,

Trifluoroacetic acid was added to the solution of silver trifluoroacetate in THF. Super hydride (triethylboron hydride in THF) was added into it. The obtained product was precipitated in methanol. It was then washed with water, methanol, and dried in vacuum (Fig. 43). The hybrid materials were characterized by using the standard methods like TEM, XRD, UV-Vis spectroscopy, DSC, and GPC.

Polystyrene-stabilized silver nanoparticles were stable against thermal and chemical degradation. Dispersions of polystyrene-stabilized nanoparticles showed exceptional stability against aggregation in organic solvents like toluene, etc. It was highly transparent with a single glass transition temperature.¹⁵⁷ The addition of nanoparticles to polymer raise the glass transition temperature. The resulting nanocomposite was used as anti bacterial coating. It has some optical applications as well.¹⁵⁸

2.8.3. With Chitosan. Triangular shaped Chitosan-coated silver nanoparticles were prepared via speed-mediated growth method.¹⁵⁹ In this method stock solution of silver nanoparticles called seed was prepared in water by the NaBH_4 reduction of AgNO_3 . One portion of the aqueous solution of AgNO_3 was mixed with aqueous solution of trisodium citrate, cooled, and stirred vigorously. Aqueous solution of NaBH_4 was added dropwise to the above reaction mixture and stored in dark. Then a mixture of aqueous solutions of trisodium citrate, ascorbic acid, chitosan in acetic acid, and seeds, a solution of AgNO_3 was added dropwise under continuous magnetic stirring.

TEM was used to determine the mean diameter of nanoparticles. The particle size of the synthesized nanosilver was obtained by dynamic light scattering measurements. Atomic absorption spectroscopy was used to measure the concentration of metal ions.

Chitosan-coated silver nanotriangle was used as a photo thermal agent. The synthesized polymer nanocomposites are excellent phototherapeutic agents in the treatment of cancer.¹⁶⁰

2.8.4. With PNIPAM-co-MMP-co-MTU. Silver nanoparticles were prepared by the sodium borohydride reduction of dilute solution of silver nitrate. PVP was used as a stabilizer which prevented the agglomeration of silver nanoparticles.

Stimuli sensitive terpolymer was synthesized by copolymerization of NIPAM, Maleimido thiourea and Maleimido methyl palmitate. 1,4 dioxane, AIBN were used as solvent and initiator, respectively.

In order to load the silver nanoparticles the terpolymer solution in water was added to the Ag nanoparticle solution and shaken well (Fig. 44).

The terpolymer showed a LCST around 25°C but the Ag nanoparticle loaded terpolymer showed decrease in LCST (22°C). This indicated the structural change during silver nanoparticle loading in solution. In the UV-Visible spectroscopic studies the blank and terpolymer loaded solution showed variation in absorption.¹⁶¹ The surface morphology of the terpolymers is studied using SEM (Fig. 45). Further studies to understand the typical packing as shown in SEM, are in progress.

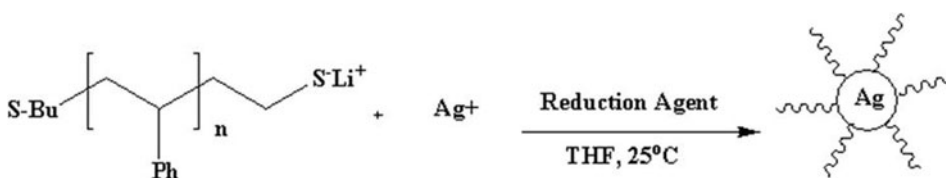


Figure 43. Synthesis of polystyrene-grafted silver nanoparticles in THF.

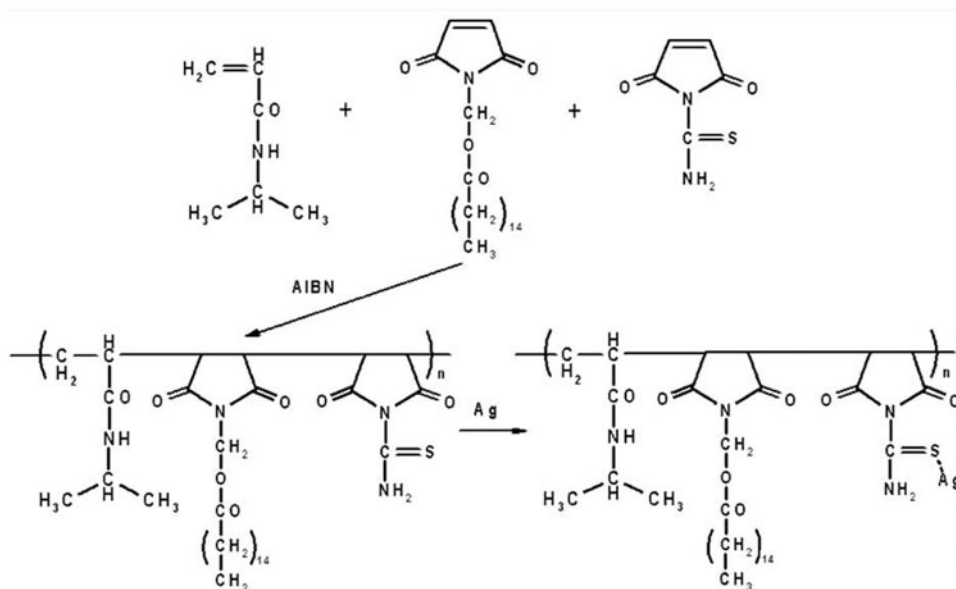


Figure 44. Synthesis of terpolymer and its loading with Ag nanoparticle.

2.8.5. With Chitlac (Lactose Modified Chitosan). Silver nanoparticles were synthesized by the reduction of silver ions using ascorbic acid in Chitlac solutions (lactose-modified chitosan)¹⁶² as reducing agent. The procedure for the synthesis is as follows: First, the aqueous solution of Chitlac was mixed with AgNO_3 solution. Ascorbic acid was added to this solution and after some time, the yellow-orange solution of silver nanoparticle was obtained.¹⁶³

BisGMA and TEGDMA were mixed and a mixture of CQ and DMAEMA was added into it. The solution was added into a Teflon mould. A hand cure light device was used for the light initiated polymerization and a photopol IR/UV Plus oven was used for post-curing. The resultant BisGMA/TEGDMA thermosets were then polished. The methacrylate esters were hydrolyzed to COO^- using HCl followed by heating. The thermoset was then removed

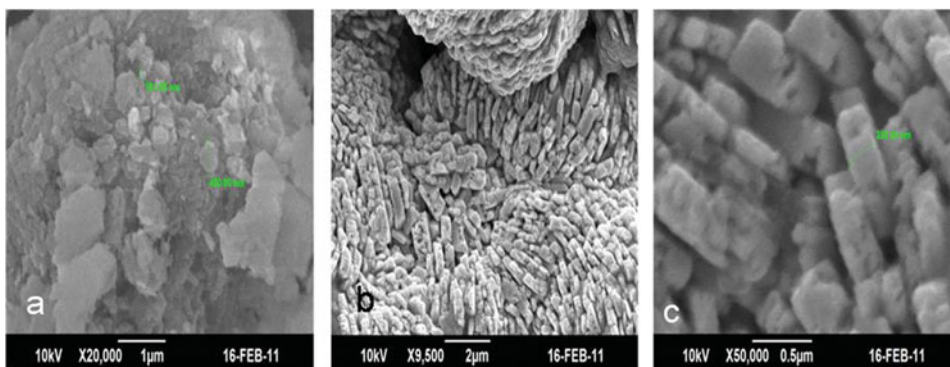


Figure 45. SEM image of (a) terpolymer (b) Ag nanoparticle loaded terpolymer (c) lower scale of (b).

and washed and air dried.¹⁶⁴ The functionalized samples were immersed in Chitlac-nAg and rinsed with deionized water. The samples were dried and sterilized by means of UV light. It was then characterized by a UV–Vis spectrophotometer, optical microscopy, TEM, and SEM. *In vitro* tests (for biological characterization) and antimicrobial efficacy tests were also carried out.

Silver–polysaccharide nanocomposite film is very effective in killing a strain of *P. aeruginosa* and a strain of *S. aureus* *in vitro*.

The nanocomposite does not show any *in vitro* cytotoxicity. It is also used for surface modification of medical devices.¹⁶⁵

2.8.6. With Polyvinylpyrrolidone Grafted Natural Rubber. Graft polymerization of NVP onto NR was done by *in situ* polymerization. Dilute aqueous solution of NRL was purged with nitrogen gas. CHP and TEPA was added to the rubber latex. The mixture was stirred and NVP and NH_3 solution was added. The product PVP-g-NR latex was centrifuged. The cream phase was collected AgNO_3 solution was added with constant stirring. The latex mixture was casted and dried. The resultant silver nanoparticles/PVP-g-NR composite films were exposed to UV light. The silver particles as well as the rubber particles were stabilized by PVP grafting.

The optical properties of the PVP-g-NR latex were measured using a spectrophotometer and were then characterized using FTIR spectra, TEM analysis, XRD, etc. Gravimetric and FTIR studies give the evidence for grafting. Face centered cubic (fcc) phase of metallic silver was obtained from XRD analysis. The core–shell structure of nanocomposite was revealed from TEM analysis.¹⁶⁶

2.8.7. With Poly(HEMA–PEGMA–MAA) Composite Hydrogel. A series of pH-responsive silver nanoparticle/poly(HEMA–PEGMA–MAA) composite hydrogel were prepared by *in situ* reduction of Ag^+ ions in hydrogel.

The PHPM (P(HEMA–PEGMA–MAA)) hydrogel was prepared by free radical polymerization of the monomers of HEMA, PEGMA, MAA, and the cross-linker MBA in water/methanol solution. APS solution and TEMED were added into the reaction system as an initiator and accelerator, respectively. At this stage polymerization took place and after the reaction, the hydrogel was washed with deionized water.

The prepared PHPM hydrogel samples were subjected to a swelling process in the $\text{H}_3\text{BO}_3/\text{CaCl}_2$ buffer solution. Then PHPM hydrogel samples were immersed in AgNO_3 solution. The excess AgNO_3 solution was removed and then again immersed in NaBH_4 solution. The prepared nanocomposite hydrogels were washed with deionized water. The PHPM1, PHPM4, PHPM10, and PHPM20 were also prepared in this manner.

Poly(HEMA–PEGMA) (PHP) hydrogel was also prepared by using the same method. By adjusting the swelling ratio of nanoparticle in hydrogels the physical state of the nanoparticle can be controlled. In order to characterize the composite system XRD, UV–Vis spectrophotometry, TEM, and electric conductivity tests were used. The Ag nanoparticle/PHPM nanocomposite hydrogel shows excellent properties such as higher swelling ratio, faster deswelling rate, and pH switchable electrical properties when compared with the pure PHPM hydrogel.¹⁶⁷

The metal nanoparticles are widely used for making functional structures. In order to improve the functions of the polymeric materials, the metal nanoparticles are incorporated into it. The combination properties of both of them are seen in the resulting composites. The composites of silver nanoparticles with the polymers like cotton fabric, PS, chitosan,

terpolymer, chitlac, PVP-g-NR, and PHPM are used in clinical paints, optics, phototherapeutic agents, anti bacterial coatings, and surface modification of medical devices, and they are also used as coatings of biomedical materials.

2.9. Gadolinium Oxide Nanoparticles

Gadolinium oxide is a basic oxide. Because of the presence of seven unpaired electrons in their 4f orbital, it shows paramagnetic character. It increases the longitudinal relaxation rate of protons in aqueous solutions and in biological tissues. Nanoparticles of gadolinium oxide act as a contrast agent for MRI analysis.

2.9.1. With Polyethylene glycol. Gadolinium oxide nanoparticles were prepared by a modified “polyol” process.¹⁶⁸ Here gadolinium chloride hexahydrate was dissolved in DEG with stirring and sodium hydroxide was added into it. A colloidal solution of gadolinium oxide nanoparticles was obtained.

Then a polysiloxane shell growth was introduced on gadolinium oxide nanoparticles. For this a DMSO solution of APTES and a mixture of dyes such as FITC, RBITC, and Cy5 mono-NHS-ester were prepared. In this step organic dyes conjugate with APTES. Gadolinium oxide nanoparticles were diluted in DEG solution. Organic dyes conjugated APTES, APTES, and TEOS were added to it and heated. Polysiloxane precursors and hydrolysis solution were added into the reaction mixtures. The polysiloxane coated nanoparticles obtained were purified.¹⁶⁹

Poly(ethylene glycol)bis(carboxymethyl)ether was added to isopropanol solution containing EDC and PFP. The mixture was then stirred to obtain pentafluorophenyl esters.^{170,171} When the pentafluorophenyl ester modified and PEG in DEG solution was added to a solution containing hybrid nanoparticles, the covalent grafting of PEG onto the hybrid nanoparticles was taking place. The colloid was then dialyzed (Fig. 46).

ICP-MS analysis was used for the determination of the gadolinium content in the sample and TEM to obtain structural and morphological information. NMR Spectroscopy was also used for their characterization. The photoluminescence spectra were recorded at the excitation wavelength of 646 nm. Magnetic measurements of the samples were also done.

PEG grafted gadolinium nanoparticles are used for molecular imaging purpose (Fig. 47) and clinical magnetic resonance imaging.¹⁷²

2.9.2. With poly(ethylene glycol)-phosphate. Gadolinium oxide nanoparticles were prepared by polyol method.^{173,174} $\text{Gd}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ was dissolved in DEG and NaOH was added into it. It was then heated, cooled, centrifuged, and filtered. The solution was dialyzed in ultrapure water or ethanol. The products were then sonicated to disperse the particles and centrifuged.

The obtained gadolinium oxide nanoparticles were deposited on flat silicon wafers. For this double-side polished silicon wafers washed in TL2 solution (a mixture of ultrapure water, 30% hydrogen peroxide, and concentrated hydrochloric acid) and dried. The dried silicon wafers were cleaned in TL1 solution (a mixture of ultrapure water, 30% H_2O_2 and 25% ammonia). A wafer was placed in a vacuum system and gadolinium(III) oxide pellets were kept directly facing the wafer. In order to coat gadolinium oxide layer on wafer an electron beam was focused onto the oxide pellets. The prepared Gd_2O_3 -coated silicon substrates were dipped into ligand solutions (solution1: ultrapure water, pH 7.0, 25°C;

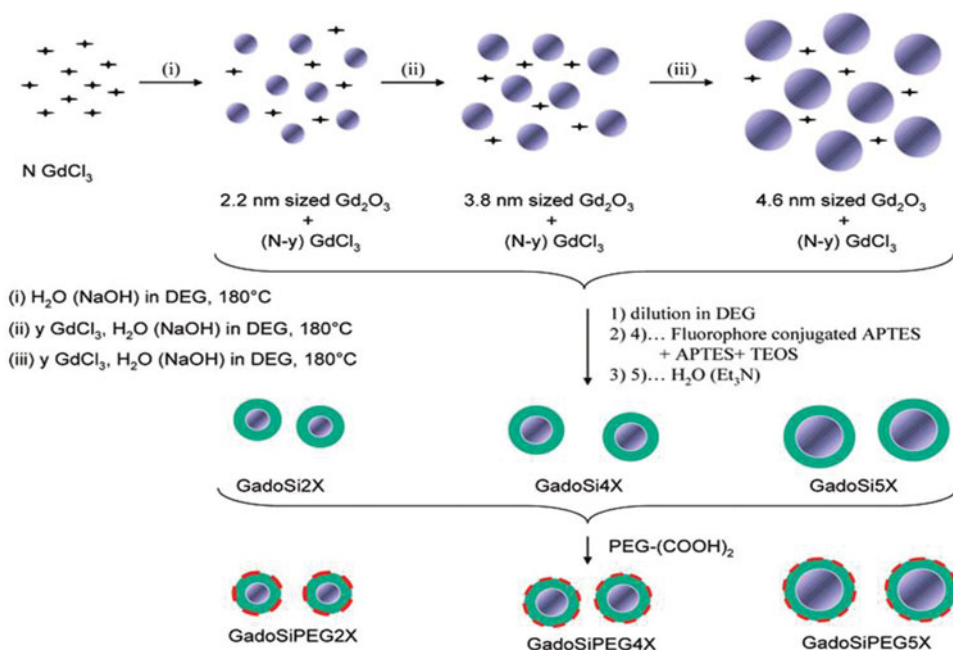


Figure 46. Synthesis and of the Surface Functionalization by PEG of Gadolinium Oxide Cores.¹⁷²
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solution 2: ethanol, 40°C) one by one. After the Gd_2O_3 -coated silicon wafers were rinsed with ultrapure water and ethanol and dried.

The dialyzed DEG- Gd_2O_3 suspension was added to PEG-phosphate solution. The mixture was sonicated and the products were collected and dialyzed. After dialysis the product was sonicated and centrifuged (Fig. 48).

Grafting of polymer on the nanoparticle was confirmed by XPS and FTIR (Fig. 49).

The PEG-phosphate grafted Gd_2O_3 nanoparticles are applied in molecular and cellular preclinical imaging purposes.¹⁷⁵

PEG, PEG-phosphate grafted Gd_2O_3 nanoparticles found applications in molecular imaging and clinical magnetic resonance imaging. It is an efficient “positive” contrast agent for molecular and cellular preclinical imaging applications.

2.10. Manganese Oxide Nanoparticles

MnO_2 nanoparticles utilized for advanced materials in batteries, image contrast agents, and water treatment. For efficient performance, these particles were grafted with polymers.

2.10.1. With Polyaniline. The surface of MnO_2 nanoparticles were modified by the hydrothermal method in two steps. In the first step a mixture of MnSO_4 and KMnO_4 was stirred, washed, dried, and heated. The dried powder was acidified with H_2SO_4 . The product was washed with distilled water and dried under vacuum.¹⁷⁶ In the second step MnO_2 nanoparticles were treated in boiling water. The hydroxylated nanoparticles were dispersed in a mixture of water and ethanol and ND42 (triethoxysilylmethyl N-substituted

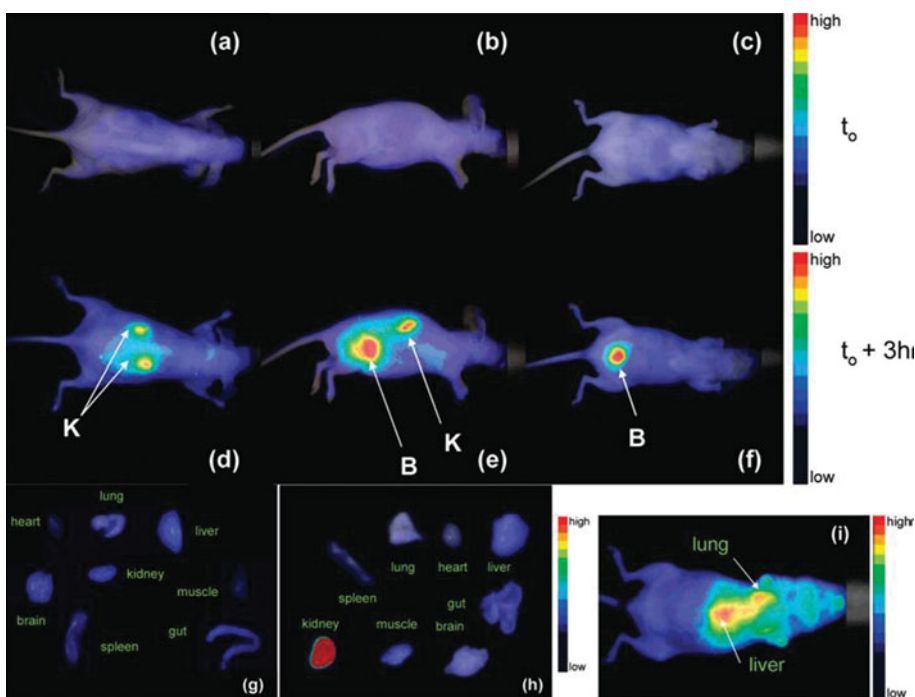


Figure 47. Fluorescence reflectance imaging of a nude mouse (a, b, c) before and (d, e, f) 3 hours after the injection of GadoSiPEG2C (K, kidneys; B, bladder). Fluorescence reflectance imaging of some organs after dissection (g) of a control mouse (no particles injection) and (h) of the nude mouse visualized on pictures (a-f). (i) Fluorescence reflectance imaging of a nude mouse after the injection of GadoSi₂C (particles without PEG). Each image is acquired with an exposure time of 200 ms.¹⁷² © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

aniline), sonicated, and stirred. The resultant modified MnO₂ nanoparticles (ND-MnO₂) were washed and then dried.

The electro-polymerization of aniline and ND-MnO₂ was carried out in a solution containing ND-MnO₂, aniline, H₂SO₄, and (NaPO₃)₆. The co-polymerization took place

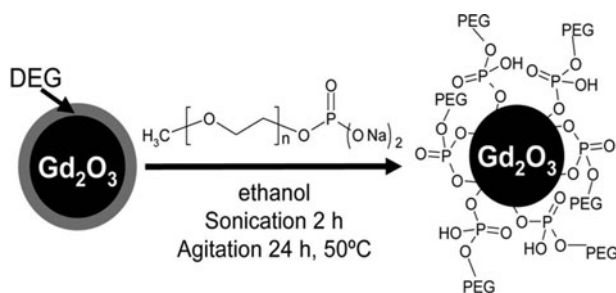


Figure 48. Reaction between PEG-phosphate and Gd₂O₃ nanoparticles.¹⁷⁵ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

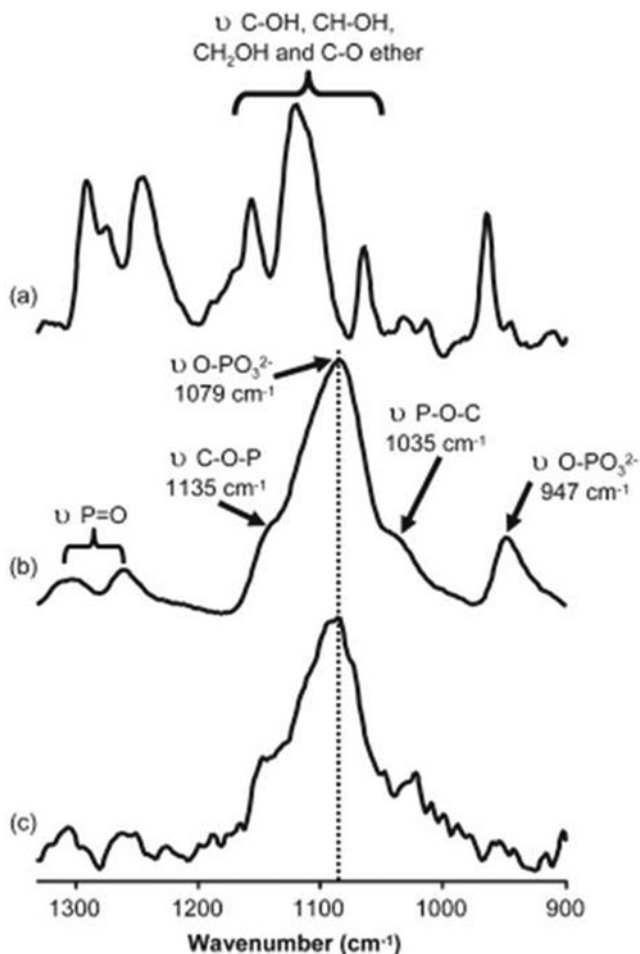


Figure 49. Infrared spectra of aqueous suspensions of DEG-Gd₂O₃ particles (a), pure PEG-phosphate (b), and PEG-phosphate-Gd₂O₃ nanoparticles after reaction in ethanol (c).¹⁷⁵ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

by repeated cyclic voltammetric scans. PANI-MnO₂ films (without ND42 linker) and PANI films were prepared in a similar way.

XRD and FTIR were used for the characterization of the above prepared compounds. The addition of MnO₂ nanoparticles promotes the one dimensional growth of PANI; it was clear from SEM images.

Specific capacitance increased in PANI-ND-MnO₂ film compared to the PANI-MnO₂ film. PANI-ND-MnO₂ film showed high intrinsic electrical conductivity, good kinetic reversibility, and very stable coulombic efficiency. This work can be helpful for developing new nanocomposite structures for super capacitors.¹⁷⁷

2.10.2. With Polyethylenimine. The reaction between MnCl₂·4H₂O and sodium oleate in 1-octadecene produces Mn-oleate complex. It was then degassed, heated, and cooled. The resultant MnO nanoparticles were purified and centrifuged. For synthesis of WMON, MnO

nanoparticles were mixed with DSPE-PEG in chloroform. Deionized water was added after evaporating the solvent. Then WMON were dispersed in phthalate buffer and stirred. The resulting HMON (Hollow manganese oxide nanoparticles) were collected by centrifugation and washed by water and centrifuged.^{178,179}

In order to synthesize DOPA conjugated branched PEI (PEI-DOPA) anhydrous DMF containing hydrocaffeic acid was taken and EDC dissolved in methanol was slowly added to it. Then branched PEI in methanol was allowed to react with this activated hydrocaffeic acid. The resultant solution was precipitated into excess ethyl ether. The polymer was purified and then lyophilized.

For the preparation of PEI-DOPA-coated hollow manganese oxide nanoparticles (HMON-PEI), a dispersion of HMON was added to PEI-DOPA. The obtained HMON-PEI was filtered and washed with PBS solution.

Herceptin (a therapeutic monoclonal antibody) -conjugated HMON-PEI (HMON-PEI-HER) was produced by reacting thiolated herceptin with maleimide-functionalized HMON-PEI. Herceptin was dissolved in PBS solution and then mixed with SATP in DMSO. PBS solution of hydroxylamine was added into it and incubated; the resulting thiolated herceptin was extracted with PBS solution. Maleimide-poly(ethylene glycol)-succinimidyl carbonate (MAL-PEG-SC) was added to HMON-PEI. The maleimide-functionalized HMON PEI by filtration was separated. It was reacted with thiolated herceptin and the resultant HMON-PEI-HER was purified.¹⁸⁰ Similarly, methoxy poly(ethylene glycol)-conjugated HMON-PEI (HMON-PEI-PEG) with mPEG-SS at the same concentration were also prepared in the same manner.

The resultant Herceptin-conjugated nanoparticles (HMON-PEI-HER) interact with siRNA (Small interfering RNA) to form nanosized polyelectrolyte complexes. These polyelectrolyte complexes readily recognized HER receptors expressed on human breast cancer cells, and then via a receptor mediated endocytosis entered in to the cells. Once internalized within cells, they would escape from the endosomal compartments and releasing their siRNA payloads into the cytoplasm. This released siRNA participate in the assembly of a RNA-induced silencing complex (RISC) for targeted gene silencing. Methods like FTIR, TEM, zeta potential measurements, etc., were used for the characterization.^{180,181}

In T₁-weighted MRI, the surface functionalized HMON helps in accurate detection of cancer cells. HMON-PEI and HMON-PEI-HER are multifunctional nanocarriers for cancer therapy.¹⁸²

The composites of MnO₂ with polymers PANI and PEI are used in super capacitors and for the detection of cancer cells in MRI.

2.11. Cerium Oxide Nanoparticles

Cerium oxide nanoparticles were prepared via three methods, viz: 1) hydrothermal crystallization with citric acid, 2) sonochemically by using the cerium nitrate, and 3) hydrolysis method.

2.11.1. With Poly Methylmethacrylate (PMMA). In order to get nano sized ceria, urea solution was dropped into Ce(NO₃)₃.6H₂O solution and refluxed. After the reaction, the mixture became opaque dispersion; the MPS in DMF were added in drop wise under nitrogen atmosphere. The resulting suspension was centrifuged, washed with ethanol, and dried in vacuum.

The MPS modified CeO₂ particles were dispersed in a mixture of MMA-toluene, sonicated and the initiator BPO was added. Three cycles of a freeze-thaw process were applied,

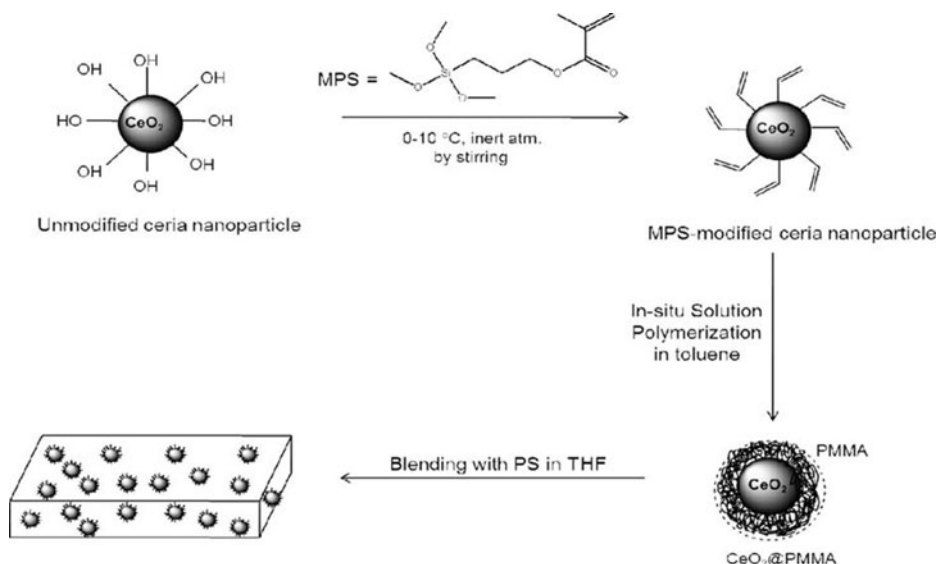


Figure 50. Preparation of optical PS composite films.¹⁸³ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

and placed in an oil bath. The resulting hybrid particles were isolated by centrifugation, washed with THF, and then dried in vacuum. Thus PMMA-Grafted CeO_2 was obtained.

For the preparation of the PS/PMMA-Grafted CeO_2 Composite Films, CeO_2 core-PMMA shell particles were dispersed in a solution PS-THF and stirred. After sonication, composite films were prepared using a Model WS 400B Spin Coater on glass or quartz substrates (Fig. 50).

The transmission of spin-coated PS/ CeO_2 composite films increased when the CeO_2 core was coated with a PMMA layer. In order to improve the optical properties of polymer/pigment systems, *in situ* solution polymerization method can be applied.

The resulting composite films were characterized by TGA, XRD, TEM, FTIR, etc. The average diameters of the PMMA-grafted CeO_2 particles were obtained by AFM (Fig. 51).¹⁸³

2.11.2. With Phosphonated-Polyethylene Glycol Oligomers. End-functional phosphonated-PEG oligomers was used for the stabilization of cerium oxide (ceria) nanoparticles. Cerium oxide nanoparticles were synthesized by thermal hydrolysis of an acidic solution of cerium-IV nitrate salt at high temperature and lower pH.¹⁸⁴ Addition of hydroxide ions changes the particles size. To increase the stability of nanoparticles, low molecular weight molecules or ligands like citric acid or poly(acrylic acid) were generally used.^{185,186} Phosphonated PEG oligomers were used as stabilizing agent and structure-directing agents to synthesize more complex structures.^{187,188} Also, PEG groups are widely used in biomedical applications to prevent nonspecific adsorption of proteins.^{189,190} The terminus phosphonate group is able to create a metal oxide hybrid sol.

CeO_2 -PPEG Hybrid Nanoparticles were prepared by mixing the solutions of CeO_2 nanoparticles and phosphonate-PEG (Fig. 52). The pH of the PPEG solution was controlled

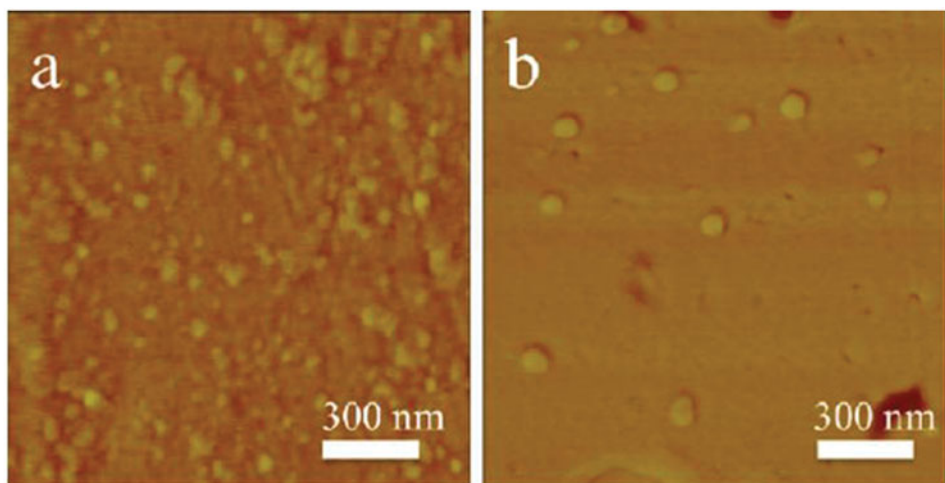


Figure 51. AFM phase images of PS composites loaded with (a) MPS-capped CeO_2 particles, (b) PMMA-grafted CeO_2 .¹⁸³ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

by adding HNO_3 . At pH 1.4, phosphonate groups are ionized. NH_4OH was used to adjust the pH of CeO_2 -PPEG dispersions.

As the pH increased, the solutions became destabilized. Here the cerium oxide particles were covered with the PEG layer. The functionalized nanoparticles can be used as an anti-UV protection agent. After freeze-drying the hybrid solution, the powder was able to redisperse in different solvents without losing its properties.¹⁹¹

Absorbance of the bare and polymer coated nanoparticle was measured by UV-visible spectroscopy (Fig. 53).

Rare earth cerium oxide (ceria) nanoparticles are extensively used in various technological fields. When it is stabilized with the polymers PMMA and PPEG, it becomes an anti-UV protection agent.

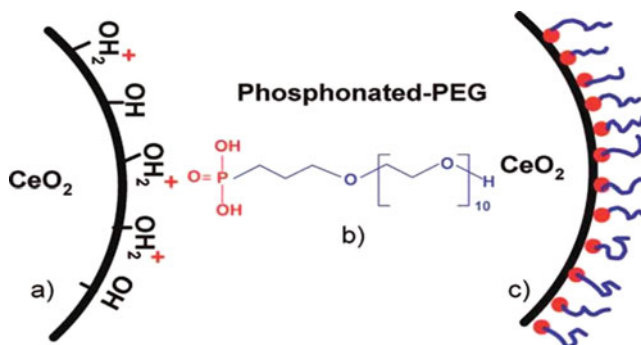


Figure 52. Preparation of CeO_2 -PPEG hybrid nanoparticles.¹⁹¹ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

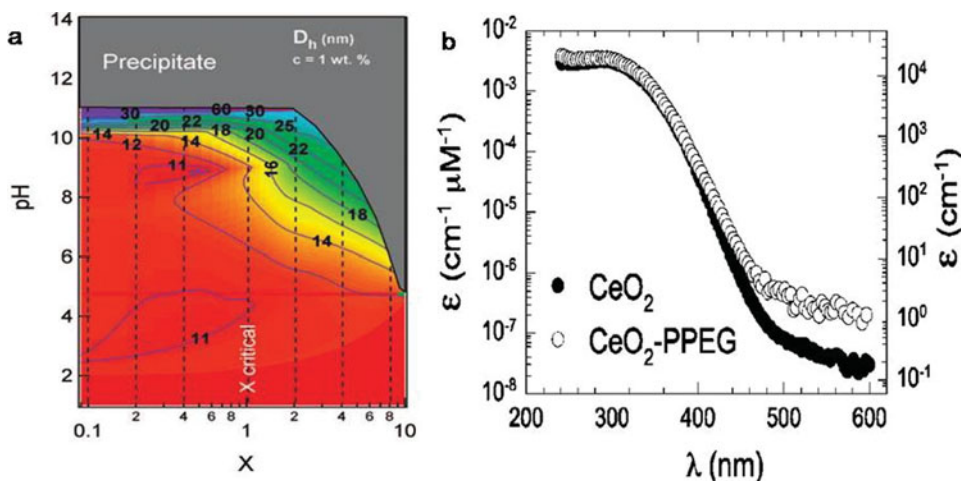


Figure 53. (a) Stability phase diagram of the PPEG-CeO₂ nanoparticle; (b) UV-visible spectra of bare and coated nanoceria.¹⁹¹ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

2.12. Copper Nanoparticle

Copper nanoparticles are conductive in nature and easy to prepare. It has high surface activity, excellent electro conductivity, and heat conductivity.

2.12.1. With Polystyrene. Thiolate/thiol-functionalized polystyrene prepared in a dry Schlenk flask with a magnetic stir bar containing dry THF. Styrene, ethylene sulfide, and macroinitiator solution were added in to it. The product PS-SH was obtained and stored under cold conditions.

A THF solution of thiolate-functionalized polystyrene added to the Copper (II)-acetylacetonate in THF. Then hydrazine was added drop wise into it and filtered. The filtrate was added into the excess methanol. The precipitated polystyrene stabilized copper nanoparticles were filtered, washed with water, methanol, and dried in vacuum. The hybrid materials were characterized by using the standard methods like TEM, XRD, UV-Vis spectroscopy, DSC, and GPC.¹⁴⁷

2.12.2. With Polyacrylamide grafted Poly(vinyl alcohol). The PAM-g-PVA graft copolymer was synthesized through free radical polymerization reaction initiated by a redox reaction. The AM monomer was added into the aqueous solution of PVA. After purging the ammonium cerium nitrate solution was poured into the mixture with stirring. The reaction mixture was heated, rinsed and dried. Polymer was dissolved in water, CuCl₂ and Na₂H₂PO₂ · H₂O were added to the mixture. After heating, the solution was coated and deposited on electrode to obtain an electrically conductive PAM-g-PVA/Cu film. The reaction of the PAM-g-PVA/Cu films to solvent vapors was accounted for by a swelling theory. This thin film has some electrical resistance properties.

PAM-g-PVA/Cu has some interaction among solvents. The solvent has significant role in the electrical properties of the film.¹⁹² For apolar or weak polar solvent vapors, this reduces the distance between Cu nanoparticles, that lead to a decrease in its resistance. For polar solvent vapors, the gap between Cu nanoparticles was increased, which made a metal

contact probability diminished and the metal conducting network broken, resulting in a large increase in resistance.^{193,194} PAM-g-PVA/Cu nanoconductive films could be utilized as sensing materials by adjusting the factors such as the ratio of PVA to AM monomer and Cu nanoparticle content.

FTIR was used for the structural characterization of PAM-g-PVA/Cu conductive thin films. Morphologies and size of Cu nanoparticles were obtained from TEM and average molecular weight of the graft copolymer was measured by using GPC.¹⁹⁵

Copper nanoparticles are applied in the fields of mechanics, magnetics, electrics, and chemistry. They are used as sensing material when grafted with PS and PAM-g-PVA.

2.13. Platinum Nanoparticles

Platinum nanoparticles have a high surface-to-volume ratio and as a result, a large portion of the metal atoms exposed at the surface are easy to get to the reactant molecules and accessible for catalysis.

2.13.1. With polybutyl methacrylate. Polymer grafted Pt nanoparticles were prepared by “grafting from” technique and controlled radical polymerization (atom transfer radical polymerization). 4-mercaptoaniline-functionalized PtNPs were prepared¹⁹⁶ and dissolved in DMAc. To this mixture DMAP and 2-bromoisobutyrate bromide were added and precipitated with acetonitrile, centrifuged, and purified.

For the Atom Transfer Radical Polymerization, initiator-derivatized PtNPs in DMAc, Cu(I)Br, and PMDETA were mixed under nitrogen and heated. Then monomer (n-butyl methacrylate) was added into it. The polymer grafted platinum nanoparticles were obtained (Fig. 54).

From small-angle neutron scattering studies it was clear that the radius of gyration of the objects was found to increase with the polymer conversion. This study also indicates that the overall structure is closer to a star architecture. The polymer layer was characterized using SANS technique (Fig. 55). TEM analysis shows the array formation.

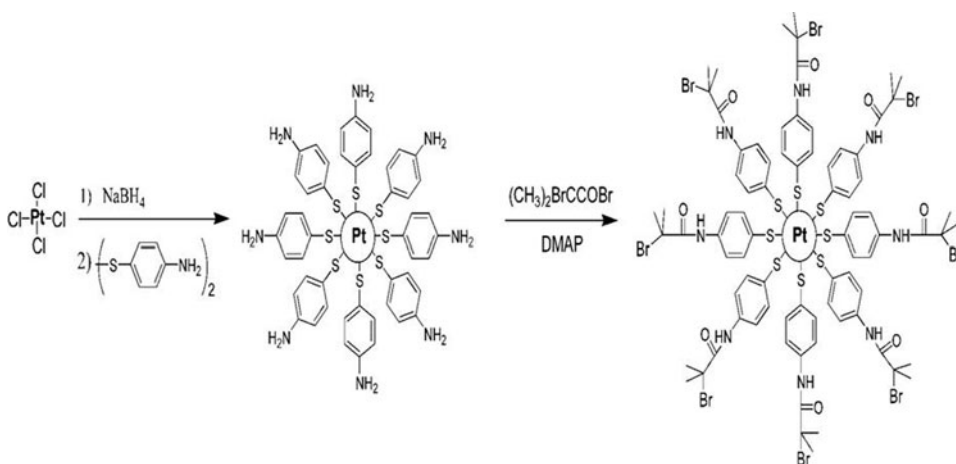


Figure 54. ATRP initiator functionalization of the PtNPs.¹⁹⁷ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

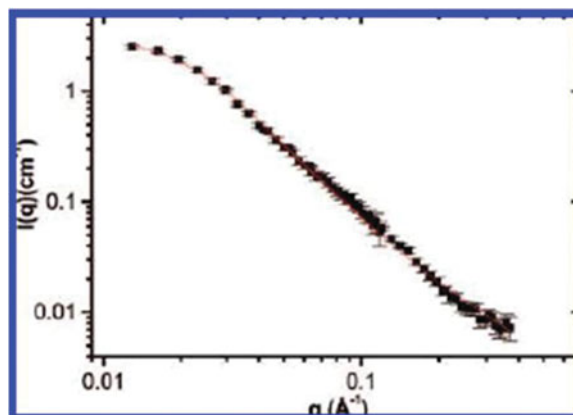


Figure 55. SANS spectrum of polybutyl methacrylate grafted PtNPs.¹⁹⁷ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

The polymer-grafted platinum nanoparticles are used in fuel cells or as biosensors.¹⁹⁷

2.13.2. With Polystyrene. Here cross linked organic and hybrid multilayer thin films based on polystyrene-grafted platinum nanoparticles and azidomethyl-functionalized polystyrenes are built-up by sequential spin-coating and UV cross linking processes.

Ligand mediated technique and derivatization or over-grafting reactions are used for the preparation platinum nanoparticles surface containing ATRP initiators.^{198–202} Styrene chains were grown on the nanoparticle by “grafting-from” method. Chloromethyl functionalized random polystyrene was prepared by NMRP copolymerization of styrene (d_8 -styrene) and 4-chloromethyl styrene. By using NaN_3 in dimethylformamide the resulted chloromethyl functionalized polystyrenes were converted into the corresponding azidomethyl-functionalized polystyrenes.

Multilayer thin films were prepared onto a polished silicon wafer by spin coating a dilute toluene of PSH (or PSD) (hydrido (PSH) and deuterated (PSD) azidomethyl-functionalized random polystyrenes) with or without polystyrene-grafted platinum nanoparticles solution. By UV curing the thin films were then cross linked. This process was repeated ‘n’ times to get a (hybrid) multilayer assembly having ‘n’ cross linked layers. TEM and neutron reflectivity experiments were carried out on this film, which indicate that the above approach allows easy tuning of the thickness, composition, and periodicity of each layer.

This method was helpful for the preparation of hybrid multilayer architectures and their stabilization by UV cross linking. The synthesized nanoscale systems have the potential applications in materials science and nanotechnology.²⁰³

2.13.3. With Poly(acrylic acid). Crystalline Pt nanoparticles were electrolessly deposited on PAA grafted MWNTs by *in situ* polymerization. AA, acetone, AIBN, and MWNTs were reacted to form PAA functionalized MWNTs. It was repeatedly washed to remove the ungrafted polymer.

In the next step, platinum nanoparticles were deposited on the PAA-g-MWNTs. In the experiment PAA-g-MWNTs were taken in a flask containing EG (solvent and reducing agent) and H_2PtCl_6 (platinum precursor). The reaction was carried out at different temperatures. Pt/ MWNTs nanocomposites were separated by centrifugation and washing. EDS,

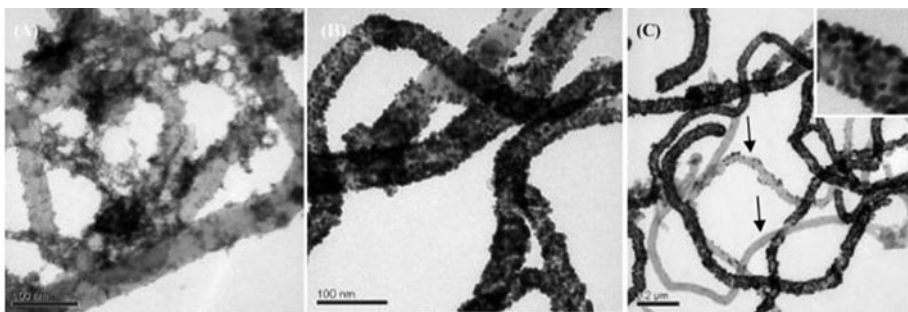


Figure 56. HR-TEM images of Pt nanoparticles deposited on different MWNTs: (A) pristine MWNTs, (B) PAA-g-MWNTs, and (C) mixture of pristine and PAA-g-MWNTs.²⁰⁴ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

TEM, and XPS were used for the confirmation of the presence of Pt on MWNTs (Figs. 56 and 57).

The synthesized nanocomposites are used as effective catalysts and are used for selective hydrogenation of α , β -unsaturated aldehydes.²⁰⁴

Here platinum nanoparticles are grafted with polybutyl methacrylate, PS, and PAA-g-MWNTs. The application of these nanocomposites are in fuel cells, materials science, nanotechnology, hydrogenation of α , β -unsaturated aldehydes, and as biosensors.

2.14. Palladium Nanoparticles

Palladium is one of the most efficient metals in catalysis.

2.14.1. With Chitosan-Grafted Graphene. CS grafted GR was synthesized as follows: Using modified Hummers method GR oxide was prepared and refluxed with SOCl_2 . To

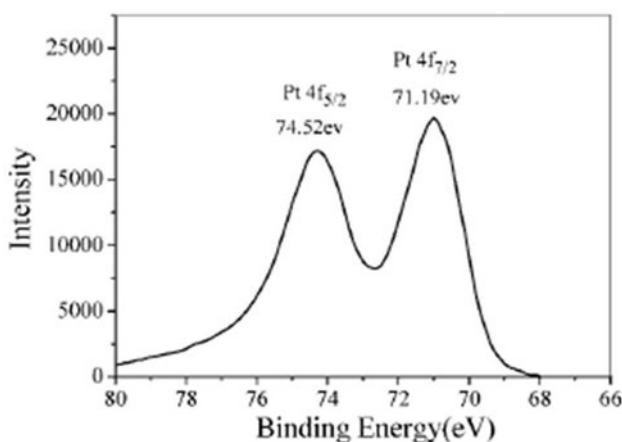


Figure 57. XPS spectrum of PAA-g-MWNTs.²⁰⁴ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

this a mixture of phthaloyl chitosan and LiCl dispersed in dimethylacetamide were added along with pyridine. The mixture was refluxed, filtered, dried, and the resulted powder was dissolved in hydrazine hydrate. After the mixture was filtered, washed, and dried to obtain CS-GR composites. The obtained product was suspended in a mixture of water and ethylene glycol and sonicated. After sonication palladium chloride in ethylene glycol was added to the reaction mixture and the pH was adjusted with HCl and NaOH. It was then stirred to attach the palladium nanoparticles on the surface of CS-GR composites. After heating and cooling the reaction mixture was filtered through a PTFE membrane, washed with ethanol and water, and dried under vacuum to obtain PdNPs/CS-GR nanocomposites.

The prepared PdNPs/CS-GR nanocomposites were coated on a GCE as follows: First, the glassy carbon electrode was polished with α -alumina slurries and sonicated. PdNPs/CS-GR nanocomposites were dispersed in aqueous solution of acetic acid by sonication. The resulting PdNPs/CS-GR suspension was coated on the surface of a glassy carbon electrode and dried.

The PdNPs/CS-GR modified electrode was then immersed in glutaric dialdehyde solution, rinsed with water, and dried. An aliquot of GOD was placed on the surface of the modified electrode and dried. The obtained GOD-immobilized electrode (glucose biosensor) was dipped in phosphate buffer and stored under cold condition.

The developed glucose biosensor has high sensitivity, wide linear range, low detection limit, and good reproducibility. The PdNPs/CS-GR nanocomposites can be applied in various oxidase-based biosensors and bioelectronic devices.²⁰⁵

2.14.2. With Polyaniline. The GCE was modified with 4-ASA to get the sulfonic group grafted GCE surface. GCE was immersed in a mixture of 4-ASA and LiClO₄ in ethanol. The 4-ASA-modified GCE (4-ASA/GCE) with rich sulfonic groups was obtained after being rinsed with ethanol and water.^{206,207}

The 4-ASA-modified GCE (4-ASA/GCE) were dipped in the positively charged PANI and negatively charged PdCl₆²⁻ solutions. Then the film was repeatedly washed with HCl to obtain the desired number of (PANI/PdCl₆²⁻)_n multilayer film and dried. Finally, applying a potential of -0.7 V in N₂-saturated KCl solution to obtain (PANI/Pd₀)_n nanocomposites.

The resulting nanocomposite film was analyzed by UV-Vis absorption spectra, XPS (gives more idea about bonding characteristics of the film, and oxidation state of Pd before and after potential reduction), AFM (for morphological characterization), FTIR, and TEM. The variation of film properties were demonstrated by using electrochemical impedance experiments.

It shows various applications in micro and nanoelectronic devices such as electrochemical sensor and biosensor due to the high catalytic activity, conductivity, redox activity, and stability of the nanocomposites film.²⁰⁸

The palladium nanoparticles are used as biosensors and also used in bioelectronics devices (micro and nano), when it is grafted with CS-GR and PANI.

2.15. Rhodium Nanoparticles

Rhodium nanoparticle is an important candidate for the catalytic application.

2.15.1. With Polyvinyl Pyrrolidone. In order to prepare of PVP coated Rhodium nanoparticle (RhNPs), RhCl₃ (precursor), and PVP (stabilizer) were first dissolved in deionized water at room temperature. Then NaBH₄ (reducing agent) solution was added to this solution with stirring. PVP coated RhNPs were formed due to the strong interaction of metal

surface and the carbonyl group of the pyrrolidone ring in PVP.^{209,210} From the resultant solution the RhNPs were collected by centrifugation, and washed with water and ethanol.

The polarity of the solvent may greatly influence the aggregating nature of PVP coated nanoparticles. The higher the polarity of the solvent, the higher the aggregation rate of RhNPs. To investigate the influence of the solvent polarity on the nature of RhNPs, ethanol, and water-ethanol mixtures were used instead of pure water as solvents. The product changed from small RhNPs in ethanol to large RhNPs in water. When water-to-ethanol ratio increases in the reaction media, RhNPs form more aggregates.

SEM and TEM were used for the characterization of RhNPs and their assemblies. The two-step formation process viz. initial rapid particle generation step followed by a slow particle aggregation step was revealed by TEM.²¹¹

2.16. Barium Titanate Nanoparticles

Barium titanate (BaTiO_3) is a ceramic material with a perovskite crystal structure and has exceptional ferroelectric and piezoelectric properties.

2.16.1. With Poly Methylmethacrylate. A dispersion of BaTiO_3 nanoparticles in a mixture of PEDOT:PSS aqueous solution was prepared by sonication. The solution was then heated and stirred magnetically. After the complete removal of water the PEDOT:PSS coated BaTiO_3 nanoparticles were obtained. It was then dried in vacuum and stored in a vial.

The functionalized nanoparticles were dispersed in chloroform solution of PMMA by a sonicator. The dispersions were coated on each of the two sides of a fiberglass patch. The solvent in between individual spreads was evaporated on a hot plate. The obtained polymer nanocomposites were compared under hot condition and then cooled.

In order to establish the successful coating of BaTiO_3 nanoparticles with the PEDOT:PSS blend, FTIR spectroscopy, TGA and SEM measurements were conducted. A good dispersion of coated nanoparticles throughout the PMMA matrix was shown by TEM analysis. Novocontrol dielectric spectrometer was used for the characterization of dielectric properties.²¹²

The BaTiO_3 filler increases the decomposition temperature and glass-transition temperature of the polymeric composites. So it can be used in high frequency alternating current and direct current applications.²¹³

2.17. Europium doped Gadolinium Oxide Nanoparticles

Here $\text{Eu:Gd}_2\text{O}_3$ nanoparticles were synthesized by spray pyrolysis; it has narrow red emission, large Stokes shift, and photostable laser-induced fluorescence with a long lifetime.

2.17.1. With Poly(L-lysine) hydrobromide. Eu^{3+} -doped nano gadolinium oxide ($\text{Eu:Gd}_2\text{O}_3$) particles were biofunctionalized with Poly(L-lysine) Hydrobromide. The functionalization process proceeds as follows: A suspension of $\text{Eu:Gd}_2\text{O}_3$ nanoparticles in Phosphate buffer were prepared using ultrasonication. PL solution was added to the suspension and incubated. The pure PL functionalized $\text{Eu:Gd}_2\text{O}_3$ nanoparticles ($\text{PL-Eu:Gd}_2\text{O}_3$) were obtained by centrifugation and redispersion.

The biofunctionalized $\text{Eu:Gd}_2\text{O}_3$ nanoparticles were then conjugated with antibodies. At first, using sodium periodate the antibody molecule was oxidized. The synthesized $\text{PL-Eu:Gd}_2\text{O}_3$ nanoparticles in carbonate buffer were added to it. In order to stabilize the Schiff bases formed during reaction sodium cyanoborohydride was added. Then ethanolamine

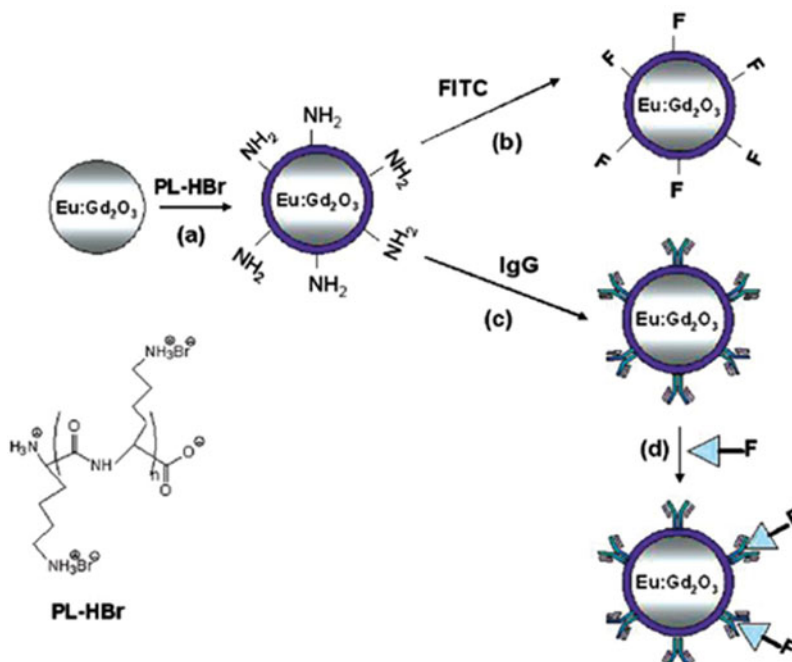


Figure 58. Preparation of biofunctional Eu:Gd₂O₃ nanoparticles: (a) encapsulation with poly(L-lysine) hydrobromide (PL), (b) quantification of amino groups by coupling to fluorescein isothiocyanate (FITC); fluorescein (F), (c) covalent coupling to oxidized antibody (IgG), and (d) quantification of antibody binding sites by fluorescein-labelled antigen.²¹⁴ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

was added to block the unreacted aldehyde sites on the antibodies. The obtained oxidized antibody-nanoparticle conjugate was subjected to purification by washing with PB. The resulting bioconjugates were washed (Fig. 58).

The TEM analysis, colloidal stability studies, and determination of the free amino groups on the surface of the encapsulated nanoparticles were used for the confirmation of the formation of the PL layer (Fig. 59).

Conventional fluorophores can be replaced with PL-Eu:Gd₂O₃ bioconjugates because of their properties like excellent photo stability, long lifetime, and low-cost production and it is also used in immunoassays. The synthesized PL-Eu:Gd₂O₃ bioconjugates find application in the field of biotechnology, and are used especially in biosensors, protein microarrays, enzyme assays, and high-throughput screening assays.²¹⁴

2.18. Alumina Zirconia Nanoparticles

Commonly, the sol-gel method is used for the preparation of Al₂O₃-ZrO₂ composite nanoparticles.

2.18.1. With Collagen. Al₂O₃-ZrO₂ composite nanoparticles were prepared by the following procedure. First, pure aluminum and zirconium were dissolved in nitric acid followed by the mixing of citric acid and water with it. It was then heated and ammonia was added

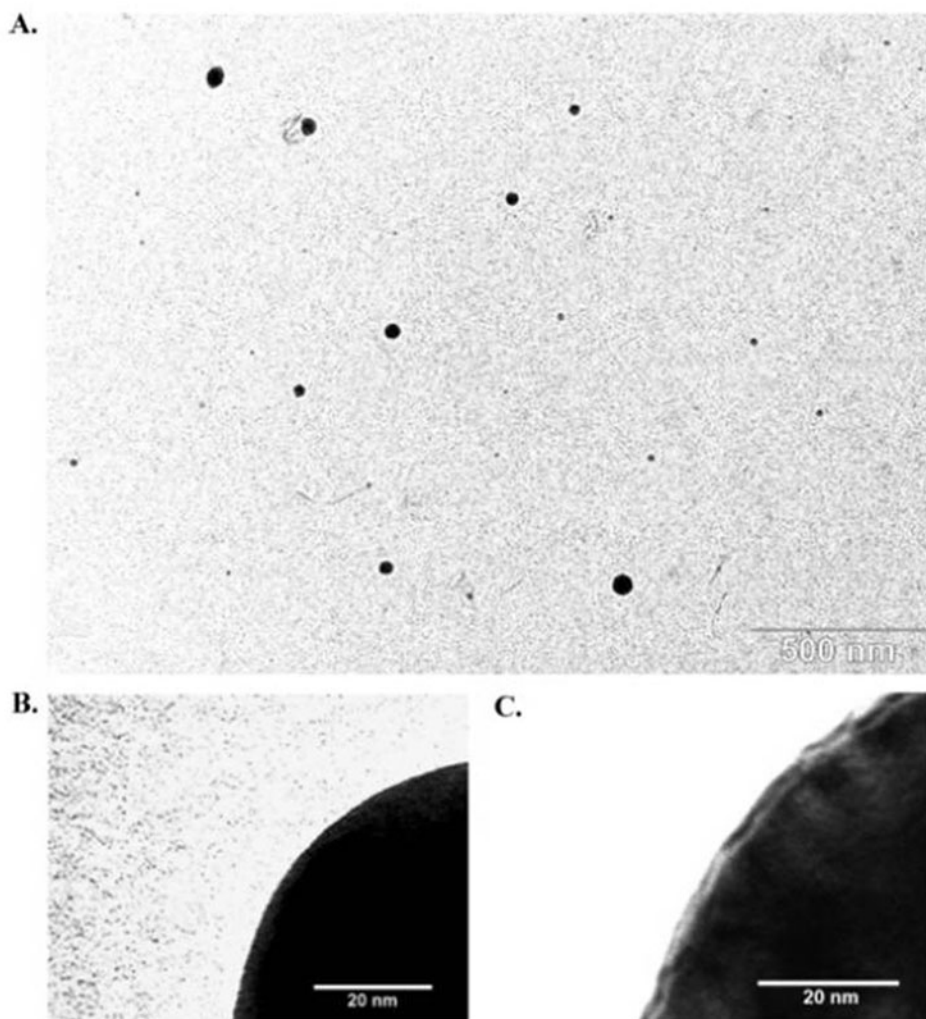


Figure 59. TEM images of Eu:Gd₂O₃ nanoparticles: (A) overview of non coated sized nanoparticles; (B) high-resolution image of the bare surface of non coated nanoparticle; (C) high-resolution image of the poly(L-lysine) shell on the surface of the poly(L-lysine)-encapsulated nanoparticle.²¹⁴ © The American Chemical Society. Reproduced by permission of the American Chemical Society. Permission to reuse must be obtained from the rightsholder.

drop wise. At this stage a sol was formed and the resultant gel particles were collected by filtration. It was then dried and converted into powder. The resultant powder was heated to obtain Al₂O₃–ZrO₂ composite nanoparticles.

In the next step the synthesized Al₂O₃–ZrO₂ composite nanoparticles were grafted to Collagen (a bioscaffold polymer). At first, the Collagen Matrix was synthesized as follows: Methanolic solution of collagen was prepared and methyl methacrylate monomer, ammonium cerium (IV) nitrate in nitric acid, and 2, 2'-azobis(2-methylpropionitrile) in methanol was added into the reaction mixture. It was then heated with stirring. The obtained solid product was collected, washed, and extracted with acetone.

Ethanol dispersion of the synthesized collagen matrix and $\text{Al}_2\text{O}_3\text{--ZrO}_2$ composite nanoparticles were prepared. The pure nanocomposites were obtained by ultrasonication and refrigeration.

Stress–strain curves of composites were recorded and their tensile tests, storage modulus tests, hardness and modulus tests were performed. Grain-like morphology and particle size were obtained from TEM analysis. SEM images show compact and disordered stacking arrangement with some agglomeration of the grafted collagen. And thermal analysis (TGA and DTA) were also carried out.

When used in medical fields, the collagen part of the synthesized polymer nanocomposites exhibits less immunogenic properties. Grafting method is used for reinforcing the mechanical and thermal properties of collagen, and therefore the resultant products can be used for biomedical and bionic applications.²¹⁵

3. Conclusion

Polymer nanohybrids are compounds formed by the combination of polymers and nanoparticles. This review presents an inclusive resource for the preparation of inorganic nanoparticles (metal, metal oxide, metal hydroxide, and metal complex), their functionalization, grafting to polymers, and properties and applications of polymer nanocomposite materials. Some common applications of polymer grafted nanoparticles are better solubility or dispersibility in various mediums, preventing the agglomeration of particles and improving its chemical and electrical properties. This grafting process has not altered its crystalline structure but shows some deviation in its crystalline size. After functionalizing the nanoparticles, their surfaces are grafted with polymers. Generally, two methods are used for the surface modification of inorganic nanoparticles. One is the physical method, in which the nanoparticle is weakly bonded to some coupling agents. The other is the chemical method, in which the nanoparticles form a chemical bond with the polymer. Sometimes both nanoparticle and polymer were functionalized to create a favorable atmosphere for grafting.

The characterization was done by several chromatographic, spectroscopic, and thermal instruments such as SEM, VSM, TEM, DSC, XRD, TGA, XPS, GPC, NMR, UV, FT-IR, etc.

Both linear and hyperbranched polymers were used for grafting. But reports for modification based on hyperbranched polymers are rare. The physical properties of these polymers are good solubility and low melt viscosity. The affinity of these hyperbranched polymers for organic solvents and polymer matrixes are due to the presence of high functional group density. So these types of polymers are used as a surface modifier and interfacial agent compared to linear polymers and they have many applications in industrial and pharmaceutical fields.

Nowadays inorganic nanoparticles grafted with polymer are an area of special interest because of the enhanced properties of both polymer and nanoparticle. This review covers the general topics of polymer grafted nanoparticles and its preparation, properties, and applications. The recent developments in the field of polymer nano-composites have been reviewed here. This review presents an opportunity for a widespread discussion on preparation, properties and applications of various polymer nanocomposites. We hope that this review would be very helpful to the research community to get an overall idea about the polymer grafted nanoparticles.

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Nomenclature

AA, AAc	Acrylic acid
ACN	Acetonitrile
ACPA	4,4'-Azobis (4-cyanopentanoic acid)
AES	Auger electron spectroscopy
AIBN	2,2' -Azobisisobutyronitrile
AM	Acrylamide
APS	Aminopropyltriethoxysilane
APSNs	γ -aminopropyl silica nanoparticles
APTES, APTS	(3-Aminopropyl)triethoxy silane
APTMS	(3-Aminopropyl)trimethoxy silane
ASA	Aminobenzene sulfonic acid
ATMS	Allyltrimethoxysilane
ATRP	Atom transfer radical polymerization
BA	α - Bromoisobutyric acid
BBEA	2-(((bromobutyryl)oxy) ethyl acrylate)
BDDA	Butanediol dimethacrylate
BIBB	2-Bromoisobutyl bromide
Bis GMA	Bisphenol A glycidylmethacrylate
Bis-MPA	2, 2-bis (hydroxymethyl) propionic acid
BL	Butyrolactam
BP	Benzophenone
BPO	Benzoyl peroxide
Bpy, bpy, bipy, Bip	2,2' -Bipyridine
BrA-ZnO	Bromo acetamide modified ZnO
BSA	Bovine serum albumin
BTPAm	2-Bromo-2-methyl-N-(3-(triethoxysilyl) propanamide
BTPT	S-Benzyl Strimethoxysilylpropyltrithiocarbonate
CAN	Acetonitrile
CCTP	Catalytic chain transfer polymerization
CDPU	Cyclodextrin-polyurethane polymer
CHP	Cumene hydroperoxide
CMS	p-Chloromethyl styrene
CoBF	Bis(aqua)bis((difluoroboryl)-dimethylglyoximato) cobalt(II)
CPA	3-chloropropionic acid
CPDB	2-cyano-prop-2-yl dithiobenzoate
CQ	Camphorquinone
CTAB	N-Cetyl trimethyl ammonium bromide
CTCS	2-(4-chlorosulfonylphenyl) ethyltrichlorosilane
CS	Chitosan
CuAAC	Copper Catalyzed Azide Alkyne Cycloaddition
DABA	3,5-Diaminobenzoic acid

DBTDL	Dibutyltin dilaurate
DCM	Dichloromethane
DDE	4,4'-Diaminodiphenyl ether
DDM	Diaminodiphenyl methane
DDS	4,4'-diaminodiphenyl sulphone
DEG	Diethylene glycol
DLS	Dynamic light scattering
DMA	Dimethylaniline
DMAAm	N,N-dimethylacrylamide
DMAc	N,N-dimethylacetamide
DMAEMA	2-N,N-dimethylamino ethyl methacrylate
DMAP	Dimethylaminopyridine
DMF	N,N-dimethyl formamide
DMS	Dimethyl siloxane
DMSO	Dimethyl sulfoxide
DOPA	L-3,4-dihydroxyphenylalanine
DPE, DPh	Diphenyl ether
DSPE-PEG	1,2-Distearoyl-sn-glycero-3-phosphoethanolamine-N-methoxy(polyethylene glycol)
DSC	Differential scanning calorimetry
DVB	Divinylbenzene
EDA-FA	N-(2-aminoethyl) folic acid
EDC	1-ethyl-3-(3-dimethylaminopropyl)carbodiimide
EDS	Energy dispersive x-ray spectroscopy
EELS	Electron energy loss spectroscopy
EG	Ethylene glycol
EGDMA	Ethylene glycol dimethacrylate
EHTES	5,6-epoxyhexyltriethoxysilane
Fe(acac) ₃	Iron (III) acetylacetonate
FITC	Fluorescein isothiocyanate
FT-IR	Fourier transforms infrared
GCE	Glassy carbon electrode
GMA	Glycidylmethacrylate
GOCF	Grafted oxidized cotton fabric
GOD	Glucose oxidase
GPC	Gel permeation chromatography
GR	Graphene
HAPE	Hyperbranched aliphatic polyester
HD	Hexadecane
HEA	Hydroxyethyl acrylate
HEMA	2-Hydroxyethyl methacrylate
HMDZ	Hexamethyldisilazane
HMON	Hollow manganese oxide nanoparticles
HPPS	High performance particle sizer
HP, HBP	Hyperbranched polymer
HRP	Horseradish peroxidase
HTPB	Hydroxyl terminated polybutadiene prepolymers
IA	Itaconic acid
ICP-MS	Inductively Coupled Plasma-Mass Spectrometry
KPS	Potassium persulfate

LCST	Lower critical solution temperature
LLS	Laser light scattering
MAA	Methacrylic acid
MASi	<i>Tert</i> -butyldimethylsilyl methacrylate
MATAC	Methacryloxyethyl- trimethyl ammonium chloride
MBA	N,N' –Methylenebisacrylamide
Me ₆ TREN	Tris(2-(dimethylamino)ethyl)amine
MIBK	Methylisobutylketone
MIP	Molecularly imprinted polymer
MMA	Methyl methacrylate
MMP	Maleimido methyl palmitate
MNP	Magnetic nanoparticle
MPC	2-methacryloyloxyethyl phosphorylcholine
MPSNP	Mesoporous silica nanoparticles
MPTES, MPS	γ -Methacryloxypropyltriethoxysilane
MPTMS	(3-mercaptopropyl)trimethoxysilane
MRI	Magnetic resonance imaging
MSMA	γ -methacryloxypropyltrimethoxy silane
MSNs	Mesoporous silica nanoparticles
MTU	Maleimido thiourea
MWNTs	Multiwalled carbon nanotubes
NHS	N-hydroxysuccinimide
NIPAM, NIPAAm	N-isopropyl acrylamide
NMP	Nitroxide mediated polymerizations
NMR	Nuclear magnetic resonance
NP	Nanoparticle
NR	Natural rubber
NRL	Natural rubber latex
NVP	N-vinyl-2-pyrrolidone
OTS	Octadecyl trichlorosilane
PAA	Poly(amic acid)
PAAc	Poly Acrylic Acid
PAM, PAAM	Polyacrylamide
PANI	Polyaniline
PB	Phosphate buffer
PBS	Phosphate-buffered saline
PCMS	Polychloromethyl styrene
PCS	Photocorrelation spectroscopy
PDMS	Poly(dimethyl siloxane)
PEDOT	Poly (3,4-ethylenedioxythiophene)
PEG	Polyethylene glycol
PEGMA	Poly(ethylene glycol) methacrylate
PEI	Polyethylenimine
PET	Poly(ethylene terephthalate)
PFP	Pentafluorophenol
PGMA	Poly(glycidylmethacrylate)
PPHM	Poly(2-Hydroxyethyl methacrylate- poly(ethylene glycol) methyl ether methacrylate- methacrylic acid)
PL	Poly(L-lysine) hydrobromide

PLLA	Poly(L-lactide)
PMDA	Pyromellitic dianhydride
PMDETA	N,N,N' N'' N'''-Pentamethyldiethylenetriamine
PMMA	Polymethyl methacrylate
PMPC	Polymethacryloyloxyethyl phosphorylcholine
PNIPAM	Poly(N-isopropyl acrylamide)
POEOMA	Poly- (oligo(ethylene oxide) monomethyl ether methacrylate)
PP	Polypropylene
PS	Poly Styrene
PSBMA	Poly(sulfobetaine methacrylate)
PSS	Polystyrene sulfonate
PTFE	Poly(tetrafluoroethylene)
p-TSA	p-Toluenesulfonic acid
PVA	Poly(vinyl alcohol)
PVP	Polyvinylpyrrolidone
P2VP	Poly(2-vinylpyridine)
RAFT	Reversible addition-fragmentation chain transfer
RBITC	Rhodamine B isothiocyanate
SANS	Small-angle neutron scattering
SATP	N-Succinimidyl-S-acetylthiopropionate
SBMA	Sulfobetaine methacrylate
SBR	Styrene butadiene rubber
SC	Succinimidyl carbonate
SCVP	Self condensing vinyl polymerization
SEC	Size exclusion chromatography
SEM	Scanning electron microscopy
SI-ATRP	Surface-initiated atom transfer radical polymerizations
SIP	Surface-initiated polymerization
SIPP	Surface-initiated photopolymerization
SNPs	Silica nanoparticles
SS	Succinimidyl Succinate
St	Styrene
STEM	Scanning transmission electron microscopy
t-BA	t-Butyl acrylate
TBHP	t-Butyl hydroperoxide
TDI	Toluene 2,4-diisocyanate
TEA	Triethylamine
TEGDMA	Triethyleneglycol dimethacrylate
TEM	Transmission electron microscopy
TEMED	N,N,N',N'-Tetramethylenediamine
TEOS	Tetraethoxysilane
TEPA	Tetraethylenepentamine
TGA	Thermogravimetric analysis
THF	Tetrahydrofuran
TFA	Trifluoro acetic acid
TMPTA	Trimethylpropane triacrylate
TMSPM, TPM	3-(Trimethoxysilyl)propyl methacrylate
TsCl	p-toluenesulfonyl chloride
UCST	Upper critical solution temperature

UV	Ultra violet
VBDC	4-vinylbenzyl N,N-diethyldithiocarbamate
VSM	Vibrating sample magnetometry
VTES	Vinyl-triethoxysilane
WAXD	Wide angle x-ray diffraction
WMON	Water dispersible manganese oxide nanoparticles
XPS	X-ray photoelectron spectroscopy
XRD	X-ray diffraction
XRF	X-ray fluorescence analysis

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